9. Clock Generation Circuit

9.1 Overview

The MCU provides a clock generation circuit. Table 9.1 and Table 9.2 list the clock generation circuit specifications. Figure 9.1 show a block diagram, and Table 9.3 lists the I/O pins.

Table 9.1 Clock generation circuit specifications for the clock sources (1 of 2)

Clock source	Description	Specification
Main clock oscillator (MOSC)	Resonator frequency	8 MHz to 48 MHz
	External clock input frequency	Up to 48 MHz
	Connectable resonator or additional circuit: ceramic resonator, crystal Connection pins: EXTAL, XTAL	Available
	Drive capability switching	Available
	Automatic Gain Control Function	Available
	Oscillation stop detection function	Available
Sub-clock oscillator (SOSC)	Resonator frequency	32.768 kHz
	Connectable resonator or additional circuit: crystal resonator Connection pin: XCIN, XCOUT	Available
	External clock input	Available
	Drive capability switching	Available
	Oscillation stop detection function	Available
PLL1 circuit PLL2 circuit	Input clock source	MOSC/HOCO
	Input pulse frequency division ratio	Selectable from 1/2/3
	Input clock frequency	8 MHz to 48 MHz
	Input clock frequency (After input frequency division)	8 MHz to 24 MHz
	Frequency multiplication ratio	Selectable from 40 to 300 (after the decimal point : 0/0.33/0.50/0.66)
	VCO frequency	960 MHz to 2400 MHz
	Number of output clocks	Output 3 different clocks
	Output pulse frequency division ratio	Number of output clocks Output 3 different clocks
	PLL Output clock P	60 to 1200 MHz (output division ratio : 2/3/4/6/8/16)
	PLL Output clock Q	106.6 to 1200 MHz (output division ratio : 1.5/2/3/4/5/6/8/9)
	PLL Output clock R	106.6 to 1200 MHz (output division ratio : 1.5/2/3/4/5/6/8/9)
High-speed on-chip oscillator (HOCO)	Oscillation frequency	16/18/20/32/48/ MHz
	FLL function	Available
	User trimming	Available
Middle-speed on-chip oscillator (MOCO)	Oscillation frequency	8 MHz
	User trimming	Available
Low-speed on-chip oscillator (LOCO)	Oscillation frequency	32.768 kHz
	User trimming	Available

Table 9.1 Clock generation circuit specifications for the clock sources (2 of 2)

Clock source	Description	Specification
External clock input for JTAG (TCK)	Input clock frequency	Up to 25 MHz
External clock input for SWD (SWCLK)	Input clock frequency	Up to 25 MHz

Table 9.2 Clock generation circuit specifications for the internal clocks (1 of 3)

Item	Clock source	Maximum Frequency of Clock Source	Clock supply	Specification
CPU0 clock (CPUCLK0)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	CPU0	Up to 1 GHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
CPU1 clock (CPUCLK1)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	CPU1	Up to 250 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
NPU clock (NPUCLK)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	NPU	Up to 500 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
MRAM bus clock (MRICLK)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	Code MRAM, MRAM code bus	Up to 250 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
MRAM clock (MRPCLK)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	Extra MRAM, MRAM peripheral bus	Up to 125 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
System clock (ICLK)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	DMAC, DTC, SRAM, System BUS, I/O Ports, ICU, DRW	Up to 250 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
Debugger clock (DCLK)	ICLK	—	Debug Subsystem	Up to 250 MHz
Peripheral module clock A (PCLKA)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	High-speed peripheral bus (ESWM, , USBHS, SCI, I3C, CANFD, CNECC, SPI, CRC, DOC, ADC16H, RSIP-E50D, GPT, PDG, DOTF AES clock, CEU, MIPI, LCDC bus)	Up to 125 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
Peripheral module clock B (PCLKB)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	Low-speed peripheral bus (CAC, ELC, POEG, RTC, WDT, IWDT, AGT, ULPT, IIC, USBFS, SSIE, SDHI, TSN, ACMPHS, OSPI, DOTF bus clock, DAC12, PDMIF)	Up to 62.5 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
Peripheral module clock C (PCLKC)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	Not used	Up to 125 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
Peripheral module clock D (PCLKD)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	Peripheral module(GPT count clock, PDG)	Up to 250 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
Peripheral module clock E (PCLKE)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	Peripheral module (CANFDRAM)	Up to 250 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
External bus clock (BCLK)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	External bus	Up to 125 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64
EBCLK pin output (EBCLK)	BCLK or 1/2 BCLK or BCLKA or 1/2 BCLKA	—	EBCLK pin	BCLK: Up to 60 MHz Division ratio: 1 or 2 BCLKA: Up to 60 MHz Division ratio: 1 or 2
SDCLK pin output (SDCLK)	BCLK or BCLKA	—	SDCLK pin	BCLK: Up to 125 MHz BCLKA: Up to 133MHz
Trace clock (TRCLK)	MOSC/SOSC/HOCO/ MOCO/PLL1P	1 GHz	CPU-OCD	Up to 125 MHz Division ratio: 1/2/3/4/6/8/12/16/24/32/64/128/256

Table 9.2 Clock generation circuit specifications for the internal clocks (2 of 3)

Item	Clock source	Maximum Frequency of Clock Source	Clock supply	Specification
SCI clock (SCICLK)	MOSC/SOSC/HOCO/ MOCO/LOCO/PLL1P/ PLL1Q/PLL1R/PLL2P/ PLL2Q/PLL2R	480 MHz	SCI	Up to 120 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
SPI clock (SPICLK)	MOSC/SOSC/HOCO/ MOCO/ LOCO/PLL1P/ PLL1Q/PLL1R/PLL2P/ PLL2Q/PLL2R	333 MHz	SPI	Up to 333.33 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
Octal-SPI clock (OCTACLK)	MOSC/SOSC/HOCO/ MOCO/ LOCO/PLL1P/ PLL1Q/PLL1R/PLL2P/ PLL2Q/PLL2R	333 MHz	Octal-SPI	Up to 333.33 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
Octal-SPI divide clock (OCTADIVCLK)	1/2 OCTACLK	—	Octal-SPI	Up to 166.67 MHz
CANFD core clock (CANFDCLK)	MOSC/SOSC/HOCO/ MOCO/ LOCO/PLL1P/ PLL1Q/PLL1R/PLL2P/ PLL2Q/PLL2R	480 MHz	CANFD	Up to 80 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
ADC clock (ADCCLK)	MOSC/SOSC/HOCO/ MOCO/LOCO/PLL1P/ PLL1Q/PLL1R/PLL2P/ PLL2Q/PLL2R	480 MHz	ADC16H	Up to 120 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
GPT clock (GPTCLK)	MOSC/HOCO/MOCO/ PLL1P/PLL1Q/PLL1R/ PLL2P/PLL2Q/PLL2R	600 MHz	GPT, PDG, ADC16H	Up to 300 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
LCD clock (LCDCLK)	MOCO/PLL1P/ PLL1Q/ PLL1R/PLL2P/PLL2Q/ PLL2R	480 MHz	GLCDC	Up to 240 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
LCD_CLK pin output (LCD_CLK) and graphic LCD pixel clock (PXCLK)	LCD_EXTCLK, LCDCLK	—	GLCDC	Up to 54 MHz (parallel RGB) Up to 60 MHz (serial RGB) LCD_CLK division ratios: 1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 16, 24, 32 LCD_CLK : PXCLK = 1:1 (parallel RGB) LCD_CLK : PXCLK = 4:1 (serial RGB)
USB clock (USBCLK)	MOSC/HOCO/MOCO/ PLL1P/ PLL1Q/PLL1R/ PLL2P/PLL2Q/PLL2R	480 MHz	USBFS, USBHS	48 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
USB clock (USB60CLK)	MOSC/HOCO/MOCO/ PLL1P/ PLL1Q/PLL1R/ PLL2P/PLL2Q/PLL2R	480 MHz	USBHS	60 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
I3C clock (I3CCLK)	MOCO/PLL1P/PLL1Q/ PLL1R/ PLL2P/PLL2Q/ PLL2R	800 MHz	I3C	Up to 200 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
Asynchronous external bus clock (BCLKA)	MOCO/PLL1P/PLL1Q/ PLL1R/PLL2P/ PLL2Q/ PLL2R	800 MHz	External bus	Up to 133 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
EtherSW clock (ESWCLK)	MOCO/PLL1P/PLL1Q/ PLL1R/PLL2P/ PLL2Q/ PLL2R	1 GHz	ESWM	Up to 250 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
EtherSW-PHY clock (ESWPHYCLK)	MOCO/PLL1P/PLL1Q/ PLL1R/PLL2P/ PLL2Q/ PLL2R	1 GHz	ESWM	500 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32

Table 9.2 Clock generation circuit specifications for the internal clocks (3 of 3)

Item	Clock source	Maximum Frequency of Clock Source	Clock supply	Specification
Ether-PHY clock (ETHPHYCLK)	MOSC/MOCO/PLL1P/ PLL1Q/PLL1R/PLL2P/ PLL2Q/PLL2R	800 MHz	ETHPHYCLK pin	25/50 MHz Division ratio: 1/2/3/4/5/6/8/10/16/32
Clock/buzzer output (CLKOUT)	MOSC/SOSC/HOCO/ MOCO/LOCO	—	CLKOUT pin	Up to 60 MHz Division ratio: 1/2/4/8/16/32/64/128
MOSC clock for USBHS (USBMCLK)	MOSC	—	USBHS	12/20/24/48 MHz No division
CAN clock (CANMCLK)	MOSC	—	CANFD	8 MHz to 48 MHz No division
MIPI clock (MIPIMCLK)	MOSC	—	MIPI	Up to 48 MHz No division
ULPT LOCO clock (ULPTLCLK)	LOCO	—	ULPT	32.768 kHz No division
ULPT Sub-clock (ULPTSCLK)	SOSC	—	ULPT	32.768 kHz No division
AGT LOCO clock (AGTLCLK)	LOCO	—	AGT	32.768 kHz No division
AGT Sub clock (AGTSCLK)	SOSC	—	AGT	32.768 kHz No division
PDMIF clock (PDMIFCLK)	MOCO	—	PDMIF	8 MHz No division
CAC Main clock (CACMCLK)	MOSC	—	CAC	Up to 48 MHz No division
CAC Sub clock (CACSCLK)	SOSC	—	CAC	32.768 kHz No division
CAC HOCO clock (CACHCLK)	HOCO	—	CAC	16/18/20/32/48 MHz No division
CAC MOCO clock (CACMOCLK)	MOCO	—	CAC	8 MHz No division
CAC LOCO clock (CACLCLK)	LOCO	—	CAC	32.768 kHz No division
RTC LOCO clock (RTCLCLK)	LOCO	—	RTC	32.768 kHz No division
RTC Sub clock (RTCSCLK)	SOCO	—	RTC	32.768 kHz No division
IWDT clock (IWDTCLK)	1/2 LOCO	—	IWDT	16.384 kHz No division
SysTick timer clock (SYSTICKCLK)	1/8 MOCO	—	SysTick timer	1 MHz No division
JTAG clock (JTAGTCK)	TCK	—	JTAG	Up to 25 MHz No division
Serial wire clock (SWCLK)	TCK	—	OCD	Up to 25 MHz No division
TCLK pin output (TCLK)	1/2 TRCLK	—	TCLK pin	Up to 60 MHz

Note: When selecting PLL as the clock source, set the PLL output frequency to 1 GHz or less.

Note: If the PLL reference clock source is HOCO, the PLL multiplication setting must be set within output frequency range of PLL in consideration of HOCO frequency (minimum/maximum).

Note: Restrictions on setting clock frequency: CPUCLK0 ≥ ICLK, CPUCLK1 ≥ ICLK, NPUCLK ≥ ICLK, MRICLK ≥ ICLK, CPUCLK0 ≥ MRICLK, CPUCLK1 ≥ MRICLK, NPUCLK ≥ MRICLK, ICLK ≥ MRPCLK, ICLK ≥ PCLKA ≥ PCLKB, ICLK ≥ BCLK, GPTCLK > PCLKA, PCLKD ≥ PCLKA ≥ PCLKB

Restrictions on clock frequency ratio: (N: integer, and up to 64)

CPUCLK0 : ICLK = N : 1, CPUCLK1 : ICLK = N : 1, NPUCLK : ICLK = N : 1, MRICLK : ICLK = N : 1, CPUCLK0 : MRICLK = N : 1, CPUCLK1 : MRICLK = N : 1, NPUCLK : MRICLK = N : 1, ICLK : MRPCLK = N : 1, ICLK : BCLK = N : 1, ICLK : PCLKA = N : 1, ICLK : PCLKB = N : 1, ICLK : PCLKC = N : 1 or 1 : N, ICLK : PCLKD = N : 1 or 1 : N, ICLK : PCLKE = N : 1 or 1 : N

When one clock selects 3, 6, 12, 24-divisions, the other clocks set by SCKDIVCR and SCKDIVCR2 registers must not select 2, 4, 8, 16, 32, 64- divisions.

If the CAN-FD is used, clock frequency ratio is constrained to be PCLKA:PCLKE = 1:2.

Note: The multiplication of PLL1 and PLL2 should be set to be within the output frequency range of PLL1 and PLL2, taking the frequency of HOCO into consideration when not using the FLL function.

Note: Clocks have a permissible frequency range (See Table 9.2).
MRAM and SRAM also have a permissible operating frequency range in each wait cycle setting. (See section 59, SRAM, section 60, MRAM)
Those clock frequency ranges must be satisfied even if the HOCO has its maximum or minimum frequency when not using FLL function. (See section 70, Electrical Characteristics).

Note: PLLCCR.PLSRCSEL bit must be set to 0 when CPUCLK0 is set to over 960 MHz.

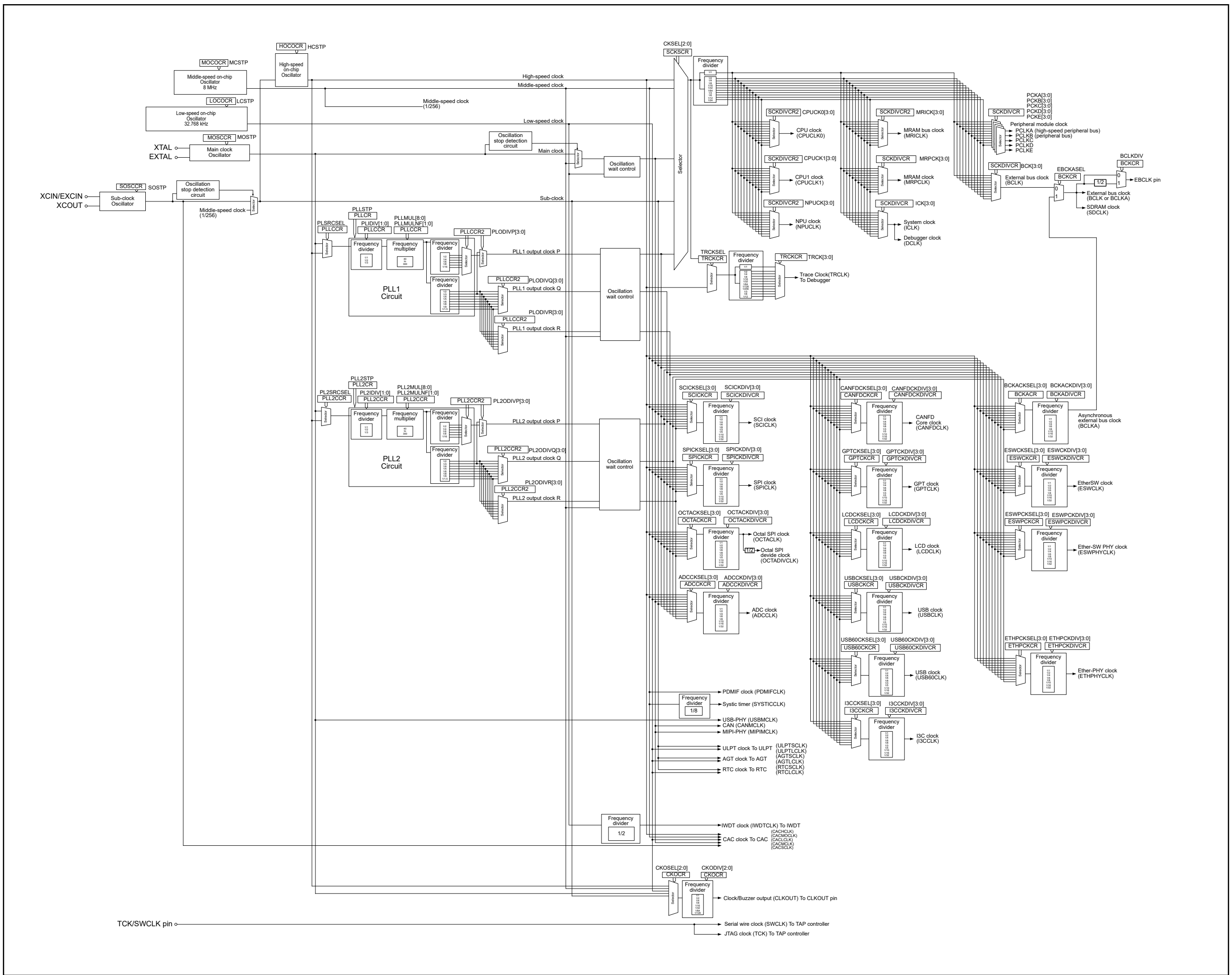


Figure 9.1 Clock generation circuit block diagram

Table 9.3 lists the input/output pins of the clock generation circuit.

Table 9.3 Input/Output Pins of Clock Generation Circuit (1 of 2)

Pin name	I/O	Description
XTAL	Output	These pins are used to connect a ceramic resonator or crystal resonator. The EXTAL pin can also be used to input an external clock. For details, see section 9.3.2. External Clock Input.
EXTAL	Input	
XCIN/EXCIN	Input	These pins are used to connect a 32.768-kHz crystal resonator. The XCIN pin can also be used to input an external clock. For details, section 9.4.2. External Clock Input.
XCOUT	Output	
TCK/SWCLK	Input	This pin is used to input the clock for the JTAG/SWD
EBCLK	Output	This pin is used to supply external devices with the external bus clock (EBCLK)
SDCLK	Output	SDRAM clock (SDCLK) supply for external devices

Table 9.3 Input/Output Pins of Clock Generation Circuit (2 of 2)

Pin name	I/O	Description
CLKOUT	Output	This pin is used to output the CLKOUT/BUZZER clock
TRCLK	Output	TRACE clock output.

9.2 Register Descriptions

9.2.1 CGFSAR : Clock Generation Function Security Attribute Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x3C0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	NONSEC28	NONSEC27	NONSEC26	NONSEC25	NONSEC24	—	NONSEC22	NONSEC21	NONSEC20	NONSEC19	NONSEC18	NONSEC17	NONSEC16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	NONSEC14	NONSEC13	NONSEC12	NONSEC11	NONSEC10	NONSEC09	NONSEC08	NONSEC07	NONSEC06	NONSEC05	NONSEC04	NONSEC03	NONSEC02	—	NONSEC00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NONSEC00	Non-secure Attribute bit 00 Target register: SCKDIVCR, SCKDIVCR2, SCKSCR Target factor: system clock control 0: Secure 1: Non-secure	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
2	NONSEC02	Non-secure Attribute bit 02 Target register: HOCOCR, HOCOCR2, FLLCR1, FLLCR2, HOCOUTCR, HOCOSCR, USBCKSELR Target factor: HOCO 0: Secure 1: Non-secure	R/W
3	NONSEC03	Non-secure Attribute bit 03 Target register: MOCOCR, MOCOUTCR, MOCOSCR Target factor: MOCO 0: Secure 1: Non-secure	R/W
4	NONSEC04	Non-secure Attribute bit 04 Target register: LOCOCR, LOCOUTCR Target factor: LOCO 0: Secure 1: Non-secure	R/W
5	NONSEC05	Non-secure Attribute bit 05 Target register: MOSCCR, MOSCWTCR, MOMCR, MOMCR2, MOSCSCR Target factor: MOSC 0: Secure 1: Non-secure	R/W
6	NONSEC06	Non-secure Attribute bit 06 Target register: OSTDCR, OSTDSR Target factor: oscillation stop detection control 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
7	NONSEC07	Non-secure Attribute bit 07 Target register: SOSCCR, SOMCR Target factor: SOSC 0: Secure 1: Non-secure	R/W
8	NONSEC08	Non-secure Attribute bit 08 Target register: PLLCCR, PLLCCR2, PLLCR Target factor: PLL 0: Secure 1: Non-secure	R/W
9	NONSEC09	Non-secure Attribute bit 09 Target register: PLL2CCR, PLL2CCR2, PLL2CR Target factor: PLL2 0: Secure 1: Non-secure	R/W
10	NONSEC10	Non-secure Attribute bit 10 Target register:SOSTDCR, SOSTDSR Target factor: Sub oscillation stop detection control 0: Secure 1: Non-secure	R/W
11	NONSEC11	Non-secure Attribute bit 11 Target register: CKOCR Target factor: CLKOUT control 0: Secure 1: Non-secure	R/W
12	NONSEC12	Non-secure Attribute bit 12 Target register: BCKCR, EBCKOCR Target factor: EBCLK 0: Secure 1: Non-secure	R/W
13	NONSEC13	Non-secure Attribute bit 13 Target register: SDCKOCR Target factor: SDCLK 0: Secure 1: Non-secure	R/W
14	NONSEC14	Non-secure Attribute bit 14 Target register: BCKADIVCR, BCKACR Target factor: BCLKA 0: Secure 1: Non-secure	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W
16	NONSEC16	Non-secure Attribute bit 16 Target register: USBCKDIVCR, USBCKCR Target factor: USBCLK 0: Secure 1: Non-secure	R/W
17	NONSEC17	Non-secure Attribute bit 17 Target register: OCTACKDIVCR, OCTACKCR Target factor: OCTACLK 0: Secure 1: Non-secure	R/W
18	NONSEC18	Non-secure Attribute bit 18 Target register: CANFDCKDIVCR, CANFDCKCR Target factor: CANFDCLK 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
19	NONSEC19	Non-secure Attribute bit 19 Target register: USB60CKDIVCR, USB60CKCR Target factor: USB60CLK 0: Secure 1: Non-secure	R/W
20	NONSEC20	Non-secure Attribute bit 20 Target register:I3CCKDIVCR, I3CCKCR Target factor: I3CCLK 0: Secure 1: Non-secure	R/W
21	NONSEC21	Non-secure Attribute bit 21 Target register: SCICKDIVCR, SCICKCR Target factor: SCICLK 0: Secure 1: Non-secure	R/W
22	NONSEC22	Non-secure Attribute bit 22 Target register: SPICKDIVCR, SPICKCR Target factor: SPICLK 0: Secure 1: Non-secure	R/W
23	—	This bit is read as 0. The write value should be 0.	R/W
24	NONSEC24	Non-secure Attribute bit 24 Target register: ADCCKDIVCR, ADCCKCR Target factor: ADCCLK 0: Secure 1: Non-secure	R/W
25	NONSEC25	Non-secure Attribute bit 25 Target register: GPTCKDIVCR, GPTCKCR Target factor: GPTCLK 0: Secure 1: Non-secure	R/W
26	NONSEC26	Non-secure Attribute bit 26 Target register: LCDCKDIVCR, LCDCKCR Target factor: LCDCLK 0: Secure 1: Non-secure	R/W
27	NONSEC27	Non-secure Attribute bit 27 Target register: ESWCKDIVCR, ESWCKCR, ESWPCKDIVCR, ESWPCKCR Target factor: ESWCLK, ESWPHYCLK 0: Secure 1: Non-secure	R/W
28	NONSEC28	Non-secure Attribute bit 28 Target register: ETHPCKDIVCR, ETHPCKCR Target factor: ETHPHYCLK 0: Secure 1: Non-secure	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1
Note: Set the PRCR.PRC4 bit to 1 (write enabled) before rewriting this register.

CGFSAR register controls the secure attribute of Clock Generation Function registers.

NONSEC00 bit (Non-secure Attribute bit 00)

This bit controls the security attribute of SCKDIVCR, SCKDIVCR2, SCKSCR.

NONSEC02 bit (Non-secure Attribute bit 02)

This bit controls the security attribute of HOCOCR, HOCOCR2, FLLCR1, FLLCR2, HOCOUTCR, HOCOSCR, USBCKSELR.

NONSEC03 bit (Non-secure Attribute bit 03)

This bit controls the security attribute of MOCOCR, MOCOUTCR, MOCOSCR.

NONSEC04 bit (Non-secure Attribute bit 04)

This bit controls the security attribute of LOCOCR, LOCOUTCR.

NONSEC05 bit (Non-secure Attribute bit 05)

This bit controls the security attribute of MOSCCR, MOSCWTCR, MOMCR, MOMCR2, MOSCSCR.

NONSEC06 bit (Non-secure Attribute bit 06)

This bit controls the security attribute of OSTDCR, OSTDSR.

NONSEC07 bit (Non-secure Attribute bit 07)

This bit controls the security attribute of SOSCCR, SOMCR.

NONSEC08 bit (Non-secure Attribute bit 08)

This bit controls the security attribute of PLLCCR, PLLCCR2, PLLCR.

NONSEC09 bit (Non-secure Attribute bit 09)

This bit controls the security attribute of PLL2CCR, PLL2CCR2, PLL2CR.

NONSEC10 bit (Non-secure Attribute bit 10)

This bit controls the security attribute of SOSTDCR, SOSTDSR.

NONSEC11 bit (Non-secure Attribute bit 11)

This bit controls the security attribute of CKOCR.

NONSEC12 bit (Non-secure Attribute bit 12)

This bit controls the security attribute of BCKCR, EBCKOCR.

NONSEC13 bit (Non-secure Attribute bit 13)

This bit controls the security attribute of SDCKOCR.

NONSEC14 bit (Non-secure Attribute bit 14)

This bit controls the security attribute of BCKADIVCR, BCKACR.

NONSEC16 bit (Non-secure Attribute bit 16)

This bit controls the security attribute of USBCKDIVCR, USBCKCR.

NONSEC17 bit (Non-secure Attribute bit 17)

This bit controls the security attribute of OCTACKDIVCR, OCTACKCR.

NONSEC18 bit (Non-secure Attribute bit 18)

This bit controls the security attribute of CANFDCKDIVCR, CANFDCKCR.

NONSEC19 bit (Non-secure Attribute bit 19)

This bit controls the security attribute of USB60CKDIVCR, USB60CKCR.

NONSEC20 bit (Non-secure Attribute bit 20)

This bit controls the security attribute of I3CCKDIVCR, I3CCKCR.

NONSEC21 bit (Non-secure Attribute bit 21)

This bit controls the security attribute of SCICKDIVCR, SCICKCR.

NONSEC22 bit (Non-secure Attribute bit 22)

This bit controls the security attribute of SPICKDIVCR, SPICKCR.

NONSEC24 bit (Non-secure Attribute bit 24)

This bit controls the security attribute of ADCCKDIVCR, ADCCKCR.

NONSEC25 bit (Non-secure Attribute bit 25)

This bit controls the security attribute of GPTCKDIVCR, GPTCKCR.

NONSEC26 bit (Non-secure Attribute bit 26)

This bit controls the security attribute of LCDCKDIVCR, LCDCKCR.

NONSEC27 bit (Non-secure Attribute bit 27)

This bit controls the security attribute of ESWCKDIVCR, ESWCKCR, ESWPCKDIVCR, ESWPCKCR

NONSEC28 bit (Non-secure Attribute bit 28)

This bit controls the security attribute of ETHPCKDIVCR, ETHPCKCR.

9.2.2 SCKDIVCR : System Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MRPCK[3:0]				ICK[3:0]				PCKE[3:0]				BCK[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PCKA[3:0]				PCKB[3:0]				PCKC[3:0]				PCKD[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	PCKD[3:0]*4	Peripheral Module Clock D (PCLKD) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Setting prohibited.	R/W
7:4	PCKC[3:0]*4	Peripheral Module Clock C (PCLKC) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Setting prohibited.	R/W

Bit	Symbol	Function	R/W
11:8	PCKB[3:0]*3	Peripheral Module Clock B (PCLKB) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Setting prohibited.	R/W
15:12	PCKA[3:0]*3	Peripheral Module Clock A (PCLKA) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Setting prohibited.	R/W
19:16	BCK[3:0]*2	External Bus Clock (BCLK) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Settings prohibited	R/W
23:20	PCKE[3:0]*4	Peripheral Module Clock E (PCLKE) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Settings prohibited	R/W

Bit	Symbol	Function	R/W
27:24	ICK [3:0]*1*2*3*4*5*6*7*8	System Clock (ICLK) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Setting prohibited.	R/W
31:28	MRPCK[3:0]*1*9	MRAM Clock (MRPCLK) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Setting prohibited.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note: When one clock selects 3, 6, 12, 24-divisions, the other clocks must not select 2, 4, 8, 16, 32, 64- divisions.
Note 1. The following relation is required between the frequencies of the system clock (ICLK) and the MRAM Clock (MRPCLK).
ICLK:MRPCLK = N:1 (N: integer)
Note 2. The following relation is required between the frequencies of the system clock (ICLK) and the external bus clock (BCLK).
ICLK:BCLK = N:1 (N: integer)
Note 3. The following relation is required between the frequencies of the system clock (ICLK) and the peripheral module clocks (PCLKA, PCLKB)
ICLK:PCLKA = N:1, ICLK:PCLKB = N:1 (N: integer)
Note 4. The following relation is required between the frequencies of the system clock (ICLK) and the peripheral module clocks (PCLKC, PCLKD, PCLKE):
ICLK:PCLKC, PCLKD PCLKE = N:1 or 1:N (N: integer)
Note 5. The following relation is required between the frequencies of the CPU0 clock (CPUCLK0) and the system clock (ICLK):
CPUCLK0:ICLK = N:1 (N: integer)
Note 6. The following relation is required between the frequencies of the CPU1 clock (CPUCLK1) and the system clock (ICLK):
CPUCLK1:ICLK = N:1 (N: integer)
Note 7. The following relation is required between the frequencies of the NPU clock (NPUCLK) and the system clock (ICLK):
NPUCLK:ICLK = N:1 (N: integer)
Note 8. The following relation is required between the frequencies of the MRAM bus clock (MRICLK), the CPU0 clock (CPUCLK0), the CPU1 clock (CPUCLK1), the NPU clock (NPUCLK) and the system clock (ICLK):
MRICLK:ICLK = N:1 (N: integer)
CPUCLK0:MRICLK = N:1 (N: integer)
CPUCLK1:MRICLK = N:1 (N: integer)
NPUCLK:MRICLK = N:1 (N: integer)
Note 9. The frequency of the MRAM clock (MRPCLK) is limited to the MRE frequency notifications register (MREFREQ). See section 60, MRAM.

SCKDIVCR selects the frequencies of the system clock (ICLK), peripheral module clock (PCLKA, PCLKB, PCLKC, PCLKD, PCLKE), MRAM Clock (MRPCLK), and external bus clock (BCLK).

9.2.3 SCKDIVCR2 : System Clock Division Control Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x024

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MRICK[3:0]				NPUCK[3:0]				CPUCK1[3:0]				CPUCK0[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CPUCK0[3:0]*1	CPU0 Clock (CPUCLK0) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Setting prohibited.	R/W
7:4	CPUCK1[3:0]*2*6	CPU1 Clock (CPUCLK1) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Setting prohibited.	R/W
11:8	NPUCK[3:0]*3	NPU Clock (NPUCLK) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Setting prohibited.	R/W

Bit	Symbol	Function	R/W
15:12	MRICK[3:0]*4*5	MRAM bus Clock (MRICLK) Select 0 0 0 0: × 1/1 0 0 0 1: × 1/2 0 0 1 0: × 1/4 0 0 1 1: × 1/8 0 1 0 0: × 1/16 0 1 0 1: × 1/32 0 1 1 0: × 1/64 1 0 0 0: × 1/3 1 0 0 1: × 1/6 1 0 1 0: × 1/12 1 0 1 1: × 1/24 Others: Setting prohibited.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: When one clock selects 3, 6, 12, 24-divisions, the other clocks must not select 2, 4, 8, 16, 32, 64- divisions.
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note 1. The following relation is required between the frequencies of the CPU0 clock (CPUCLK0) and the system clock (ICLK): CPUCLK0:ICLK = N:1 (N: integer)
Note 2. The following relation is required between the frequencies of the CPU1 clock (CPUCLK1) and the system clock (ICLK):CPUCLK1:ICLK = N:1 (N: integer)
Note 3. The following relation is required between the frequencies of the NPU clock (NPUCLK) and the system clock (ICLK):NPUCLK:ICLK = N:1 (N: integer)
Note 4. The following relation is required between the frequencies of the MRAM bus clock (MRICLK), the CPU0 clock (CPUCLK0), the CPU1 clock (CPUCLK1), the NPU clock (NPUCLK) and the system clock (ICLK):
MRICLK:ICLK = N:1 (N: integer)
CPUCLK0:MRICLK = N:1 (N: integer)
CPUCLK1:MRICLK = N:1 (N: integer)
NPUCLK:MRICLK = N:1 (N: integer)
Note 5. The frequency of the MRAM bus clock (MRICLK) is limited to the MRC frequency notifications register (MRCFREQ). See section 60, MRAM.
Note 6. Set the same value as MRICK[3:0] for single core product.

SCKDIVCR2 selects the frequencies of the CPU0 clock (CPUCLK0), the CPU1 clock (CPUCLK1), the NPU clock (NPUCLK) and the MRAM bus clock (MRICLK).

9.2.4 SYRACCR : System Register Access Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xCC

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	BUSY
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BUSY	Access Ready monitor 0: Ready to read/write access 1: Writing in progress	R
7:1	—	These bits are read as 0. The write value should be 0.	R

Note: S-TYPE-5, P-TYPE-5
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

SYRACCR register monitor that the target registers can be accessed.

The target registers are LOCOCR and LOCOUTCR.

BUSY bit (Access Ready monitor)

When SYRACCR.BUSY is 0, you can access the target registers.

Make sure that SYRACCR.BUSY is 0 before accessing the target registers.

When SYRACCR.BUSY is 1, the write access to one of the target registers is in progress. If SYRACCR.BUSY is 1, further writing to one of the target registers is ignored, and the read value from the target registers is not guaranteed.

9.2.5 SCKSCR : System Clock Source Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x026

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	CKSEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
2:0	CKSEL[2:0]	Clock Source Select 0 0 0: HOCO 0 0 1: MOCO (Value after reset) 0 1 0: Setting prohibited 0 1 1: Main clock oscillator (MOSC) 1 0 0: Sub-clock oscillator (SOSC) 1 0 1: PLL1 output clock P (PLL1P) 1 1 0: Setting prohibited 1 1 1: Setting prohibited	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note: S-TYPE-3, P-TYPE-2

The SCKSCR register selects the clock source for the system clock.

CKSEL[2:0] bits (Clock Source Select)

The CKSEL[2:0] bits select the source for the following modules:

- CPU0 clock (CPUCLK0)
- CPU1 clock (CPUCLK1)
- NPU clock (NPUCLK)
- MRAM bus clock (MRICLK)
- MRAM clock (MRPCLK)
- System clock (ICLK)
- Peripheral module clocks (PCLKA, PCLKB, PCLKC, PCLKD and PCLKE)
- External bus clock (BCLK)
- SDRAM clock (SDCLK)

The bits select from one of the following sources:

- Middle-speed on-chip oscillator (MOCO)
- High-speed on-chip oscillator (HOCO)
- Main clock oscillator (MOSC)
- Sub-clock oscillator (SOSC)
- PLL1 output clock P (PLL1P)

The operating state of each clock source is controlled not only by the clock oscillation enable settings but also by the operating modes of the product. Some clock sources might be forcibly stopped depending on the product operating mode being used.

Check the operation state of clock sources in each product operating mode, and do not select the clock source to be stopped in SCKSCR. The clock sources should be switched when there are no occurring internal asynchronous interrupt. For details, see section 11, Low Power Mode.

9.2.6 PLLCCR : PLL Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0AC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PLLMUL[8]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PLLMUL[7:0]								PLLMULNF[1:0]		—	PLSRCSEL	—		PLIDIV[1:0]	
Value after reset:	0	0	1	0	0	1	1	1	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PLIDIV[1:0]*1	PLL1 Input Frequency Division Ratio Select 0 0: 1/1 0 1: 1/2 1 0: 1/3 Others: Setting prohibited.	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	PLSRCSEL	PLL1 Clock Source Select 0: Main clock oscillator*3 1: HOCO*4	R/W
5	—	This bit is read as 0. The write value should be 0.	R/W
7:6	PLLMULNF[1:0]*2	PLL1 Frequency Multiplication Fractional Factor Select 0 0: 0.00 (Value after reset) 0 1: 0.33 (1/3) 1 0: 0.66 (2/3) 1 1: 0.50 (1/2)	R/W
16:8	PLLMUL[8:0]*2	PLL1 Frequency Multiplication Factor Select 0x27: × 40 (Value after reset) 0x28: × 41 0x29: × 42 ⋮ 0x12A: × 299 0x12B: × 300 Others: Setting prohibited.	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note 1. PLIDIV[1:0] should be set so that the frequency of PLL1 input signal is within the range of section 9.1. Overview.
Note 2. PLLMUL[8:0] and PLLMULNF[1:0] should be set so that the frequency of PLL1 output signal is within the range of section 9.1. Overview.
Note 3. PLSRCSEL must be set to 0 when CPUCLK0 is set to over 960MHz.
Note 4. The FLL function must be enabled when using USBCLK.

The PLLCCR register sets the operation of the PLL1 circuit.

Writing to the PLLCCR is prohibited when the PLLCR.PLLSTP bit is 0 (the PLL1 operates).

PLIDIV[1:0] bits (PLL1 Input Frequency Division Ratio Select)

These bits select the frequency division ratio of the PLL1 clock source.

PLSRCSEL bit (PLL1 Clock Source Select)

This bit selects the clock source for the PLL1.

PLLMULNF[1:0] bits (PLL1 Frequency Multiplication Fractional Factor Select)

These bits select the fractional part of the frequency multiplication factor for the PLL1 circuit.

PLLMUL[8:0] bits (PLL1 Frequency Multiplication Factor Select)

These bits select the frequency multiplication factor of the PLL1 circuit.

9.2.7 PLLCCR2 : PLL Clock Control Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x4C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—				PLODIVR[3:0]				PLODIVQ[3:0]				PLODIVP[3:0]			
Value after reset:	0	0	0	0	0	1	0	1	0	1	0	1	0	1	0	1

Bit	Symbol	Function	R/W
3:0	PLODIVP[3:0]*1	PLL1 Output Frequency Division Ratio Select for output clock P 0001: × 1/2 0010: × 1/3 0011: × 1/4 0101: × 1/6 (Value after reset) 0111: × 1/8 1111: × 1/16 Others: Setting prohibited.	R/W
7:4	PLODIVQ[3:0]*1	PLL1 Output Frequency Division Ratio Select for output clock Q 0001: × 1/2 0010: × 1/3 0011: × 1/4 0100: × 1/5 0101: × 1/6 (Value after reset) 0111: × 1/8 1000: × 1/9 1001: × 1/1.5 Others: Setting prohibited.	R/W
11:8	PLODIVR[3:0]*1	PLL1 Output Frequency Division Ratio Select for output clock R 0001: × 1/2 0010: × 1/3 0011: × 1/4 0100: × 1/5 0101: × 1/6 (Value after reset) 0111: × 1/8 1000: × 1/9 1001: × 1/1.5 Others: Setting prohibited.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note 1. It must be set so that the frequency of PLL1 output signal is within the range listed in Table 9.1.

The PLLCCR2 register sets up the operation of the PLL1 circuit. Writing to the PLLCCR2 is prohibited when the PLLCR.PLLSTP bit is 0 (the PLL1 operates).

Even if only one of the PLL1 output clock P, Q, or R is used, PLLCCR2 must be set so that the frequency of PLL1 output signal is within the range listed in Table 9.1.

PLODIVP[3:0] bit (PLL1 Output Frequency Division Ratio Select for output clock P)

These bits select the output frequency division ratio for PLL1 output clock P (PLL1P).

PLODIVQ[3:0] bit (PLL1 Output Frequency Division Ratio Select for output clock Q)

These bits select the output frequency division ratio for PLL1 output clock Q (PLL1Q).

PLODIVR[3:0] bit (PLL1 Output Frequency Division Ratio Select for output clock R)

These bits select the output frequency division ratio for PLL1 output clock R (PLL1R).

9.2.8 PLLCR : PLL Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x02A

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PLLSTP
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	PLLSTP	PLL1 Stop Control 0: PLL1 is operating 1: PLL1 is stopped.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The PLLCR register controls the operation of the PLL1 circuit.

PLLSTP bit (PLL1 Stop Control)

This bit runs or stops the PLL1 circuit.

If the main clock oscillator is to be selected as the clock source for the PLL1 by the PLLCCR.PLSRCSEL bit, the Main Clock Oscillator Wait Control Register (MOSCWTCR) must be set.

After the PLLSTP bit setting is changed to run the PLL1, only use the PLL1 clock after confirming that the OSCSF.PLLSF bit is set to 1. That is, a fixed time for stabilization is required after starting the PLL1 operation. A fixed time is also required for oscillation to stop after stopping the PLL1 operation. Additionally, apply the following limitations when starting and stopping the PLL1 operation by the PLLSTP bit:

- After stopping the PLL1, confirm that the OSCSF.PLLSF bit is 0 before restarting the PLL1.
- Confirm that the PLL1 is operating and that the OSCSF.PLLSF bit is 1 before stopping the PLL1.
- Regardless of whether the PLL1 clock is selected as the system clock source, confirm that the OSCSF.PLLSF is set to 1 before executing a WFI instruction to place the MCU in Software Standby or Deep Software Standby mode after operating the PLL1.
- When transitioning to Software Standby or Deep Software Standby mode after stopping the PLL1, confirm that the OSCSF.PLLSF bit is cleared to 0 before executing a WFI instruction.

Writing 1 to the PLLSTP bit is prohibited when SCKSCR.CKSEL[2:0] = 101 (system clock source = PLL1 output clock P (PLL1P)).

Confirm the following conditions before writing 0 to PLLSTP:

- When PLL1 source clock = MOSC: MOSCCR.MOSTP = 0 (MOSC is enabled)
- When PLL1 source clock = HOCO: HOCOCR.HCSTP = 0 (HOCO is enabled).

9.2.9 PLL2CCR : PLL2 Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0C8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PLL2MUL[8]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PLL2MUL[7:0]								PLL2MULNF[1:0]		—	PL2SRCSEL	—		PL2IDIV[1:0]	
Value after reset:	0	0	1	0	0	1	1	1	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PL2IDIV[1:0][*1]	PLL2 Input Frequency Division Ratio Select 0 0: 1/1 (Value after reset) 0 1: 1/2 1 0: 1/3 Others: Setting prohibited.	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	PL2SRCSEL	PLL2 Clock Source Select 0: Main clock oscillator 1: HOCO[*3]	R/W
5	—	This bit is read as 0. The write value should be 0.	R/W
7:6	PLL2MULNF[1:0][*2]	PLL2 Frequency Multiplication Fractional Factor Select 00: 0.00 (Value after reset) 01: 0.33 (1/3) 10: 0.66 (2/3) 11: 0.50 (1/2)	R/W
16:8	PLL2MUL[8:0][*2]	PLL2 Frequency Multiplication Factor Select 0x27: × 40 (Value after reset) 0x28: × 41 0x29: × 42 ⋮ 0x12A: × 299 0x12B: × 300 Others: Setting prohibited.	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note 1. PL2IDIV[1:0] should be set so that the frequency of PLL2 input signal is within the range of section 9.1. Overview.
Note 2. PLL2MUL[8:0] and PLL2MULNF[1:0] should be set so that the frequency of PLL2 output signal is within the range of section 9.1. Overview.
Note 3. The FLL function must be enabled when using USBCLK.

The PLL2CCR register sets the operation of the PLL2 circuit.

Writing to the PLL2CCR register is prohibited when the PLL2CR.PLL2STP bit is 0 (the PLL2 operates).

PL2IDIV[1:0] bits (PLL2 Input Frequency Division Ratio Select)

These bits select the frequency division ratio of the PLL2 clock source.

PL2SRCSEL bit (PLL2 Clock Source Select)

This bit selects the clock source for the PLL2.

PLL2MULNF[1:0] bit (PLL2 Frequency Multiplication Fractional Factor Select)

These bits select the fractional part of the frequency multiplication factor for the PLL2 circuit.

PLL2MUL[8:0] bits (PLL2 Frequency Multiplication Factor Select)

These bits select the frequency multiplication factor of the PLL2 circuit.

9.2.10 PLL2CCR2 : PLL2 Clock Control Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x4E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—				PL2ODIVR[3:0]				PL2ODIVQ[3:0]				PL2ODIVP[3:0]			
Value after reset:	0	0	0	0	0	1	0	1	0	1	0	1	0	1	0	1

Bit	Symbol	Function	R/W
3:0	PL2ODIVP[3:0]*1	PLL2 Output Frequency Division Ratio Select for output clock P 0001: × 1/2 0010: × 1/3 0011: × 1/4 0101: × 1/6 (Value after reset) 0111: × 1/8 1111: × 1/16 Others: Setting prohibited.	R/W
7:4	PL2ODIVQ[3:0]*1	PLL2 Output Frequency Division Ratio Select for output clock Q 0001: × 1/2 0010: × 1/3 0011: × 1/4 0100: × 1/5 0101: × 1/6 (Value after reset) 0111: × 1/8 1000: × 1/9 1001: × 1/1.5 Others: Setting prohibited.	R/W
11:8	PL2ODIVR[3:0]*1	PLL2 Output Frequency Division Ratio Select for output clock R 0001: × 1/2 0010: × 1/3 0011: × 1/4 0100: × 1/5 0101: × 1/6 (Value after reset) 0111: × 1/8 1000: × 1/9 1001: × 1/1.5 Others: Setting prohibited.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note 1. It must be set so that the frequency of PLL2 output signal is within the range listed in Table 9.1.

The PLL2CCR2 register sets up the operation of the PLL2 circuit. Writing to the PLL2CCR2 is prohibited when the PLL2CR.PLL2STP bit is 0 (the PLL2 operates).

Even if only one of the PLL2 output clock P, Q, or R is used, PLL2CCR2 must be set so that the frequency of PLL2 output signal is within the range listed in Table 9.1.

PL2ODIVP[3:0] bit (PLL2 Output Frequency Division Ratio Select for output clock P)

These bits select the output frequency division ratio for PLL2 output clock P (PLL2P).

PL2ODIVQ[3:0] bit (PLL2 Output Frequency Division Ratio Select for output clock Q)

These bits select the output frequency division ratio for PLL2 output clock Q (PLL2Q).

PL2ODIVR[3:0] bit (PLL2 Output Frequency Division Ratio Select for output clock R)

These bits select the output frequency division ratio for PLL2 output clock R (PLL2R).

9.2.11 PLL2CR : PLL2 Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x04A

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PLL2STP
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	PLL2STP	PLL2 Stop Control 0: PLL2 is operating 1: PLL2 is stopped.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The PLL2CR register controls the operation of the PLL2 circuit.

PLL2STP bit (PLL2 Stop Control)

This bit runs or stops the PLL2 circuit.

If the main clock oscillator is to be selected as the clock source for the PLL2 by the PLL2CCR.PL2SRCSEL bit, the Main Clock Oscillator Wait Control Register (MOSCWTCR) must be set.

After the PLL2STP bit setting is changed to run the PLL2, only use the PLL2 clock after confirming that the OSCSF.PLL2SF bit is set to 1. That is, a fixed time for stabilization is required after starting the PLL2 operation. A fixed time is also required for oscillation to stop after stopping the PLL2 operation. Additionally, apply the following limitations when starting and stopping the PLL2 operation by the PLL2STP bit:

- After stopping the PLL2, confirm that the OSCSF.PLL2SF bit is 0 before restarting the PLL2.
- Confirm that the PLL2 is operating and that the OSCSF.PLL2SF bit is 1 before stopping the PLL2.
- Confirm that the OSCSF.PLL2SF bit is set to 1 before executing a WFI instruction to place the MCU in Software Standby or Deep Software Standby mode after operating the PLL2.
- When transitioning to Software Standby or Deep Software Standby mode after stopping the PLL2, confirm that the OSCSF.PLL2SF bit is cleared to 0 before executing a WFI instruction.

Confirm the following conditions before writing 0 to PLL2STP:

- When the PLL2 source clock = MOSC: MOSCCR.MOSTP = 0 (MOSC is enabled)
- When the PLL2 source clock = HOCO: HOCOCR.HCSTP = 0 (HOCO is enabled).

9.2.12 BCKCR : External Bus Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x030

Bit position:	7	6	5	4	3	2	1	0
Bit field:	EBCKASEL	—	—	—	—	—	—	BCLKDIV
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BCLKDIV	BCLK Pin Output Select 0: BCLK or BCLKA 1: BCLK/2 or BCLKA/2	R/W
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	EBCKASEL*1	External Bus Asynchronous Select 0: Synchronous (BCLK) 1: Asynchronous (BCLKA)	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note 1. Before changing this bit, set BCKACR.BCKACKSREQ = 1 and then change the value while BCKACR.BCKACKSRDY = 1.

The BCKCR register controls the external bus clock.

This register must be selected with BCKACR. For detail, see section 9.2.64. BCKACR : Asynchronous External Bus Clock Control Register.

BCLKDIV bit (BCLK Pin Output Select)

This bit selects the clock signal for output from the BCLK pin.

Either the BCLK clock with the frequency selected by the BCK[2:0] bits in SCKDIVCR or the BCLK clock divided by 2 can be selected.

EBCKASEL bit (External Bus Asynchronous Select)

This bit selects the clock signal for External Bus, EBCLK and SDCLK.

9.2.13 MOSCCR : Main Clock Oscillator Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x032

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MOSTP
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	MOSTP	Main Clock Oscillator Stop 0: Operate the main clock oscillator*1 1: Stop the main clock oscillator	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note: S-TYPE-3, P-TYPE-2
Note 1. MOMCR register must be set before setting MOSTP to 0.

The MOSCCR register controls the main clock oscillator.

MOSTP bit (Main Clock Oscillator Stop)

The MOSTP bit starts or stops the main clock oscillator.

When changing the value of the MOSTP bit, execute subsequent instructions only after reading the bit to check that the value is updated.

When using the main clock, the Main Clock Oscillator Mode Oscillation Control Register (MOMCR) and the Main Clock Oscillator Wait Control Register (MOSCWTCR) must be set before setting MOSTP to 0. After setting the MOSTP bit to 0, confirm that the OSCSF.MOSCSF bit is set to 1 before using the main clock oscillator.

A fixed stabilization wait time is required after setting the main clock oscillator to start operation. A fixed wait time is also required for oscillation to stop after stopping the main clock oscillator.

The following restrictions apply when starting and stopping operation:

- After stopping the main clock oscillator, confirm that the OSCSF.MOSCSF bit is 0 before restarting the main clock oscillator
- Confirm that the main clock oscillator operates and that the OSCSF.MOSCSF bit is 1 before stopping the main clock oscillator
- Regardless of whether the main clock oscillator is selected as the system clock source, confirm that the OSCSF.MOSCSF bit is set to 1 before executing a WFI instruction to place the MCU in Software standby after operating the main clock oscillator or Deep Software Standby mode.
- When a transition to Software Standby or Deep Software Standby mode is to follow the setting to stop the main clock oscillator, confirm that the OSCSF.MOSCSF bit is set to 0 before executing the WFI instruction.

Writing 1 to MOSTP is prohibited under the following condition:

- SCKSCR.CKSEL[2:0] = 011b (system clock source = MOSC).
- PLLCCR.PLSRCSEL = 0 (PLL1 source clock = MOSC) and SCKSCR.CKSEL[2:0] = 101b (system clock source = PLL1P)
- PLLCCR.PLSRCSEL = 0 (PLL1 source clock = MOSC) and PLLCR.PLLSTP = 0 (PLL1 is operating)
- PLL2CCR.PL2SRCSEL = 0 (PLL2 source clock = MOSC) and PLL2CR.PLL2STP = 0 (PLL2 is operating)

9.2.14 SOSCCR : Sub-Clock Oscillator Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SOSTP
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	SOSTP	Sub-Clock Oscillator Stop 0: Operate the sub-clock oscillator*1 1: Stop the sub-clock oscillator	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note 1. The SOMCR register must be set before setting SOSTP to 0.

The SOSCCR register controls the sub-clock oscillator.

SOSTP bit (Sub-Clock Oscillator Stop)

The SOSTP bit starts or stops the sub-clock oscillator. When changing the value of the SOSTP bit, only execute subsequent instructions after reading the bit to check that the value is updated. Use the SOSTP bit when using the sub-clock oscillator as the source for a peripheral module, for example the RTC. When using the sub-clock oscillator, set the Sub-Clock Oscillator Mode Control Register (SOMCR) before setting SOSTP to 0.

When selecting External clock input by SOMCR.SOSEL, set SOMCR.SOSEL to 1 and wait 50us, then set SOSTP to 0.

The following restrictions apply when starting and stopping the operation:

- After stopping the sub-clock oscillator, allow a stop interval of at least 5 SOSC clock cycles before restarting it
- After setting the SOSTP bit to 0, use the sub-clock only after the sub-clock oscillation stabilization time ($t_{SUBOSCWT}$) has elapsed.
- Regardless of whether the sub-clock oscillator is selected as the system clock, confirm that the sub-clock oscillation is stable before executing a WFI instruction to place the MCU in Software Standby or Deep Software Standby mode
- When a transition to Software Standby or Deep Software Standby mode is to follow the setting to stop the sub-clock oscillator, wait for at least 3 SOSC clock cycles before executing the WFI instruction.

Writing 1 to SOSTP is prohibited under the following condition:

- SCKSCR.CKSEL[2:0] = 100b (system clock source = SOSC).

9.2.15 LOCOCR : Low-Speed On-Chip Oscillator Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x400

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	LCSTP
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LCSTP	LOCO Stop 0: Operate the LOCO clock 1: Stop the LOCO clock	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note: Make sure that SYRACCR.BUSY is 0 before accessing LOCOCR. When SYRACCR.BUSY is 1, the write access is ignored and the read value is not guaranteed.

The LOCOCR register controls the LOCO clock.

LCSTP bit (LOCO Stop)

The LCSTP bit starts or stops the LOCO clock.

When changing the value of the bit, only execute subsequent instructions after reading the bit and checking that its value has actually been updated.

9.2.16 HOCOCR : High-Speed On-Chip Oscillator Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x036

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	HCSTP
Value after reset:	0	0	0	0	0	0	0	0/1*1

Bit	Symbol	Function	R/W
0	HCSTP	HOCO Stop 0: Operate the HOCO clock *2 *3 1: Stop the HOCO clock	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note 1. The HCSTP bit value after a reset is 0 when the OFS1(_SEC).HOCOEN bit is 0. It is 1 when the OFS1(_SEC).HOCOEN bit is 1.
Note 2. If you are using the HOCO (HCSTP = 0), set the OFS1(_SEC).HOCOFRQ0[2:0] bit to an optimum value.
Note 3. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] even if OFS1(_SEC).HOCOFRQ0[2:0] is not appropriate value.

The HOCOCR register controls the HOCO clock.

HCSTP bit (HOCO Stop)

The HCSTP bit starts or stops the HOCO clock.

After setting the HCSTP bit to 0 to start the HOCO clock, confirm that the OSCSF.HOCOSF is set to 1 before using the clock. When OFS1(_SEC).HOCOEN is set to 0, confirm that OSCSF.HOCOSF is also set to 1 before using the HOCO clock. A fixed stabilization wait time is required after setting the HOCO clock to start operation. A fixed wait time is also required after setting the HOCO clock to stop.

The following limitations apply when starting and stopping operation:

- After stopping the HOCO clock, wait more than HOCO stop width time and confirm that the OSCSF.HOCOSF is 0 before restarting the HOCO clock. See section 70, Electrical Characteristics for HOCO stop width time.
- Confirm that the HOCO clock operates and that the OSCSF.HOCOSF is 1 before stopping the HOCO clock.
- Regardless of whether the HOCO clock is selected as the system clock source, confirm that the OSCSF.HOCOSF is set to 1 before executing a WFI instruction to place the MCU in Software Standby or Deep Software Standby mode after setting HOCO operation with the HCSTP bit.
- When a transition to Software Standby or Deep Software Standby mode is to follow the setting of the HOCO clock to stop, confirm that the OSCSF.HOCOSF is set to 0 after setting the HOCO clock and before executing the WFI instruction.

Writing 1 to HCSTP is prohibited under the following conditions:

- SCKSCR.CKSEL[2:0] = 000b (system clock source = HOCO).
- PLLCCR.PLSRCSEL = 1 (PLL1 source clock = HOCO) and SCKSCR.CKSEL[2:0] = 101b (system clock source = PLL1P)
- PLLCCR.PLSRCSEL = 1 (PLL1 source clock = HOCO) and PLLCR.PLLSTP = 0 (PLL1 is operating)
- PLL2CCR.PL2SRCSEL = 1 (PLL2 source clock = HOCO) and PLL2CR.PLL2STP = 0 (PLL2 is operating)

9.2.17 HOCOCR2 : High-Speed On-Chip Oscillator Control Register2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x037

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	HCFRQ0[2:0]		
Value after reset:	0	0	0	0	0	0/1*1	0/1*1	0/1*1

Bit	Symbol	Function	R/W
2:0	HCFRQ0[2:0]	HOCO Frequency Setting 0 0 0 0: 16 MHz 0 0 1: 18 MHz 0 1 0: 20 MHz 1 0 0: 32 MHz 1 1 1: 48 MHz Others: Setting prohibited.	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note 1. Value after reset of the HCFRQ0[2:0] bits depend on OFS1(_SEC).HOCOFRQ0[2:0] bits.

The HOCOCR2 register controls the HOCO clock.

Writing to the HOCOCR2 is prohibited when the HOCOCR.HCSTP bit is 0 (the HOCO operates)

HCFRQ0[2:0] bits (HOCO Frequency Setting 0)

These bits select the frequency of HOCO.

9.2.18 MOCOCR : Middle-Speed On-Chip Oscillator Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x038

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MCSTP
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MCSTP	MOCO Stop 0: MOCO clock is operating 1: MOCO clock is stopped	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The MOCOCR register controls the MOCO clock.

MCSTP bit (MOCO Stop)

This bit starts or stops the MOCO clock.

When changing the value of the bit, only execute subsequent instructions after reading the bit and checking that its value has actually been updated.

There are restrictions on MOCO operation conditions, MOCO stop conditions and the setting of the MCSTP bit. For details, see section 9.6. Middle Speed On-chip Oscillator (MOCO).

9.2.19 FLLCR1 : FLL Control Register1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x039

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	FLLEN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FLLEN	FLL Enable 0: FLL function is disabled 1: FLL function is enabled.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: HOCO must be stopped (HOCOCR.HCSTP = 1) before FLLCR1.FLLEN is modified.
Note: SOSC must be operating with stabilization while FLL is enabled (FLLCR1.FLLEN = 1).
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The FLLCR1 register controls the FLL function of the HOCO.

FLLEN bit (FLL Enable)

This bit enables or disables the FLL function of the HOCO.

If FLL is enabled, the frequency accuracy is guaranteed after FLL is stabilized. The FLL stabilization can be checked by the CAC frequency measurement, but it must be executed after HOCO stabilization.

In addition, you must disable FLL by setting the FLLEN bit to 0 before transitioning to Software Standby mode.

Table 9.4 show an example flow of the FLL setting for each case.

Table 9.4 FLL setting flow

Step		Operation
After reset release/ Deep Software Standby cancellation	1	Start (After reset release / Deep Software Standby cancellation)
	2	FLL setting (FLLCR2.FLLCNTL)
	3	Enable FLL (FLLCR1.FLLEN = 1) Note: SOSC must be running with the oscillation stabilization.
	4	Enable HOCO (HOCOCR.HCSTP = 0)
	5	Wait for the FLL stabilization (t_{FLLWT})
	6	Check the HOCO stabilization (OSCSF.HOCOSF = 1)
	7	End (HOCO can be used.)
Software standby transition/ cancellation	1	Start (FLL is being used.)
	2	Stop HOCO (HOCOCR.HCSTP = 1) Note: If HOCO is used as the system clock or the PLL reference clock, these clock source must be changed to another clock before HOCO is stopped.
	3	Disable FLL (FLLCR1.FLLEN = 0)
	4	WFI instruction
	5	Software standby mode
	6	Software standby cancellation
	7	Enable FLL (FLLCR1.FLLEN = 1)
	8	Enable HOCO (HOCOCR.HCSTP = 0)
	9	Wait for the FLL stabilization (t_{FLLWT})
	10	Check the HOCO stabilization (OSCSF.HOCOSF = 1)
	11	End (HOCO can be used.)

9.2.20 FLLCR2 : FLL Control Register2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x03A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	FLLCNTL[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	FLLCNTL[10:0]	FLL Multiplication Control When OFS1(_SEC).HOCOFRQ0[2:0] is 000b (16 MHz) or 100b (32 MHz), these bits must be set to 0x1E8. When OFS1(_SEC).HOCOFRQ0[2:0] is 001b (18 MHz), these bits must be set to 0x225. When OFS1(_SEC).HOCOFRQ0[2:0] is 010b (20 MHz), these bits must be set to 0x262. When OFS1(_SEC).HOCOFRQ0[2:0] is 111b (48 MHz), these bits must be set to 0x1E8. Settings other than above are prohibited.	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note: The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

The FLLCR2 register controls the FLL function of the HOCO.

FLLCNTL[10:0] bits (FLL Multiplication Control)

These bits select the multiplication ratio of the FLL reference clock.

These bits must be set before FLL is enabled (FLLCR1.FLLEN=1).

9.2.21 OSCSF : Oscillation Stabilization Flag Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x03C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	PLL2SF	PLLSF	—	MOSCSF	—	—	HOCOSF
Value after reset:	0	0	0	0	0	0	0	0/1*1

Bit	Symbol	Function	R/W
0	HOCOSF	HOCO Clock Oscillation Stabilization Flag 0: The HOCO clock is stopped or is not yet stable 1: The HOCO clock is stable, so is available for use as the system clock source	R
2:1	—	These bits are read as 0.	R
3	MOSCSF	Main Clock Oscillation Stabilization Flag 0: The main clock oscillator is stopped or is not yet stable*2 1: The main clock oscillator is stable, so is available for use as the system clock source	R
4	—	This bit is read as 0.	R
5	PLLSF	PLL1 Clock Oscillation Stabilization Flag 0: The PLL1 clock is stopped or is not yet stable. 1: The PLL1 clock is stable, so is available for use as the system clock source	R
6	PLL2SF	PLL2 Clock Oscillation Stabilization Flag 0: The PLL2 clock is stopped or is not yet stable. 1: The PLL2 clock is stable	R
7	—	This bit is read as 0.	R

Note: S-TYPE-5, P-TYPE-5

Note 1. The value after reset depends on the OFS1(_SEC).HOCOEN setting.
When OFS1(_SEC).HOCOEN = 1 (disable HOCO), the value after reset of HOCOSF is 0.
When OFS1(_SEC).HOCOEN = 0 (enable HOCO), the HOCOSF value is set to 0 immediately after reset is released, and the HOCOSF value is set to 1 after the HOCO oscillation stabilization wait time elapses.

Note 2. This is true when an appropriate value is set in the Wait Control register for the main clock oscillator. If the wait time value is not sufficient, the oscillation stabilization flag is set to 1 and supply of the clock signal to the internal circuits starts before oscillation is stable.

This register is not controlled by CGFSAR register.

The OSCSF register contains flags that indicate the operating status of the oscillation stabilization wait circuit of each oscillator. After oscillation starts, these circuits generate a waiting time until the output clock of each oscillator stabilizes. The status of each oscillation stabilization wait flag set to 1 indicates that the clock supply is stable and can be used in related circuits.

HOCOSF flag (HOCO Clock Oscillation Stabilization Flag)

The HOCOSF flag indicates the operating status of the counter that measures the wait time for the high-speed clock oscillator (HOCO). When OFS1(_SEC).HOCOEN is set to 0, confirm that OSCSF.HOCOSF is set to 1 before using the HOCO clock.

[Setting condition]

- After the HOCO clock stops and the HOCOCR.HCSTP bit is set to 0, HOCOSF is set to 1 after HOCO clock is stable to ±10% or less. If the HOCO clock is used at 1.5%, it must wait for wait time specified in the electrical characteristics. (For details, see section 70, Electrical Characteristics)
  The HOCO clock is supplied when HOCOCR.HCSTP bit is set to 0, regardless of the state of HOCOSF.

[Clearing condition]

- When the HOCO clock is operating and then is deactivated because the HOCOCR.HCSTP bit is set to 1.

Note: Even if OFS1(_SEC).HOCOEN has been set to 0, confirm that the OSCSF.HOCOSF has been set to 1 before starting to use the HOCO clock.

Note: In debugging mode, even if HOCO is oscillating, this flag is masked to 0 for the user program debugging when all of following conditions are met simultaneously:

- The Trace Clock is operating enabled by setting TRCKCR.TRCKEN bit to 1.
- The Trace Clock source is selected HOCO for oscillation in debug mode by setting TRCKCR.TRCKSEL bit to 1.
- HOCOCR.HCSTP bit is set to 1.

MOSCSF flag (Main Clock Oscillation Stabilization Flag)

The MOSCSF flag indicates the operating status of the counter that measures the wait time for the main clock oscillator.

[Setting condition]

- When the main clock oscillator is stopped and the MOSCCR.MOSTP bit is set to 0, and then the number of LOCO clock cycles corresponding to the setting of the MOSCWTCR register is counted and supply of the main clock within the MCU is started.

[Clearing condition]

- When the main clock oscillator is operating and then is deactivated because the MOSCCR.MOSTP bit is set to 1.

PLLSF flag (PLL1 Clock Oscillation Stabilization Flag)

The PLLSF flag indicates the oscillation stabilization status for the PLL1.

[Setting condition]

- After the PLL1 stops and the PLLCR.PLLSTP bit is set to 0, supply of the PLL1 clock starts after PLL1 clock is stable. If oscillation by the PLL1 clock source selected in the PLLCCR.PLSRCSEL bit is not stable when the PLLSTP bit is set to 0, waiting for the PLL1 oscillation stabilization after the PLL1 clock source oscillation is stabilized.

[Clearing condition]

- When the PLL1 is operating and then is deactivated because the PLLCR.PLLSTP bit is set to 1.

PLL2SF flag (PLL2 Clock Oscillation Stabilization Flag)

This flag indicates the oscillation stabilization status for the PLL2.

[Setting condition]

- After the PLL2 stops and the PLL2CR.PLL2STP bit is set to 0, supply of the PLL2 clock starts after PLL2 clock is stable.
  If oscillation by the PLL2 clock source selected in the PLLCCR.PLSRCSEL is not stable when the PLL2STP bit is set to 0, waiting for the PLL2 oscillation stabilization after the oscillation of the PLL2 clock source is stabilized.

[Clearing condition]

- When the PLL2 is operating and then is deactivated because the PLL2CR.PLL2STP bit is set to 1.

9.2.22 OSCMONR : Oscillator Monitor Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x43

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	LOCOMON	MOCOMON	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	MOCOMON	MOCO operation monitor 0: MOCO is set to operate. 1: MOCO is set to stop.	R
2	LOCOMON	LOCO operation monitor 0: LOCO is set to operate. 1: LOCO is set to stop.	R
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5, P-TYPE-5

This register is not controlled by CGFSAR register.

MOCOMON bit (MOCO operation monitor)

MOCOMON indicates the operation or stop setting status for MOCO.

[Clearing condition]

- When MOCO operating conditions are met. For details, see section 9.6.1. MOCO Operating Conditions.

[Setting condition]

- When MOCO stop conditions are met. For details, see section 9.6.2. MOCO Stop Conditions.

LOCOMON bit (LOCO operation monitor)

LOCOMON indicates the operation or stop setting status for LOCO.

[Clearing condition]

- When LOCO operating conditions are met. For details, see section 9.5.1. LOCO Operating Conditions.

[Setting condition]

- When LOCO stop conditions are met. For details, see section 9.5.2. LOCO Stop Conditions.

9.2.23 OSTDCR : Oscillation Stop Detection Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x040

Bit position:	7	6	5	4	3	2	1	0
Bit field:	OSTDE	—	—	—	—	—	—	OSTDIE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OSTDIE	Oscillation Stop Detection Interrupt Enable 0: Disable oscillation stop detection interrupt (do not notify the POEG) 1: Enable oscillation stop detection interrupt (notify the POEG)	R/W
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	OSTDE	Oscillation Stop Detection Function Enable 0: Disable oscillation stop detection function 1: Enable oscillation stop detection function	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The OSTDCR register controls main-clock oscillation stop detection function.

OSTDIE bit (Oscillation Stop Detection Interrupt Enable)

The OSTDIE bit enables the oscillation stop detection function interrupt. It also controls whether oscillation stop detection is reported to the POEG.

If the Oscillation Stop Detection flag in the Oscillation Stop Detection Status Register (OSTDSR.OSTDF) requires clearing, set the OSTDIE bit to 0 before clearing OSTDF. Wait for at least 2 PCLKB cycles before setting the OSTDIE bit to 1. By reading the I/O register whose access cycle number is defined by PCLKB, it is possible to secure waiting time of 2 or more cycles of PCLKB.

OSTDE bit (Oscillation Stop Detection Function Enable)

The OSTDE bit enables the oscillation stop detection function.

When the OSTDE bit is 1 (enabled), the MOCO stop bit (MOCOCR.MCSTP) is set to 0 and the MOCO operation starts. The MOCO clock cannot be stopped while the oscillation stop detection function is enabled. Writing 1 to the MOCOCR.MCSTP bit (MOCO stopped) is invalid.

When the Oscillation Stop Detection flag in the Oscillation Stop Detection Status Register (OSTDSR.OSTDF) is 1 (main clock oscillation stop detected), writing 0 to the OSTDE bit is invalid.

9.2.24 OSTDSR : Oscillation Stop Detection Status Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x041

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	OSTDF
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OSTDF	Oscillation Stop Detection Flag 0: Main clock oscillation stop not detected 1: Main clock oscillation stop detected	R/W*1
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note 1. This bit can only be set to 0. This bit is cleared to 0 by writing 0 after reading it as 1.

The OSTDSR register indicates the stop detection status of the main clock oscillator.

OSTDF flag (Oscillation Stop Detection Flag)

The OSTDF flag indicates the main clock oscillator status. When this flag is 1, it indicates that the main clock oscillation stop was detected. After this stop is detected, the OSTDF flag is not set to 0 even when the main clock oscillation is restarted. The OSTDF bit is cleared to 0 by writing 0 after reading it as 1.

At least 3 ICLK cycles of wait time are required between writing 0 to OSTDF and reading it as 0. If the OSTDF flag is set to 0 when the main clock oscillation is stopped, the OSTDF flag becomes 0 then returns to 1.

The OSTDF flag cannot be set to 0 under the following conditions:

- SCKSCR.CKSEL[2:0] = 011b (system clock source = MOSC).
- PLLCCR.PLSRCSEL = 0 (PLL1 source clock = MOSC) and SCKSCR.CKSEL[2:0] = 101b (System clock source = PLL1P)

The OSTDF flag must be set to 0 after switching the clock source to sources other than the main clock oscillator and PLL1P.

[Setting condition]

- The main clock oscillator is stopped when OSTDCR.OSTDE = 1 (oscillation stop detection function enabled).

[Clearing condition]

- 1 is read and then 0 is written when the SCKSCR.CKSEL[2:0] bits are neither 011b (system clock source is MOSC) nor, 101b (system clock source is PLL1P) and PLLCCR.PLSRCSEL bit is not 0 (PLL1 source clock is MOSC).

9.2.25 SOSTDCR : Sub-clock Oscillation Stop Detection Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC04

Bit position:	7	6	5	4	3	2	1	0
Bit field:	SOSTDE	—	—	—	—	—	—	SOSTDIE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SOSTDIE	Sub-clock Oscillation Stop Detection Interrupt Enable 0: The oscillation stop detection interrupt is disabled. 1: The oscillation stop detection interrupt is enabled.	R/W
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	SOSTDE	Sub-clock Oscillation Stop Detection Function Enable 0: Oscillation stop detection function is disabled. 1: Oscillation stop detection function is enabled.	R/W

Note: S-TYPE-3, P-TYPE-2

SOSTDCR Register controls Sub-clock oscillation-stop detection function.

This register is only initialized by VBATT_POR reset.

SOSTDIE bit (Sub-clock Oscillation Stop Detection Interrupt Enable)

This bit enables or disables the Sub-clock oscillation stop detection function interrupt.

If the Sub-clock oscillation stops detection flag in the Sub-clock oscillation stop detection status register (SOSTDSR.SOSTDF) requires clearing, clear the SOSTDF after the SOSTDIE bit is cleared to 0. Make sure SOSTDF is cleared before setting the SOSTDIE bit to 1 again.

SOSTDE bit (Sub-clock Oscillation Stop Detection Function Enable)

This bit enables or disables the sub-clock oscillation stop detection function.

When the SOSTDE bit is 1 (sub-clock oscillation stop detection function enabled), the MOCO stop bit (MOCOCR.MCSTP) is cleared to 0 and the MOCO operation is started. The MOCO cannot be stopped while the oscillation stop detection function is enabled except in deep software standby mode and VBAT mode. Writing 1 to the MOCOCR.MCSTP bit (MOCO stopped) is invalid.

When the sub-clock oscillation stops detection flag in the sub-clock oscillation stop detection status register (SOSTDSR.SOSTDF) is 1 (sub-clock oscillation stop detected), writing 0 to the SOSTDE bit is invalid.

To enable the sub-clock oscillation stop detection function, the SOSTDSR.SOSTDF flag must be cleared to 0 after setting the SOSTDCR.SOSTDE bit to 1. However, after setting the SOSTDCR.SOSTDE bit to 1, be sure to wait for the start-up time of the sub-clock oscillation stop detector (Tsosdup) before clearing the SOSTDSR.SOSTDF flag. For SOSTDSR, please see section 9.2.26. SOSTDSR : Sub-clock Oscillation Stop Detection Status Register.

Be sure to set the system clock to other clock sources than the sub-clock oscillator before enabling the sub-clock oscillation stop detection function.

9.2.26 SOSTDSR : Sub-clock Oscillation Stop Detection Status Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC05

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SOSTDF
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SOSTDF	Sub-clock Oscillation Stop Detection Flag 0: The sub-clock oscillation stop has not been detected. 1: The sub-clock oscillation stop has been detected.	R/W*1
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note 1. This bit can only be set to 0.

SOSTDSR Register indicates status of the sub-clock oscillation stop detection.

This register is only initialized by VBATT_POR reset.

SOSTDF flag (Sub-clock Oscillation Stop Detection Flag)

This bit is a flag to indicate the sub-clock status. When the SOSTDF flag is 1, it indicates that the sub-clock oscillation stop has been detected.

Once the sub-clock oscillation stop is detected, the OSTDF bit is not cleared to 0 even though the sub-clock oscillation is restarted. The SOSTDF bit is cleared to 0 by reading 1 from the bit and then writing 0.

At least 3 PCLKB cycles of wait time is necessary between writing 0 to SOSTDF and reading SOSTDF as 0.

If the SOSTDF bit is cleared to 0 while the sub-clock oscillation is stopped, the SOSTDF bit is immediately set to 1.

SOSTDSR.SOSTDF cannot be cleared to 0 while the following conditions:

- SCKSCR.CKSEL[2:0] = 100 (System clock source = SOSC)

[Setting condition]

- The sub-clock oscillation is stopped with the SOSTDCR.SOSTDE being 1 (sub-clock oscillation stop detection function enabled).

[Clearing condition]

- 1 is read and then 0 is written when the SCKSCR.CKSEL[2:0] bits are not 100b (System clock is SOSC).

9.2.27 MOSCWTCR : Main Clock Oscillator Wait Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0A2

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	MSTS[3:0]			
Value after reset:	0	0	0	0	0	1	0	1

Bit	Symbol	Function	R/W
3:0	MSTS[3:0]	Main Clock Oscillator Wait Time Setting 0x0: Wait time = 3 cycles (11.4 μs) 0x1: Wait time = 35 cycles (133.5 μs) 0x2: Wait time = 67 cycles (255.6 μs) 0x3: Wait time = 131 cycles (499.7 μs) 0x4: Wait time = 259 cycles (988.0 μs) 0x5: Wait time = 547 cycles (2086.6 μs) 0x6: Wait time = 1059 cycles (4039.8 μs) 0x7: Wait time = 2147 cycles (8190.2 μs) 0x8: Wait time = 4291 cycles (16368.9 μs) 0x9: Wait time = 8163 cycles (31139.4 μs) Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

MSTS[3:0] bits (Main Clock Oscillator Wait Time Setting)

The MSTS[3:0] bits specify the oscillation stabilization wait time for the main clock oscillator.

Set the main clock oscillation stabilization time to a period longer than or equal to the stabilization time recommended by the oscillator manufacturer. When the main clock is input externally, set these bits to 0x0 because the oscillation stabilization time is not required.

The wait time set in these bits is counted using: 1 cycle (μs) = $1/(f_{LOCO}[MHz] \times 8) = 1/(0.032768 \times 8) = 3.81$ (μs) (min.) The LOCO clock automatically oscillates when necessary, regardless of the value of the LOCO.LCSTP bit. After the specified wait time elapses, supply of the main clock starts internally in the MCU, and the OSCSF.MOSCSF flag is set to 1. If the specified wait time is short, supply of the main clock starts before oscillation of the clock becomes stable.

Only rewrite the MOSCWTCR register when the MOSCCR.MOSTP bit is 1 and the OSCSF.MOSCSF flag is 0. Do not rewrite this register under any other conditions.

9.2.28 MOMCR : Main Clock Oscillator Mode Oscillation Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA50

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	MOSEL	—	—	MODRV0[2:0]			—
Value after reset:	0	0	0	1	1	0	1	0

Bit	Symbol	Function	R/W
0	—	These bits are read as 0. The write value should be 0.	R/W
3:1	MODRV0[2:0]	Main Clock Oscillator Drive Capability 0 Switching 0 0 0: 8 MHz 0 1 1: 8MHz to 24MHz 1 0 1: 8MHz to 48MHz Others: Setting prohibited	R/W
4	—	This bit is read as 1. The write value should be 1.	R/W
5	—	These bits are read as 0. The write value should be 0.	R/W
6	MOSEL	Main Clock Oscillator Switching 0: Resonator 1: External clock input	R/W
7	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: The EXTAL/XTAL pins are also used as ports. In the initial state, the pin is set as a port.
Note: The MOSCCR.MOSTP bit must be 1 (MOSC is stopped) before changing this register.
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

MODRV0[2:0] bit (Main Clock Oscillator Drive Capability 0 Switching)

The MODRV0[2:0] bit switches the drive capability of the main clock oscillator. It is recommended to set 101b, other setting is possible to reduce current consumption. See section 70, Electrical Characteristics for current consumption.

MOSEL bit (Main Clock Oscillator Switching)

The MOSEL bit switches the source for the main clock oscillator.

9.2.29 SOMCR : Sub-Clock Oscillator Mode Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC01

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	SOSEL	—	—	—	—	SODRV[1:0]	
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	SODRV[1:0]	Sub-Clock Oscillator Drive Capability Switching 0 0: Standard (12.5pf) 0 1: Lowpower mode 1 (9pf) 1 0: Lowpower mode 2 (7pf) 1 1: Lowpower mode 3 (4pf)	R/W
5:2	—	These bits are read as 0. The write value should be 0.	R/W
6	SOSEL	Sub-Clock Oscillator Switching 0: Resonator 1: External clock input	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The SOMCR register must be modified when SOSCCR.SOSTP is 1 (SOSC is stopped).

This register is initialized by only VBATT_POR reset.

SODRV[1:0] bits (Sub-Clock Oscillator Drive Capability Switching)

The SODRV[1:0] bits switch the drive capability of the sub-clock oscillator.

SOSEL bit (Sub-Clock Oscillator Switching)

The SOSEL bit switches the source for the Sub-clock oscillator.

For details, see section 9.4. Sub-Clock Oscillator.

9.2.30 CKOCR : Clock Out Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x03E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	CKOEN	CKODIV[2:0]			—	CKOSEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
2:0	CKOSEL[2:0]	Clock Out Source Select 0 0 0: HOCO 0 0 1: MOCO (Value after reset) 0 1 0: LOCO 0 1 1: MOSC 1 0 0: SOSC Others: Setting prohibited	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
6:4	CKODIV[2:0]	Clock Output Frequency Division Ratio 0 0 0: × 1/1 0 0 1: × 1/2 0 1 0: × 1/4 0 1 1: × 1/8 1 0 0: × 1/16 1 0 1: × 1/32 1 1 0: × 1/64 1 1 1: × 1/128	R/W
7	CKOEN	Clock Out Enable 0: Disable clock out 1: Enable clock out	R/W

Note: S-TYPE-3, P-TYPE-2

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

CKOSEL[2:0] bits (Clock Out Source Select)

The CKOSEL[2:0] bits select the source of the clock to be output from the CLKOUT pin. When changing the clock source, set the CKOEN bit to 0.

CKODIV[2:0] bits (Clock Output Frequency Division Ratio)

The CKODIV[2:0] bits specify the clock division ratio. Set the CKOEN bit to 0 when changing the division ratio.

CKOEN bit (Clock Out Enable)

The CKOEN bit enables output from the CLKOUT pin.

When this bit is set to 1, the selected clock is output. When this bit is set to 0, low is output. When changing this bit, confirm that the clock out source clock selected in the CKOSEL[2:0] bits is stable. Otherwise, a glitch might be generated in the output.

Clear this bit before entering Software Standby or Deep Software Standby mode if the selecting clock out source clock is stopped in that mode.

9.2.31 EBCKOCR : External Bus Clock Output Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x052

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	EBCKOEN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EBCKOEN	EBCLK Pin Output Control 0: EBCLK pin output is disabled (fixed high) 1: EBCLK pin output is enabled.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

9.2.32 SDCKOCR : SDRAM Clock Output Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x053

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SDCK OEN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SDCKOEN	SDCLK Pin Output Control 0: SDCLK pin output is disabled. (Fixed high) 1: SDCLK pin output is enabled.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

SDCKOEN bit (SDCLK Pin Output Control)

This bit selects controls the output of the SDCLK pin.

9.2.33 LOCOUTCR : LOCO User Trimming Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x402

Bit position:	7	6	5	4	3	2	1	0
Bit field:	LOCOUTRM[7:0]							
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	LOCOUTRM[7:0]	LOCO User Trimming 0x80: -128 0x81: -127 ⋮ 0xFF: -1 0x00: Center Code 0x01: +1 ⋮ 0x7E: +126 0x7F: +127	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note: Make sure that SYRACCR.BUSY is 0 before accessing LOCOUTCR. When SYRACCR.BUSY is 1, the write access is ignored and the read value is not guaranteed.
Note: MCU operation is not guaranteed when LOCOUTCR is set to a value that causes the LOCO frequency to be outside of the specification range.
Note: After changing LOCOUTCR, it takes certain time until the frequency to be stable. The time is same as the oscillation stabilization time in startup.

The LOCOUTCR register is added to the original LOCO trimming data.

Changing LOCOUTCR during RTC operation is prohibited.

9.2.34 MOCOUTCR : MOCO User Trimming Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x061

Bit position:	7	6	5	4	3	2	1	0
Bit field:	MOCOUTRM[7:0]							
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	MOCOUTRM[7:0]	MOCO User Trimming 0x80: -128 0x81: -127 ⋮ 0xFF: -1 0x00: Center Code 0x01: +1 ⋮ 0x7E: +126 0x7F: +127	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note: MCU operation is not guaranteed when MOCOUTCR is set to a value that causes the MOCO frequency to be outside of the specification range.
Note: After changing MOCOUTCR, it takes certain time until the frequency to be stable. The time is same as the oscillation stabilization time in startup.

The MOCOUTCR register is added to the original MOCO trimming data.

9.2.35 HOCOUTCR : HOCO User Trimming Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x062

Bit position:	7	6	5	4	3	2	1	0
Bit field:	HOCOUTRM[7:0]							
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	HOCOUTRM[7:0]	HOCO User Trimming 0x80: -128 0x81: -127 ⋮ 0xFF: -1 0x00: Center Code 0x01: +1 ⋮ 0x7E: +126 0x7F: +127	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note: MCU operation is not guaranteed when HOCOUTCR is set to a value that causes the HOCO frequency to be outside of the specification range.
Note: After changing HOCOUTCR, it takes certain time until the frequency to be stable. The time is same as the oscillation stabilization time in startup.
Note: These bits must be 0x00 when FLL is enabled (FLLCR1.FLLEN = 1).

The HOCOUTCR register is added to the original HOCO trimming data.

9.2.36 MOSCSCR : Main Clock Oscillator Standby Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x07C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MOSCSOKP
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MOSCSOKP	Main Clock Oscillator Standby Oscillation Keep select 0: Disable 1: Enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The MOSCSCR must be changed while MOSCCR.MOSTP bit is 1 (MOSC is stopped).

MOSCSOKP bit (Main Clock Oscillator Standby Oscillation Keep select)

This bit controls that Main Clock Oscillator keeps oscillating in Software Standby mode.

If this bit is disabled and MOSCCR.MOSTP bit is 0 (MOSC is run), the Main Clock Oscillator stops in Software Standby mode.

If this bit is enabled and MOSCCR.MOSTP bit is 0 (MOSC is run), the Main Clock Oscillator will keep oscillating in Software Standby mode.

If MOSCCR.MOSTP bit is 1 (MOSC is stopped), the Main Clock Oscillator stops in Software Standby mode regardless of the setting of this bit.

9.2.37 HOCOSCR : High-Speed On-Chip Oscillator Standby Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x07D

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	HOCOSOKP
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	HOCOSOKP	HOCO Standby Oscillation Keep select. 0: Disable 1: Enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

HOCOSOKP bit (HOCO Standby Oscillation Keep select.)

This bit controls that HOCO keeps oscillating in Software Standby mode.

If this bit is disabled and HOCOCR.HCSTP bit is 0 (HOCO is run), the HOCO stops in Software Standby mode.

If this bit is enabled and HOCOCR.HCSTP bit is 0 (HOCO is run), the HOCO will keep oscillating in Software Standby mode.

If HOCOCR.HCSTP bit is 1 (HOCO is stopped), the HOCO stops in Software Standby mode regardless of the setting of this bit.

9.2.38 MOCOSCR : Middle-Speed On-Chip Oscillator Standby Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x084

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MOCO SOKP
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MOCOSOKP	MOCO Standby Oscillation Keep select 0: Disable 1: Enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

MOCOSOKP bit (MOCO Standby Oscillation Keep select)

This bit controls that MOCO keeps oscillating in Software Standby mode.

If this bit is disabled and MOCOCR.MCSTP bit is 0 (MOCO is run), the MOCO stops in Software Standby mode.

If this bit is enabled and MOCOCR.MCSTP bit is 0 (MOCO is run), the MOCO will keep oscillating in Software Standby mode.

9.2.39 USBCKDIVCR : USB Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x06C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	USBCKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	USBCKDIV[3:0]	USB Clock (USBCLK) Division Select 0 0 0 0: 1/1 (value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 0 1: 1/3 0 1 1 0: 1/5 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The USBCKDIVCR register controls the USB clock.

USBCKDIV[3:0] bits (USB Clock (USBCLK) Division Select)

These bits select the frequency of the USB clock (USBCLK) and must be modified when USBCKCR.USBCKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRB.MSTPB11 = 1 and MSTPCRB.MSTPB12 = 1 before setting USBCKCR.USBCKSREQ = 1.

9.2.40 OCTACKDIVCR : Octal-SPI Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x06D

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	OCTACKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	OCTACKDIV[3:0]	Octal-SPI Clock (OCTACLK) Division Select 0 0 0 0: 1/1 (Value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 0 1: 1/3 0 1 1 0: 1/5 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

OCTACKDIVCR controls the Octal-SPI clock.

OCTACKDIV[3:0] bits (Octal-SPI Clock (OCTACLK) Division Select)

These bits select the frequency of the Octal-SPI clock (OCTACLK) and must be modified when OCTACKCR.OCTACKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRB.MSTPB16 = 1 and MSTPCRB.MSTPB17 = 1 before setting OCTACKCR.OCTACKSREQ = 1.

9.2.41 CANFDCKDIVCR : CANFD Core Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x06E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	CANFDCKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CANFDCKDIV[3:0]	CANFD core clock (CANFDCLK) Division Select 0 0 0 0: 1/1 (Value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 0 1: 1/3 0 1 1 0: 1/5 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

CANFDCKDIVCR controls the CANFD core clock.

CANFDCKDIV[3:0] bits (CANFD core clock (CANFDCLK) Division Select)

These bits select the frequency of the CANFD core clock (CANFDCLK) and must be modified when CANFDCKCR.CANFDCKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRC.MSTPC26 = 1 and MSTPCRC.MSTPC27 = 1 before setting CANFDCKCR.CANFDCKSREQ = 1.

9.2.42 USB60CKDIVCR : USB60 Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x06F

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	USB60CKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	USB60CKDIV[3:0]	USB clock (USB60CLK) Division Select 0 0 0 0: 1/1 (Value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 0 1: 1/3 0 1 1 0: 1/5 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

USB60CKDIVCR controls the USB clock.

USB60CKDIV[3:0] bits (USB clock (USB60CLK) Division Select)

These bits select the frequency of the USB clock USB60CKDIV and must be modified when USB60CKCR.USB60CKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRB.MSTPB12 = 1 before setting USB60CKCR.USB60CKSREQ = 1.

9.2.43 I3CCKDIVCR : I3C Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x070

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	I3CCKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	I3CCKDIV[3:0]	I3C clock (I3CCLK) Division Select 0 0 0 0: 1/1 (Value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 0 1: 1/3 0 1 1 0: 1/5 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

I3CCKDIVCR controls the I3C clock.

I3CCKDIV[3:0] bits (I3C clock (I3CCLK) Division Select)

These bits select the frequency of the I3C clock (I3CCLK) and must be modified when I3CCKCR.I3CCKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRB.MSTPB4 = 1 before setting I3CCKCR.I3CCKSREQ = 1.

9.2.44 USBCKCR : USB Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x074

Bit position:	7	6	5	4	3	2	1	0
Bit field:	USBCKSRDY	USBCKSREQ	—	—	USBCKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	USBCKSEL[3:0]	USB Clock (USBCLK) Source Select 0000: HOCO 0001: MOCO (Value after reset) 0011: Main clock oscillator 0101: PLL1P 0110: PLL2P 0111: PLL1Q 1000: PLL1R 1001: PLL2Q 1010: PLL2R Others: Setting prohibited.	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
6	USBCKSREQ	USB Clock (USBCLK) Switching Request 0: No request 1: Request switching.	R/W
7	USBCKSRDY	USB Clock (USBCLK) Switching Ready state flag 0: Impossible to Switch 1: Possible to Switch	R

Note: S-TYPE-3, P-TYPE-2

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The USBCKCR register controls the USB clock.

When switching the clock source, ensure that the clock before the switch and the clock after the switch generate stable output. To change the set value of USBCKDIVCR.USBCKDIV[3:0] and USBCKSEL[3:0], use the following procedure:

1. Write 1 to MSTPCRB.MSTPB11 and MSTPCRB.MSTPB12 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two USBCLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to USBCKSREQ.
4. Poll until USBCKSRDY is read as 1. While USBCKSRDY = 1, no clock is output to USBCLK.
5. Write to USBCKDIVCR.USBCKDIV[3:0] and USBCKSEL[3:0].
6. Write 0 to USBCKSREQ.
7. Poll until USBCKSRDY is read as 0.
8. When USBCKSRDY becomes 0, USBCLK starts to output. Clock switching is complete.

When transitioning to Software Standby or Deep Software Standby mode, do not execute the WFI instruction while performing clock switching. That is, do not execute the WFI instruction when USBCKSREQ = 1 and USBCKSRDY = 0, or when USBCKSREQ = 0 and USBCKSRDY = 1.

USBCKSEL[3:0] bits (USB Clock (USBCLK) Source Select)

These bits select the clock source of the USB clock (USBCLK) and must be modified when USBCKCR.USBCKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

USBCKSREQ bit (USB Clock (USBCLK) Switching Request)

This bit selects the USBCLK switching request.

USBCKSRDY flag (USB Clock (USBCLK) Switching Ready state flag)

This flag indicates the state of switching ready for the USBCLK. When USBCKSRDY = 1, no clock is output to USBCLK.

9.2.45 OCTACKCR : Octal-SPI Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x075

Bit position:	7	6	5	4	3	2	1	0
Bit field:	OCTACKSRDY	OCTACKSREQ	—	—	OCTACKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	OCTACKSEL[3:0]	Octal-SPI Clock (OCTACLK) Source Select 0 0 0 0: HOCO 0 0 0 1: MOCO (Value after reset) 0 0 1 0: LOCO 0 0 1 1: Main clock oscillator 0 1 0 0: Sub-clock oscillator 0 1 0 1: PLL1P 0 1 1 0: PLL2P 0 1 1 1: PLL1Q 1 0 0 0: PLL1R 1 0 0 1: PLL2Q 1 0 1 0: PLL2R Others: Setting prohibited.	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	OCTACKSREQ	Octal-SPI Clock (OCTACLK) Switching Request 0: No request 1: Request switching.	R/W
7	OCTACKSRDY	Octal-SPI Clock (OCTACLK) Switching Ready state flag 0: Impossible to Switch 1: Possible to Switch	R

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The OCTACKCR register controls the Octal-SPI clock.

When switching the clock source, ensure that the clock before the switch and the clock after the switch generate stable output.

To change the set value of OCTACKDIVCR.OCTACKDIV[3:0] and OCTACKSEL[3:0], use the following procedure:

1. Write 1 to MSTPCRB.MSTPB16 and MSTPCRB.MSTPB17 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Confirm the reading value of both MSTPCRB.MSTPB16 and MSTPCRB.MSTPB17 is 1 only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to OCTACKSREQ.
4. Poll until OCTACKSRDY is read as 1. While OCTACKSRDY = 1, no clock is output to OCTACLK.
5. Write to OCTACKDIVCR.OCTACKDIV[3:0] and OCTACKSEL[3:0].
6. Write 0 to OCTACKSREQ.
7. Poll until OCTACKSRDY is read as 0.
8. When OCTACKSRDY becomes 0, OCTACLK starts to output. Clock switching is complete.

When transitioning to Software Standby or Deep Software Standby mode, do not execute the WFI instruction while performing clock switching. That is, do not execute the WFI instruction when OCTACKSREQ = 1 and OCTACKSRDY = 0, or when OCTACKSREQ = 0 and OCTACKSRDY = 1.

OCTACKSEL[3:0] bits (Octal-SPI Clock (OCTACLK) Source Select)

These bits select the clock source of the Octal-SPI clock (OCTACLK) and must be modified when OCTACKCR.OCTACKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

OCTACKSREQ bit (Octal-SPI Clock (OCTACLK) Switching Request)

This bit selects the OCTACLK switching request.

OCTACKSRDY flag (Octal-SPI Clock (OCTACLK) Switching Ready state flag)

This flag indicates the state of switching ready for the OCTACLK. When OCTACKSRDY = 1, no clock is output to OCTACLK.

9.2.46 CANFDCKCR : CANFD Core Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x076

Bit position:	7	6	5	4	3	2	1	0
Bit field:	CANFDCKSRDY	CANFDCKSREQ	—	—	CANFDCKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	CANFDCKSEL[3:0]	CANFD Core Clock (CANFDCLK) Source Select 0 0 0 0: HOCO 0 0 0 1: MOCO (Value after reset) 0 0 1 0: LOCO 0 0 1 1: Main clock oscillator 0 1 0 0: Sub-clock oscillator 0 1 0 1: PLL1P 0 1 1 0: PLL2P 0 1 1 1: PLL1Q 1 0 0 0: PLL1R 1 0 0 1: PLL2Q 1 0 1 0: PLL2R Others: Setting prohibited.	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	CANFDCKSREQ	CANFD Core Clock (CANFDCLK) Switching Request 0: No request 1: Request switching	R/W
7	CANFDCKSRDY	CANFD Core Clock (CANFDCLK) Switching Ready state flag 0: Impossible to Switch 1: Possible to Switch	R

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The CANFDCKCR register controls the CANFD Core Clock (CANFDCLK).

When switching the clock source, ensure that the clock before the switch and the clock after the switch generate stable output. To change the set value of CANFDCKDIVCR.CANFDCKDIV[3:0] and CANFDCKSEL[3:0], use the following procedure:

1. Write 1 to MSTPCRC.MSTPC26 and MSTPCRC.MSTPC27 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two CANFDCLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to CANFDCKSREQ.
4. Poll until CANFDCKSRDY is read as 1. While CANFDCKSRDY = 1, no clock is output to CANFDCLK.
5. Write to CANFDCKDIVCR.CANFDCKDIV[3:0] and CANFDCKSEL[3:0].
6. Write 0 to CANFDCKSREQ.
7. Poll until CANFDCKSRDY is read as 0.
8. When CANFDCKSRDY becomes 0, CANFDCLK starts to output. Clock switching is complete.

When transitioning to Software Standby or Deep Software Standby mode, do not execute the WFI instruction while performing clock switching. That is, do not execute the WFI instruction when CANFDCKSREQ = 1 and CANFDCKSRDY = 0, or when CANFDCKSREQ = 0 and CANFDCKSRDY = 1.

CANFDCKSEL[3:0] bits (CANFD Core Clock (CANFDCLK) Source Select)

These bits select the clock source of the CANFD Core clock (CANFDCLK) and must be modified when CANFDCKCR.CANFDCKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

CANFDCKSREQ bit (CANFD Core Clock (CANFDCLK) Switching Request)

This bit selects the CANFDCLK switching request.

CANFDCKSRDY flag (CANFD Core Clock (CANFDCLK) Switching Ready state flag)

This flag indicates the state of switching ready for the CANFDCLK. When CANFDCKSRDY = 1, no clock is output to CANFDCLK.

9.2.47 USB60CKCR : USB60 Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x077

Bit position:	7	6	5	4	3	2	1	0
Bit field:	USB60CKSRDY	USB60CKSREQ	—	—	USB60CKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	USB60CKSEL[3:0]	USB60 Clock (USB60CK) Source Select 0 0 0 0: HOCO 0 0 0 1: MOCO (Value after reset) 0 0 1 1: Main clock oscillator 0 1 0 1: PLL1P 0 1 1 0: PLL2P 0 1 1 1: PLL1Q 1 0 0 0: PLL1R 1 0 0 1: PLL2Q 1 0 1 0: PLL2R Others: Setting prohibited.	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	USB60CKSREQ	USB60 Clock (USB60CK) Switching Request 0: No request 1: Request switching	R/W
7	USB60CKSRDY	USB60 Clock (USB60CK) Switching Ready state flag 0: Impossible to Switch 1: Possible to Switch	R

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

The USB60CKCR register controls the USB60 Clock.

To change the set value of USB60CKDIVCR. USB60CKDIV[3:0] and USB60CKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure:

1. Write 1 to MSTPCRB.MSTPB12 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two USB60CLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to USB60CKSREQ.
4. Poll until USB60CKSRDYY is read as 1. While USB60CKSRDY = 1, no clock is output to USB60CLK.
5. Write to USB60CKDIVCR.USB60CKDIV[3:0] and USB60CKSEL[3:0].

6. Write 0 to USB60CKSREQ.
7. Poll until USB60CKSRDY is read as 0.
8. When USB60CKSRDY becomes 0, USB60CLK starts to output. Clock switching is complete.

When transitioning to Software Standby or Deep Software Standby mode, do not execute the WFI instruction while performing clock switching. That is, do not execute the WFI instruction when USB60CKSREQ = 1 and USB60CKSRDY = 0, or when USB60CKSREQ = 0 and USB60CKSRDY = 1.

USB60CKSEL[3:0] bits (USB60 Clock (USB60CK) Source Select)

These bits select the clock source of the USB Clock (USB60CLK) and must be modified when USB60CKCR.USB60CKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

USB60CKSREQ bit (USB60 Clock (USB60CK) Switching Request)

This bit selects the USB60CLK switching request.

USB60CKSRDY flag (USB60 Clock (USB60CK) Switching Ready state flag)

This flag indicates the state of switching ready for the USB60CLK. When USB60CKSRDY = 1, no clock is output to USB60CLK.

9.2.48 I3CCKCR : I3C Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x078

Bit position:	7	6	5	4	3	2	1	0
Bit field:	I3CCK SRDY	I3CCK SREQ	—	—	I3CCKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	I3CCKSEL[3:0]	I3C Clock (I3CCLK) Source Select 0001: MOCO (value after reset) 0101: PLL1P 0110: PLL2P 0111: PLL1Q 1000: PLL1R 1001: PLL2Q 1010: PLL2R Others: Setting prohibited	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	I3CCKSREQ	I3C Clock (I3CCLK) Switching Request 0: No request 1: Request switching	R/W
7	I3CCKSRDY	I3C Clock (I3CCLK) Switching Ready state flag 0: Impossible to switch 1: Possible to switch	R

Note: S-TYPE-3, P-TYPE-2

I3CCKCR controls the I3C clock.

To change the set value of I3CCKDIVCR.I3CCKDIV[3:0] and I3CCKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure:

1. Write 1 to MSTPCRB.MSTPB4 only when switching the division ratio setting from 1/n (n ≠ 1).

2. Wait for two I3CCLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to I3CCKSREQ.
4. Polling until I3CCKSRDY is read as 1. While I3CCKSRDY = 1, no clock is output to I3CCLK.
5. Write the setting value to I3CCKDIVCR.I3CCKDIV[3:0] and I3CCKSEL[3:0].
6. Write 0 to I3CCKSREQ.
7. Polling until I3CCKSRDY is read as 0.
8. When I3CCKSRDY becomes 0, I3CCLK starts outputting. Clock switching complete.

When a transition to Software Standby or Deep Software Standby, do not execute the WFI instruction while performing clock selection switching. In other words, do not execute the WFI instruction with I3CCKSREQ = 1 and I3CCKSRDY = 0, or I3CCKSREQ = 0 and I3CCKSRDY = 1.

I3CCKSEL[3:0] bits (I3C Clock (I3CCLK) Source Select)

These bits select the clock source of the I3C clock (I3CCLK) and must be modified when I3CCKCR.I3CCKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

I3CCKSREQ bit (I3C Clock (I3CCLK) Switching Request)

This bit selects the I3CCLK switching request.

I3CCKSRDY bit (I3C Clock (I3CCLK) Switching Ready state flag)

This flag indicates the state of switching ready for the I3CCLK.

While I3CCKSRDY = 1, no clock is output to I3CCLK.

9.2.49 SCICKDIVCR : SCI Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x054

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	SCICKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	SCICKDIV[3:0]	SCI clock (SCICLK) Division Select 0 0 0 0: 1/1 (Value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 0 1: 1/3 0 1 1 0: 1/5 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

SCICKDIVCR controls the SCI clock.

SCICKDIV[3:0] bits (SCI clock (SCICLK) Division Select)

These bits select the frequency of the SCI clock (SCICLK) and must be modified when SCICKCR.SCICKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set all MSTPCRB.MSTPB22 to MSTPB31 = 1 before setting SCICKCR.SCICKSREQ = 1.

9.2.50 SPICKDIVCR : SPI Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x056

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	SPICKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	SPICKDIV[3:0]	SPI clock (SPICLK) Division Select 0 0 0 0: 1/1 (Value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 0 1: 1/3 0 1 1 0: 1/5 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

SPICKDIVCR controls the SPI clock.

SPICKDIV[3:0] bits (SPI clock (SPICLK) Division Select)

These bits select the frequency of the SPI clock (SPICLK) and must be modified when SPICKCR.SPICKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRB.MSTPB18 = 1 and MSTPCRB.MSTPB19 = 1 before setting SPICKCR.SPICKSREQ = 1.

9.2.51 ADCCKDIVCR : ADC Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x05A

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	ADCCKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	ADCCKDIV[3:0]	ADC Clock (ADCCLK) Division Select 0 0 0 0: 1/1 (value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

ADCCKDIVCR controls the ADC clock.

ADCCKDIV[3:0] bits (ADC Clock (ADCCLK) Division Select)

These bits select the frequency of the ADC clock (ADCCLK).

These bits must change when ADCCKCR.ADCCKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRD.MSTPD21 = 1 before setting ADCCKCR.ADCCKSREQ = 1.

9.2.52 GPTCKDIVCR : GPT Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x05C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	GPTCKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	GPTCKDIV[3:0]	GPT Clock (GPTCLK) Division Select 0 0 0 0: 1/1 (value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 0 1: 1/3*1 0 1 1 0: 1/5*1 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note 1. Do not set when GPT clock is used for ADC16H or PDG.

GPTCKDIVCR controls the GPT clock.

GPTCKDIV[3:0] bits (GPT Clock (GPTCLK) Division Select)

These bits select the frequency of the GPT clock (GPTCLK).

These bits must change when GPTCKCR.GPTCKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set all MSTPCRE.MSTPE18 to MSTPE21 and MSTPE27 to MSTPE31 = 1 before setting GPTCKCR.GPTCKSREQ = 1.

9.2.53 LCDCKDIVCR : LCD Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x05E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	LCDCKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	LCDCKDIV[3:0]	LCD Clock (LCDCLK) Division Select 0 0 0 0: 1/1 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 0 1: 1/3 0 1 1 0: 1/5 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

LCDCKDIVCR controls the LCD clock.

LCDCKDIV[3:0] bits (LCD Clock (LCDCLK) Division Select)

These bits select the frequency of the LCD clock (LCDCLK) and must be modified when LCDCKCR.LCDCKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRC.MSTPC4 = 1 before setting LCDCKCR.LCDCKSREQ = 1.

9.2.54 SCICKCR : SCI Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x055

Bit position:	7	6	5	4	3	2	1	0
Bit field:	SCICK SRDY	SCICK SREQ	—	—	SCICKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	SCICKSEL[3:0]	SCI Clock (SCICLK) Source Select 0000: HOCO 0001: MOCO (value after reset) 0010: LOCO 0011: Main clock oscillator 0100: Sub-clock oscillator 0101: PLL1P 0110: PLL2P 0111: PLL1Q 1000: PLL1R 1001: PLL2Q 1010: PLL2R Others: Setting prohibited	R/W

Bit	Symbol	Function	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	SCICKSREQ	SCI Clock (SCICLK) Switching Request 0: No request 1: Request switching	R/W
7	SCICKSRDY	SCI Clock (SCICLK) Switching Ready state flag 0: Impossible to switch 1: Possible to switch	R

Note: S-TYPE-3, P-TYPE-2

SCICKCR controls the SCI clock.

To change the set value of SCICKDIVCR.SCICKDIV[3:0] and SCICKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure:

1. Write 1 to all MSTPCRB.MSTPB22 to MSTPB31 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two SCICLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to SCICKSREQ.
4. Polling until SCICKSRDY is read as 1. While SCICKSRDY = 1, no clock is output to SCICLK.
5. Write the setting value to SCICKDIVCR.SCICKDIV[3:0] and SCICKSEL[3:0].
6. Write 0 to SCICKSREQ.
7. Polling until SCICKSRDY is read as 0.
8. When SCICKSRDY becomes 0, SCICLK starts outputting. Clock switching complete.

When a transition to Software Standby or Deep Software Standby, do not execute the WFI instruction while performing clock selection switching. In other words, do not execute the WFI instruction with SCICKSREQ = 1 and SCICKSRDY = 0, or SCICKSREQ = 0 and SCICKSRDY = 1.

SCICKSEL[3:0] bits (SCI Clock (SCICLK) Source Select)

These bits select the clock source of the SCI clock (SCICLK) and must be modified when SCICKCR.SCICKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

SCICKSREQ bit (SCI Clock (SCICLK) Switching Request)

This bit selects the SCICLK switching request.

SCICKSRDY bit (SCI Clock (SCICLK) Switching Ready state flag)

This flag indicates the state of switching ready for the SCICLK.

While SCICKSRDY = 1, no clock is output to SCICLK.

9.2.55 SPICKCR : SPI Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x057

Bit position:	7	6	5	4	3	2	1	0
Bit field:	SPICK SRDY	SPICK SREQ	—	—	SPICKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	SPICKSEL[3:0]	SPI Clock (SPICLK) Source Select 0000: HOCO 0001: MOCO (value after reset) 0010: LOCO 0011: Main clock oscillator 0100: Sub-clock oscillator 0101: PLL1P 0110: PLL2P 0111: PLL1Q 1000: PLL1R 1001: PLL2Q 1010: PLL2R Others: Setting prohibited	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	SPICKSREQ	SPI Clock (SPICLK) Switching Request 0: No request 1: Request switching	R/W
7	SPICKSRDY	SPI clock (SPICLK) Switching Ready State Flag 0: Impossible to switch 1: Possible to switch	R

Note: S-TYPE-3, P-TYPE-2

SPICKCR controls the SPI clock.

To change the set value of SPICKDIVCR.SPICKDIV[3:0] and SPICKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure:

1. Write 1 to MSTPCRB.MSTPB18 and MSTPCRB.MSTPB19 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two SPICLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to SPICKSREQ.
4. Polling until SPICKSRDY is read as 1. While SPICKSRDY = 1, no clock is output to SPICLK.
5. Write the setting value to SPICKDIVCR.SPICKDIV[3:0] and SPICKSEL[3:0].
6. Write 0 to SPICKSREQ.
7. Polling until SPICKSRDY is read as 0.
8. When SPICKSRDY becomes 0, SPICLK starts outputting. Clock switching complete.

When a transition to Software Standby or Deep Software Standby, do not execute the WFI instruction while performing clock selection switching. In other words, do not execute the WFI instruction with SPICKSREQ = 1 and SPICKSRDY = 0, or SPICKSREQ = 0 and SPICKSRDY = 1.

SPICKSEL[3:0] bits (SPI Clock (SPICLK) Source Select)

These bits select the clock source of the SPI clock (SPICLK) and must be modified when SPICKCR.SPICKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

SPICKSREQ bit (SPI Clock (SPICLK) Switching Request)

This bit selects the SPICLK switching request.

SPICKSRDY bit (SPI clock (SPICLK) Switching Ready State Flag)

This flag indicates the state of switching ready for the SPICLK.

While SPICKSRDY = 1, no clock is output to SPICLK.

9.2.56 ADCCKCR : ADC Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x05B

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ADCCKSRDY	ADCCKSREQ	—	—	ADCCKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	ADCCKSEL[3:0]	ADC Clock (ADCCLK) Source Select 0 0 0 0: HOCO 0 0 0 1: MOCO (value after reset) 0 0 1 0: LOCO 0 0 1 1: Main clock oscillator 0 1 0 0: Sub-clock oscillator 0 1 0 1: PLL1P 0 1 1 0: PLL2P 0 1 1 1: PLL1Q 1 0 0 0: PLL1R 1 0 0 1: PLL2Q 1 0 1 0: PLL2R Others: Setting prohibited	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	ADCCKSREQ	ADC Clock (ADCCLK) Switching Request 0: No request 1: Request switching	R/W
7	ADCCKSRDY	ADC Clock (ADCCLK) Switching Ready State Flag 0: Impossible to switch 1: Possible to switch	R[*1]

Note: S-TYPE-3, P-TYPE-2
Note 1. This bit is read only.

ADCCKCR controls the ADCCLK.

To change the set value of ADCCKDIVCR.ADCCKDIV[3:0] and ADCCKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure:

1. Write 1 to MSTPCRD.MSTPD21 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two ADCCLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to ADCCKSREQ.
4. Polling until ADCCKSRDY is read as 1.
   While ADCCKSRDY = 1, no clock is output to the ADCCLK.
5. Write the setting value to ADCCKDIVCR.ADCCKDIV[3:0] and ADCCKSEL[3:0].
6. Write 0 to ADCCKSREQ.
7. Polling until ADCCKSRDY is read as 0.
   When ADCCKSRDY becomes 0, the ADCCLK starts outputting.
8. Clock switching complete.

When a transition to Software Standby or Deep Software Standby, do not execute the WFI instruction while performing clock selection switching. In other words, do not execute the WFI instruction with ADCCKSREQ = 1 and ADCCKSRDY = 0, or ADCCKSREQ = 0 and ADCCKSRDY = 1.

ADCCKSEL[3:0] bits (ADC Clock (ADCCLK) Source Select)

These bits select the clock source of the ADCCLK.

These bits must change when ADCCKCR.ADCCKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

ADCCKSREQ bit (ADC Clock (ADCCLK) Switching Request)

This bit selects the ADCCLK switching request.

ADCCKSRDY flag (ADC Clock (ADCCLK) Switching Ready State Flag)

This flag indicates the state of switching ready for the ADCCLK.

While ADCCKSRDY = 1, no clock is output to the ADCCLK.

9.2.57 GPTCKCR : GPT Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x05D

Bit position:	7	6	5	4	3	2	1	0
Bit field:	GPTCKSRDY	GPTCKSREQ	—	—	GPTCKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	GPTCKSEL[3:0]	GPT Clock (GPTCLK) Source Select 0 0 0 0: HOCO 0 0 0 1: MOCO (value after reset) 0 0 1 1: Main clock oscillator 0 1 0 1: PLL1P 0 1 1 0: PLL2P 0 1 1 1: PLL1Q 1 0 0 0: PLL1R 1 0 0 1: PLL2Q 1 0 1 0: PLL2R Others: Setting prohibited.	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	GPTCKSREQ	GPT Clock (GPTCLK) Switching Request 0: No request 1: Request switching	R/W
7	GPTCKSRDY	GPT Clock (GPTCLK) Switching Ready State Flag 0: Impossible to switch 1: Possible to switch	R/W[*1]

Note: S-TYPE-3, P-TYPE-2
Note 1. This bit is read only.

GPTCKCR controls the GPTCLK.

To change the set value of GPTCKDIVCR.GPTCKDIV[3:0] and GPTCKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure:

1. Write 1 to all MSTPCRE.MSTPE18 to MSTPE21 and MSTPE27 to MSTPE31 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two GPTCLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to GPTCKSREQ.

4. Polling until GPTCKSRDY is read as 1.
   While GPTCKSRDY = 1, no clock is output to the GPTCLK.
5. Write the setting value to GPTCKDIVCR.GPTCKDIV[3:0] and GPTCKSEL[3:0].
6. Write 0 to GPTCKSREQ.
7. Polling until GPTCKSRDY is read as 0.
   When GPTCKSRDY becomes 0, the GPTCLK starts outputting.
8. Clock switching complete.

When a transition to Software Standby or Deep Software Standby, do not execute the WFI instruction while performing clock selection switching. In other words, do not execute the WFI instruction with GPTCKSREQ = 1 and GPTCKSRDY = 0, or GPTCKSREQ = 0 and GPTCKSRDY = 1.

GPTCKSEL[3:0] bits (GPT Clock (GPTCLK) Source Select)

These bits select the clock source of GPTCLK.

These bits must change when GPTCKCR.GPTCKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

GPTCKSREQ bit (GPT Clock (GPTCLK) Switching Request)

This bit selects the GPTCLK switching request.

GPTCKSRDY flag (GPT Clock (GPTCLK) Switching Ready State Flag)

This flag indicates the state of switching ready for the GPTCLK.

While GPTCKSRDY = 1, no clock is output to the GPTCLK.

9.2.58 LCDCKCR : LCD Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x05F

Bit position:	7	6	5	4	3	2	1	0
Bit field:	LCDCKSRDY	LCDCKSREQ	—	—	LCDCKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	LCDCKSEL[3:0]	LCD Clock (LCDCLK) Source Select 0001: MOCO (value after reset) 0101: PLL1P 0110: PLL2P 0111: PLL1Q 1000: PLL1R 1001: PLL2Q 1010: PLL2R Others: Setting prohibited	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	LCDCKSREQ	LCD Clock (LCDCLK) Switching Request 0: No request 1: Request switching	R/W
7	LCDCKSRDY	LCD Clock (LCDCLK) Switching Ready State Flag 0: Impossible to switch 1: Possible to switch	R

Note: S-TYPE-3, P-TYPE-2

LCDCKCR controls the LCD clock.

To change the set value of LCDCKDIVCR.LCDCKDIV[3:0] and LCDCKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure:

1. Write 1 to MSTPCRC.MSTPC4 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two LCDCLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to LCDCKSREQ.
4. Polling until LCDCKSRDY is read as 1. While LCDCKSRDY = 1, no clock is output to LCDCLK.
5. Write the setting value to LCDCKDIVCR.LCDCKDIV[3:0] and LCDCKSEL[3:0].
6. Write 0 to LCDCKSREQ.
7. Polling until LCDCKSRDY is read as 0.
8. When LCDCKSRDY becomes 0, LCDCLK starts outputting. Clock switching complete.

When a transition to Software Standby or Deep Software Standby, do not execute the WFI instruction while performing clock selection switching. In other words, do not execute the WFI instruction with LCDCKSREQ = 1 and LCDCKSRDY = 0, or LCDCKSREQ = 0 and LCDCKSRDY = 1.

LCDCKSEL[3:0] bits (LCD Clock (LCDCLK) Source Select)

These bits select the clock source of the LCD clock (LCDCLK) and must be modified when LCDCKCR.LCDCKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

LCDCKSREQ bit (LCD Clock (LCDCLK) Switching Request)

This bit selects the LCDCLK switching request.

LCDCKSRDY bit (LCD Clock (LCDCLK) Switching Ready State Flag)

This flag indicates the state of switching ready for the LCDCLK.

While LCDCKSRDY = “1”, no clock is output to LCDCLK.

9.2.59 TRCKCR : Trace Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x03F

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TRCKEN	—	—	TRCKSEL	TRCK[3:0]			
Value after reset:	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
3:0	TRCK[3:0]	Trace Clock Operating Frequency Select 0000: 1/1 0001: 1/2 0010: 1/4 (value after reset) 0011: 1/8 0100: 1/16 0101: 1/32 0110: 1/64 0111: 1/128 1000: 1/256 1001: 1/3 1010: 1/6 1011: 1/12 Others: Setting prohibited	R/W
4	TRCKSEL	Trace Clock Source Select 0: System clock source (value after reset) 1: HOCO (oscillation in debug mode)	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	TRCKEN	Trace Clock Operating Enable 0: Stop 1: Operation enable	R/W

Note: S-TYPE5, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note: Trace clock must be enabled when accessing TPIU registers.

Trace Clock Control Register controls switching the trace clock.

This register is not controlled by CGFSAR register.

When changing the frequency of the Trace Clock, the trace clock must be stopped (TRCKCR.TRCKEN = 0).

TRCK[3:0] bit (Trace Clock Operating Frequency Select)

These bits select the frequency of the trace clock.

This bit must change when TRCKCR.TRCKEN = 0.

TRCKSEL bit (Trace Clock Source Select)

This bit selects the clock source of the trace clock. According to this bit, the trace clock is as follows:

[When this bit = 0]

- The system clock source is selected as the clock source for the trace clock.
  The system clock source is selected by SCKSCR.CKSEL[2:0] bits.
- The clock supply to the trace clock is selected of MOCO divided by 1 when in Software Standby mode or Deep Software Standby mode.

[When this bit = 1]

- The HOCO is selected as the clock source for the trace clock.
- If TRCKSEL is set to 1, HOCO is operated depending on the state of HOCOCR.HCSTP. However, OSCSF does not confirm that the swing is stable, so wait for the stable wait time.
- HOCO continues to oscillate and supply the trace clock even if in Software Standby mode or Deep Software Standby mode.

This bit must change when TRCKCR.TRCKEN = 0.

TRCKEN bit (Trace Clock Operating Enable)

This bit enables or disables the clock supply of the trace clock.

TRCK[3:0] bits and TRCKSEL bit must be set before changing this bit from 0 (stop) to 1 (enable).

When changing this bit from 1 (enable) to 0 (stop), the trace function must be disabled.

The trace clock is output with this bit set to 1 and under the following conditions:

- CDBGPWRUPREQ = 1 and AL ≠ AL0
- DBGCTRL0.REGEN = 1 and OFS1.SWDBG = 0.

9.2.60 BCKADIVCR : Asynchronous External Bus Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0D4

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	BCKACKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	BCKACKDIV[3:0]	Asynchronous External Bus Clock (BCLKA) Division Select 0 0 0 0: 1/1 (value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

BCKADIVCR controls the asynchronous external bus clock.

BCKACKDIV[3:0] bits (Asynchronous External Bus Clock (BCLKA) Division Select)

These bits select the frequency of the asynchronous external bus clock (BCLKA).

These bits must change when BCKACR.BCKACKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set EBCKOCR.EBCKOEN = 0 and SDCKOCR.SDCKOEN = 0 before setting BCKACR.BCKACKSREQ = 1. In addition, set BCKCR.EBCKASEL = 0 to select the synchronous (BCLK) side before switching the division ratio.

9.2.61 ESWCKDIVCR : EtherSW Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0D5

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	ESWCKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	ESWCKDIV[3:0]	EtherSW Clock (ESWCLK) Division Select 0 0 0 0: 1/1 (value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

ESWCKDIVCR controls the EtherSW clock.

ESWCKDIV[3:0] bits (EtherSW Clock (ESWCLK) Division Select)

These bits select the frequency of the EtherSW clock (ESWCLK).

These bits must change when ESWCKCR.ESWCKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRC.MSTPC30 = 1 before setting ESWCKCR.ESWCKSREQ = 1.

9.2.62 ESWPCKDIVCR : EtherSW-PHY Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0D6

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	ESWPCKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	ESWPCKDIV[3:0]	EtherSW-PHY Clock (ESWPHYCLK) Division Select 0 0 0 0: 1/1 (value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

ESWPCKDIVCR controls the EtherSW-PHY clock.

ESWPCKDIV[3:0] bits (EtherSW-PHY Clock (ESWPHYCLK) Division Select)

These bits select the frequency of the EtherSW-PHY clock (ESWPHYCLK).

These bits must change when ESWPCKCR.ESWPCKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRC.MSTPC30 = 1 before setting ESWPCKCR.ESWPCKSREQ = 1.

9.2.63 ETHPCKDIVCR : Ether-PHY Clock Division Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0D8

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	ETHPCKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	ETHPCKDIV[3:0]	Ether-PHY Clock (ETHPHYCLK) Division Select 0 0 0 0: 1/1 (value after reset) 0 0 0 1: 1/2 0 0 1 0: 1/4 0 0 1 1: 1/6 0 1 0 0: 1/8 0 1 1 1: 1/10 1 0 0 0: 1/16 1 0 0 1: 1/32 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

ETHPCKDIVCR controls the Ether-PHY clock.

ETHPCKDIV[3:0] bits (Ether-PHY Clock (ETHPHYCLK) Division Select)

These bits select the frequency of the Ether-PHY clock (ETHPHYCLK).

These bits must change when ETHPCKCR.ETHPCKSRDY = 1.

When switching the division ratio setting from 1/n (n ≠ 1), set MSTPCRC.MSTPC28 = 1 before setting ETHPCKCR.ETHPCKSREQ = 1.

9.2.64 BCKACR : Asynchronous External Bus Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0DA

Bit position:	7	6	5	4	3	2	1	0
Bit field:	BCKACKSRDY	BCKACKSREQ	—	—	BCKACKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	BCKACKSEL[3:0]	Asynchronous External Bus Clock (BCLKA) 0 0 0 1: MOCO (value after reset) 0 1 0 1: PLL1P 0 1 1 0: PLL2P 0 1 1 1: PLL1Q 1 0 0 0: PLL1R 1 0 0 1: PLL2Q 1 0 1 0: PLL2R Others: Setting prohibited.	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	BCKACKSREQ	Asynchronous External Bus Clock (BCLKA) Switching Request 0: No request 1: Request switching	R/W

Bit	Symbol	Function	R/W
7	BCKACKSRDY	Asynchronous External Bus Clock (BCLKA) Switching Ready State Flag 0: Impossible to switch 1: Possible to switch	R*1

Note: S-TYPE-3, P-TYPE-2
Note 1. This bit is read only.

BCKACR controls the BCLKA.

To change the set value of BCKADIVCR.BCKACKDIV[3:0] and BCKACKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure:

1. Write 0 to EBCKOCR.EBCKOEN and SDCKOCR.SDCKOEN only when switching the division ratio setting from 1/n (n ≠ 1).
2. Write 1 to BCKACKSREQ.
3. Polling until BCKACKSRDY is read as 1.
   While BCKACKSRDY = 1, no clock is output to the BCLKA.
4. Write 0 to BCKCR.EBCKASEL only when switching the division ratio setting from 1/n (n ≠ 1).
5. Write the setting value to BCKCR.BCLKDIV, BCKCR.EBCKASEL, BCKADIVCR.BCKACKDIV[3:0] and BCKACKSEL[3:0].
6. Write 0 to BCKACKSREQ.
7. Polling until BCKACKSRDY is read as 0.
   When BCKACKSRDY becomes 0, the BCLKA starts outputting.
8. Clock switching complete.

When a transition to Software Standby or Deep Software Standby, do not execute the WFI instruction while performing clock selection switching. In other words, do not execute the WFI instruction with BCKACKSREQ = 1 and BCKACKSRDY = 0, or BCKACKSREQ = 0 and BCKACKSRDY = 1.

BCKACKSEL[3:0] bits (Asynchronous External Bus Clock (BCLKA))

These bits select the clock source of the BCLKA.

These bits must change when BCKACR.BCKACKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected

BCKACKSREQ bit (Asynchronous External Bus Clock (BCLKA) Switching Request)

This bit selects the BCLKA switching request.

BCKACKSRDY flag (Asynchronous External Bus Clock (BCLKA) Switching Ready State Flag)

This flag indicates the state of switching ready for the BCLKA.

While BCKACKSRDY = 1, no clock is output to the BCLKA.

9.2.65 ESWCKCR : EtherSW Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0DB

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ESWCKSRDY	ESWCKSREQ	—	—	ESWCKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	ESWCKSEL[3:0]	EtherSW Clock (ESWCLK) Source Select 0 0 0 1: MOCO (value after reset) 0 1 0 1: PLL1P 0 1 1 0: PLL2P 0 1 1 1: PLL1Q 1 0 0 0: PLL1R 1 0 0 1: PLL2Q 1 0 1 0: PLL2R Others: Setting prohibited	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	ESWCKSREQ	EtherSW Clock (ESWCLK) Switching Request 0: No request 1: Request switching	R/W
7	ESWCKSRDY	EtherSW Clock (ESWCLK) Switching Ready State Flag 0: Impossible to switch 1: Possible to switch	R*1

Note: S-TYPE-3, P-TYPE-2
Note 1. This bit is read only.

ESWCKCR controls the ESWCLK.

To change the set value of ESWCKDIVCR.ESWCKDIV[3:0] and ESWCKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure:

1. Write 1 to MSTPCRC.MSTPC30 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two ESWCLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to ESWCKSREQ.
4. Polling until ESWCKSRDY is read as 1.
   While ESWCKSRDY = 1, no clock is output to the ESWCLK.
5. Write the setting value to ESWCKDIVCR.ESWCKDIV[3:0] and ESWCKSEL[3:0].
6. Write 0 to ESWCKSREQ.
7. Polling until ESWCKSRDY is read as 0.
   When ESWCKSRDY becomes 0, the ESWCLK starts outputting.
8. Clock switching complete.

When a transition to Software Standby or Deep Software Standby, do not execute the WFI instruction while performing clock selection switching. In other words, do not execute the WFI instruction with ESWCKSREQ = 1 and ESWCKSRDY = 0, or ESWCKSREQ = 0 and ESWCKSRDY = 1.

ESWCKSEL[3:0] bits (EtherSW Clock (ESWCLK) Source Select)

These bits select the clock source of the ESWCLK.

These bits must change when ESWCKCR.ESWCKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

ESWCKSREQ bit (EtherSW Clock (ESWCLK) Switching Request)

This bit selects the ESWCLK switching request.

ESWCKSRDY flag (EtherSW Clock (ESWCLK) Switching Ready State Flag)

This flag indicates the state of switching ready for the ESWCLK.

While ESWCKSRDY = 1, no clock is output to the ESWCLK.

9.2.66 ESWPCKCR : EtherSW-PHY Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0DC

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ESWPCKSRDY	ESWPCKSREQ	—	—	ESWPCKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	ESWPCKSEL[3:0]	EtherSW-PHY Clock (ESWPHYCLK) Source Select 0 0 0 1: MOCO (value after reset) 0 1 0 1: PLL1P 0 1 1 0: PLL2P 0 1 1 1: PLL1Q 1 0 0 0: PLL1R 1 0 0 1: PLL2Q 1 0 1 0: PLL2R Others: Setting prohibited	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	ESWPCKSREQ	EtherSW-PHY Clock (ESWPHYCLK) Switching Request 0: No request 1: Request switching	R/W
7	ESWPCKSRDY	EtherSW-PHY Clock (ESWPHYCLK) Switching Ready State Flag 0: Impossible to switch 1: Possible to switch	R*1

Note: S-TYPE-3, P-TYPE-2
Note 1. This bit is read only.

ESWPCKCR controls the ESWPHYCLK.

To change the set value of ESWPCKDIVCR.ESWPCKDIV[3:0] and ESWPCKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure

1. Write 1 to MSTPCRC.MSTPC30 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two ESWPHYCLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to ESWPCKSREQ.
4. Polling until ESWPCKSRDY is read as 1.
   While ESWPCKSRDY = 1, no clock is output to the ESWPHYCLK.
5. Write the setting value to ESWPCKDIVCR.ESWPCKDIV[3:0] and ESWPCKSEL[3:0].
6. Write 0 to ESWPCKSREQ.
7. Polling until ESWPCKSRDY is read as 0.
   When ESWPCKSRDY becomes 0, the ESWPHYCLK starts outputting.
8. Clock switching complete.

When a transition to Software Standby or Deep Software Standby, do not execute the WFI instruction while performing clock selection switching. In other words, do not execute the WFI instruction with ESWPCKSREQ = 1 and ESWPCKSRDY = 0, or ESWPCKSREQ = 0 and ESWPCKSRDY = 1.

ESWPCKSEL[3:0] bits (EtherSW-PHY Clock (ESWPHYCLK) Source Select)

These bits select the clock source of the ESWPHYCLK.

These bits must change when ESWPCKCR.ESWPCKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

ESWPCKSREQ bit (EtherSW-PHY Clock (ESWPHYCLK) Switching Request)

This bit selects the ESWPHYCLK switching request.

ESWPCKSRDY flag (EtherSW-PHY Clock (ESWPHYCLK) Switching Ready State Flag)

This flag indicates the state of switching ready for the ESWPHYCLK.

While ESWPCKSRDY = 1, no clock is output to the ESWPHYCLK.

9.2.67 ETHPCKCR : Ether-PHY Clock Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0DE

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ETHPCKSRDY	ETHPCKSREQ	—	—	ETHPCKSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	ETHPCKSEL[3:0]	Ether-PHY Clock (ETHPHYCLK) Source Select 0 0 0 1: MOCO (value after reset) 0 0 1 1: Main clock oscillator 0 1 0 1: PLL1P 0 1 1 0: PLL2P 0 1 1 1: PLL1Q 1 0 0 0: PLL1R 1 0 0 1: PLL2Q 1 0 1 0: PLL2R Others: Setting prohibited	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	ETHPCKSREQ	Ether-PHY Clock (ETHPHYCLK) Switching Request 0: No request 1: Request switching	R/W
7	ETHPCKSRDY	Ether-PHY Clock (ETHPHYCLK) Switching Ready State Flag 0: Impossible to switch 1: Possible to switch	R*1

Note: S-TYPE-3, P-TYPE-2
Note 1. This bit is read only.

ETHPCKCR controls the ETHPHYCLK.

To change the set value of ETHPCKDIVCR.ETHPCKDIV[3:0] and ETHPCKSEL[3:0], follow the procedure below.

When switching the clock source, it is necessary that the clock before switching and the clock after switching are stably output.

Clock selection switching procedure:

1. Write 1 to MSTPCRC.MSTPC28 only when switching the division ratio setting from 1/n (n ≠ 1).
2. Wait for two ETHPHYCLK only when switching the division ratio setting from 1/n (n ≠ 1).
3. Write 1 to ETHPCKSREQ.
4. Polling until ETHPCKSRDY is read as 1.
   While ETHPCKSRDY = 1, no clock is output to the ETHPHYCLK.
5. Write the setting value to ETHPCKDIVCR.ETHPCKDIV[3:0] and ETHPCKSEL[3:0].
6. Write 0 to ETHPCKSREQ.
7. Polling until ETHPCKSRDY is read as 0.

When ETHPCKSRDY becomes 0, the ETHPHYCLK starts outputting.

8. Clock switching complete.

When a transition to Software Standby or Deep Software Standby, do not execute the WFI instruction while performing clock selection switching. In other words, do not execute the WFI instruction with ETHPCKSREQ = 1 and ETHPCKSRDY = 0, or ETHPCKSREQ = 0 and ETHPCKSRDY = 1.

ETHPCKSEL[3:0] bits (Ether-PHY Clock (ETHPHYCLK) Source Select)

These bits select the clock source of the ETHPHYCLK.

These bits must change when ETHPCKCR.ETHPCKSRDY = 1.

Do not stop the oscillator selected by these bits except when MOCO is selected.

ETHPCKSREQ bit (Ether-PHY Clock (ETHPHYCLK) Switching Request)

This bit selects the ETHPHYCLK switching request.

ETHPCKSRDY flag (Ether-PHY Clock (ETHPHYCLK) Switching Ready State Flag)

This flag indicates the state of switching ready for the ETHPHYCLK.

While ETHPCKSRDY = 1, no clock is output to the ETHPHYCLK.

9.2.68 USBCKSELR : USB Clock Selection Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0F0

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	UCKSEL
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	UCKSEL	USB Clock (USBCLK) Selection 0: Disable (value after reset) 1: Enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Rewrite the USBCKSELR register while the SYSCFG.SCKE bit is 0.

UCKSEL bit (USB Clock (USBCLK) Selection)

Set the UCKSEL bit depending on other register settings shown in Table 9.5.

The UCKSEL bit can only be set to 1 when USBFS is used as the device controller. Set the UCKSEL bit to 0 to use the host controller and the On-The-Go (OTG) function.

The HOCO user trimming function cannot be used when the UCKSEL bit is 1. Set the HOCOUTCR.HOCOUTRIM[7:0] bits to 0x00.

Table 9.5 UCKSEL bit setting

UCKSEL	USBCKCR.USBCKSEL[3:0]	FLLCR1.FLLEN
0	Other than 0000 (Other than HOCO)	—
	0000 (HOCO)	1 (FLL enabled)
1	0000 (HOCO)	0 (FLL disabled)

9.3 Main Clock Oscillator

To supply the clock signal to the main clock oscillator, use one of the following ways:

- Connect an oscillator
- Connect the input of an external clock signal.

9.3.1 Connecting a Crystal Resonator

Figure 9.2 shows an example of connecting a crystal resonator. A damping resistor (Rd) can be added, if required.

Because the resistor values vary according to the resonator and the oscillation drive capability, use values recommended by the resonator manufacturer. If the manufacturer recommends using an external feedback resistor (Rf), insert an Rf between EXTAL and XTAL by following the instructions.

When connecting a resonator to supply the clock, the frequency of the resonator must be in the frequency range of the resonator for the main clock oscillator as described in Table 9.1.

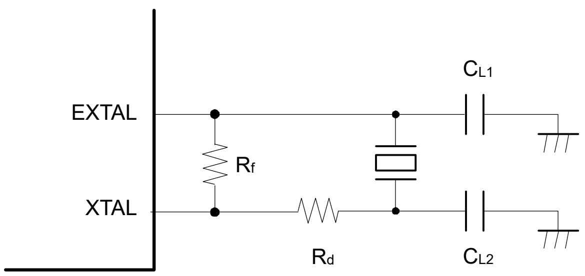


Figure 9.2 Example of crystal resonator connection

Figure 9.3 shows an equivalent circuit of the crystal resonator.

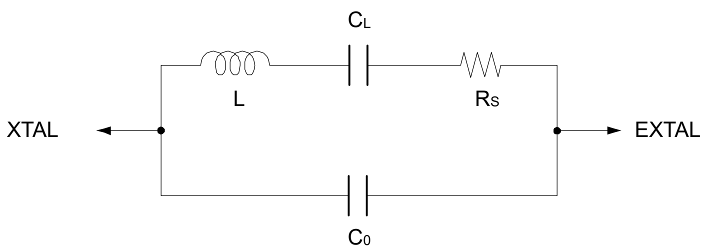


Figure 9.3 Equivalent circuit of the crystal resonator

9.3.2 External Clock Input

Figure 9.4 shows an example of connecting an external clock input. To operate the oscillator with an external clock signal, set the MOMCR.MOSEL bit to 1. The XTAL pin becomes high impedance.

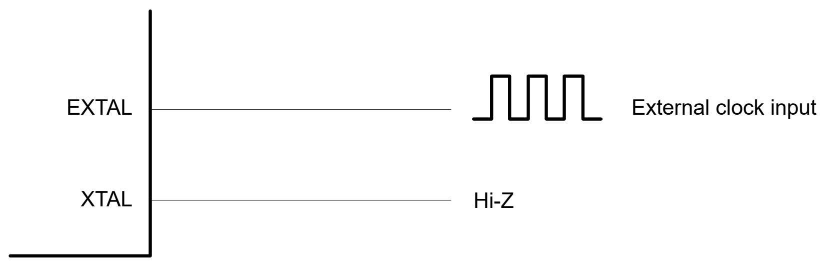


Figure 9.4 Equivalent circuit for external clock

9.3.3 Notes on External Clock Input

The frequency of the external clock input can only be changed when the main clock oscillator is stopped. Do not change the frequency of the external clock input when the setting of the Main Clock Oscillator Stop bit (MOSCCR.MOSTP) is 0.

9.3.4 Note on Oscillation Keep in Software Standby Mode

By using the oscillation keep function of the Main Clock Oscillator in Software Standby mode, the Main Clock Oscillator stabilization wait time after canceling Software standby mode can be shortened.

If MCU enters Software Standby mode under the MOSCSCR.MOSCSOKP bit is set to enabled (MOSCSCR.MOSCSOKP bit is 1) and under the Main Clock Oscillator is operating, the Main Clock Oscillator will keep oscillating in Software Standby mode. When canceling Software Standby mode in this state, the Main clock can be used without the oscillator stabilization wait time.

If using this function, confirm that the setting value is written to the register before entering the Software standby mode.

However, even if this function is enabled, the clock supply to the system and peripheral modules during Software Standby mode is stopped.

9.4 Sub-Clock Oscillator

To supply the clock signal to the sub-clock oscillator, use one of the following ways:

- Connect an oscillator
- Connect the input of an external clock signal.

9.4.1 Connecting a 32.768-kHz Crystal Resonator

To supply a clock to the sub-clock oscillator, connect a 32.768-kHz crystal resonator as shown in Figure 9.5. A damping resistor (Rd) can be added, if necessary. Because the resistor values vary according to the resonator and the oscillation drive capability, use values recommended by the resonator manufacturer. If the resonator manufacturer recommends the use of an external feedback resistor (Rf), insert an Rf between XCIN and XCOUT by following the instructions. When connecting a resonator to supply the clock, the frequency of the resonator must be in the frequency range of the resonator for the sub-clock oscillator as described in Table 9.1.

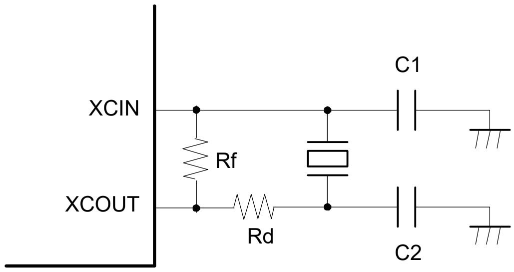


Figure 9.5 Connection example of 32.768-kHz crystal resonator

Figure 9.6 shows an equivalent circuit for the 32.768-kHz crystal resonator.

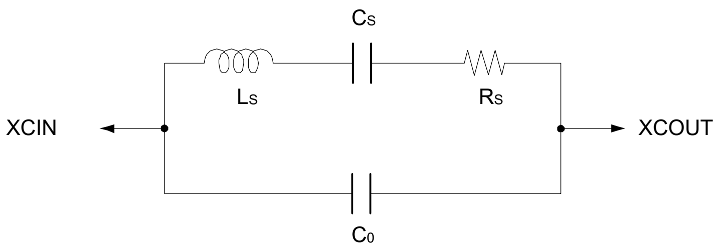


Figure 9.6 Equivalent circuit for the 32.768-kHz crystal resonator

9.4.2 External Clock Input

In the external clock input mode, clock is input from EXCIN pin.

9.5 Low Speed On-chip Oscillator (LOCO)

Low Speed On-chip Oscillator (LOCO) is the internal oscillator that supplies a low-speed clock.

9.5.1 LOCO Operating Conditions

There are restriction on LOCO operation setting. For detail, see section 9.5.3. Restrictions on LOCO Operation and Stopping

LOCO oscillates when any of the following conditions is met.*1

[LOCO operating conditions]

- When LOCO is set to operate in the LOCOCR register (LOCOCR.LCSTP bit is 0).
- When IWDT is set to the auto start mode. (OFS0.IWDTSTRT bit is 0)*2
- When IWDT is set to the register start mode (OFS0.IWDTSTRT bit is 1) and counting is started.*3
- \When waiting for the oscillation stabilization of the Main Clock Oscillator (while MOCOCR.MCSTP bit is 0 and OSCSF.MOSCSF bit is 0).*4

Note 1. LOCO stops at Deep Software Standby mode 2 or Deep Software Standby mode 3.
Note 2. When IWDT is set to the auto start mode, regardless of the setting of IWDTSTPCTL bit in OFS0 register, LOCO continues to oscillate.
Note 3. When IWDT is set to the register start mode, regardless of the setting of SLCSTP bit in IWDTCSTPR register, LOCO continues to oscillate once IWDT starts counting.
Note 4. Even if the Main Clock Oscillator is set to the external clock input mode, LOCO operates because the oscillation stabilization wait circuit operates when the Main Clock Oscillator starts operating.

9.5.2 LOCO Stop Conditions

There are restriction on LOCO stop setting. For detail, see section 9.5.3. Restrictions on LOCO Operation and Stopping.

There are two ways to top LOCO: stop by transition to Low Power mode or stop by LOCOCR register.

(A) LOCO stop conditions related to Low Power mode

LOCO stops when transitioning to the following Low Power mode

- Deep Software Standby mode 2
- Deep Software Standby mode 3

(B) LOCO stop conditions related to LOCOCR register

When LOCO is stopped using the LOCOCR register, even if the LCSTP bit set to stop (LOCOCR.LCSTP bit is 1), LOCO may not be stopped depending on the operating status of IWDT and oscillator.

LOCO can be stopped when all of the following conditions are met.

- LOCO is set to stop using the LOCOCR register (LOCOCR.LCSTP bit is 1)
- IWDT is not operating. (Include the case of stopping in CPU Sleep mode, CPU Deep Sleep mode, Software Standby mode, and Deep Software Standby mode 1 by setting the OFS0.IWDTSTPCTL bit or the IWDTCSTPR.SLCSTP bit.).
- Main Clock Oscillator is stopped or Main Clock oscillation is stable. (MOCOCR.MCSTP bit is 1 or OSCSF.MOSCSF bit is 1)

9.5.3 Restrictions on LOCO Operation and Stopping

After setting the LCSTP bit in LOCOCR to 0 to start the LOCO clock, only use the clock after the LOCO clock oscillation stabilization wait time (tLOCOWT) elapses. A fixed stabilization wait is required after setting the LOCO clock to start operation. A fixed wait for the oscillation to stop is also required.

The following restrictions apply when starting and stopping operation:

- After stopping the LOCO clock, read OSCMONR.LOCOMON, make sure LOCO is stopped, and then set the operation.
- Confirm that LOCO oscillation is stable before stopping the LOCO clock
- Confirm that LOCO oscillation is stable before executing a WFI instruction to place the MCU in Software Standby or Deep Software Standby mode.
- When a transition to Software Standby mode or Deep Software Standby mode after setting LOCO to stop, read OSCMONR.LOCOMON, make sure LOCO is stopped, and executing the WFI instruction.

The LOCO clock is used to measure the oscillation stabilization wait time of the main clock oscillator. While the main clock oscillator is waiting for oscillation stabilization, the LOCO clock oscillates regardless of the LCSTP bit setting. At this time, it is necessary to prevent conflicts between the LOCO stop (or operation) by the LCSTP bit in LOCOCR and the LOCO clock oscillation for the Main clock Oscillator stabilization wait time (or LOCO clock stop after Main clock oscillation stabilization). Therefore, do not change the LOCOCR.LCSTP bit during the period from the 3 LOCO clock cycle before setting the start of main clock oscillator operation until the main clock oscillator stabilization can be confirmed with the oscillation stabilization flag in the OSCSF register.

When using IWDT in the register start mode (OFS0.IWDTSTRT bit is 1), set the LOCO to run (LOCOCR.LCSTP = 0) before IWDT count operation, and use IWDT after LOCO oscillation has stabilized. When IWDT is used in register start mode, do not change the LOCOCR.LCSTP bit during the period from the 3 LOCO clock cycles before IWDT count operation starts until the start of IWDT count operation can be confirmed.

When using the Digital filter or the Reset negate select for Voltage monitor m (m = 1 to 2) in Programmable Voltage Detection (PVDm), set the LOCO to run (LOCOCR.LCSTP = 0) before enabling PVDm and use PVDm after LOCO oscillation has stabilized.

9.6 Middle Speed On-chip Oscillator (MOCO)

Middle Speed On-chip Oscillator (MOCO) is the internal oscillator that supplies a middle-speed clock.

9.6.1 MOCO Operating Conditions

There are restriction on MOCO operation setting. For detail, see section 9.6.3. Restrictions on MOCO Operation and Stopping.

MOCO oscillates when any of the following conditions is met.

[MOCO operating conditions]

- When MOCO is set to operate in the LOCOCR register (MOCOCR.MCSTP = 0)
- When waiting for the HOCO oscillation stabilization (while OFS1.HOCOEN bit is 0 or HOCOCR.HCSTP bit is 0, and OSCSF.HOCOSF bit is 0).
- When waiting for the PLL1 oscillation stabilization (while PLLCR.PLLSTP bit is 0, and OSCSF.PLLSF bit is 0).
- When waiting for the PLL2 oscillation stabilization (while PLL2CR.PLL2STP bit is 0, and OSCSF.PLL2SF bit is 0).
- When MOSC oscillation stop detection function is set to enable.
- When SOSC oscillation stop detection function is set to enable.

Note: MOCO stops at Software Standby mode or Deep Software Standby mode.

9.6.2 MOCO Stop Conditions

There are restriction on MOCO stop setting. For detail, see section 9.6.3. Restrictions on MOCO Operation and Stopping.

There are two ways to top MOCO: stop by transition to Low Power mode or stop by MOCOCR register.

(A) MOCO stop conditions related to Low Power mode

MOCO stops when transitioning to the following Low Power mode.

- Software Standby mode (MOCOSCR.MOCOSOKP=0)
- Deep Software Standby mode

Note: When the On-Chip Debugger function is enabled, MOCO does not stop when transitioning to each standby mode.

(B) MOCO stop conditions related to MOCOCR register

When MOCO is stopped using the MOCOCR register, even if the MCSTP bit set to stop (MOCOCR.MCSTP bit is 1), MOCO may not be stopped depending on the operating status of the system.

MOCO can be stopped when all of the following conditions are met.

[MOCO stop conditions]

- MOCO is set to stop using the MOCOCR register (MOCOCR.MCSTP bit is 1)
- When HOCO is stopped or HOCO oscillation is stable. (OFS1(_SEC).HOCOEN bit is 1 and HOCOCR.HCSTP bit is 1, or OSCSF.HOCOSF bit is 1)
- PLL1 circuit is stopped or PLL1 oscillation is stable. (PLLCR.PLSTP bit is 1 or OSCSF.PLLSF bit is 1)
- PLL2 circuit is stopped or PLL2 oscillation is stable. (PLL2CR.PL2STP bit is 1 or OSCSF.PLL2SF bit is 1)

9.6.3 Restrictions on MOCO Operation and Stopping

After setting the MCSTP bit to 0 to start the MOCO clock, only use the clock after the MOCO clock oscillation stabilization time (tMOCOWT) elapses. A fixed stabilization wait is required after setting the MOCO clock to start operation. A fixed wait for oscillation to stop is also required.

The following restrictions apply when starting and stopping operation.

- After stopping the MOCO clock, read OSCMONR.MOCOMON, make sure MOCO is stopped, and then set the operation.
- Confirm that MOCO oscillation is stable before stopping the MOCO clock.
- Regardless of whether the MOCO clock is selected as the system clock source, confirm that MOCO oscillation is stable before executing a WFI instruction to place the MCU in Software Standby or Deep Software Standby mode. If MOCO is stopped, operate the MOCO.

- Regardless of whether the MOCO clock is selected as the system clock source, confirm that MOCO oscillation is stable before executing the wait instruction to place the CPU in the CPU Deep Sleep mode. If MOCO is stopped, operate the MOCO.
- Confirm that MOCO oscillation is stable before executing a Power Control for each power domain by Software. If MOCO is stopped, operate the MOCO.

Writing 1 to MCSTP is prohibited under the following condition:

- SCKSCR.CKSEL[2:0] = 001b (system clock source = MOCO).
- Oscillation stop detection is enabled (OSTDCR.OSTDE bit is 1).
- While waiting for the HOCO oscillation stabilization (OFS1.HOCOEN bit is 0 or HOCOCR.HCSTP bit is 0, and OSCSF.HOCOSF bit is 0)
- While waiting for the PLL1 oscillation stabilization (PLLCR.PLLSTP bit is 0, and OSCSF.PLLSF bit is 0)
- While waiting for the PLL2 oscillation stabilization (PLL2CR.PLL2STP bit is 0, and OSCSF.PLL2SF bit is 0)
- While executing a Power Control for each power domain by software (During the Power Gating Control)

The MOCO clock is used to measure the oscillation stabilization wait time of the HOCO/PLL1/PLL2 clock. While waiting for the HOCO/PLL1/PLL2 clock to stabilize, the MOCO clock oscillates regardless of the MOCOCR.MCSTP bit setting. At this time, it is necessary to prevent conflicts between the MOCO stop (or operation) by the MCSTP bit in MOCOCR and the MOCO clock oscillation for HOCO/PLL1/PLL2 clock oscillation stabilization wait time (or MOCO clock stop after HOCO/PLL1/PLL2 clock oscillation stabilization). Therefore, do not change the MOCOCR.MCSTP bit during the period from the 3 MOCO clock cycles before the start of HOCO/PLL1/PLL2 clock operation setting until the HOCO/PLL1/PLL2 clock oscillation stability can be confirmed by the HOCOSF/PLLSF/PLL2SF flag in the OSCSF register.

9.7 High Speed On-chip Oscillator (HOCO)

High Speed On-chip Oscillator (HOCO) is the internal oscillator that supplies a high-speed clock.

9.7.1 Frequency Locked Loop Function

HOCO supports the Frequency Locked Loop (FLL) function that corrects the frequency accuracy of the HOCO clock using the Sub-Clock Oscillator. When the FLL function is enabled, the frequency accuracy of the HOCO clock can be improved compared to when the FLL function is disabled.

When using the FLL function, the oscillation of the Sub-Clock Oscillator must be stable. In addition, the sub-clock oscillator must meet the frequency accuracy specified in section 70, Electrical Characteristics.

When enabling the FLL function and using HOCO, enable the FLL function in advance and enable HOCO (HOCOCR.HCTSP=0) after SOSC has stabilized.

Set FLLCR2.FLLCNTL[10:0] as follows for each HOCO frequency setting. And the HOCO clock reaches the desired frequency after tFLLWT after enabling HOCO.

Table 9.6 FLLCNTL setting for each HOCO frequency setting

FLLCR2.FLLCNTL[10:0]	OFS1(_SEC).HOCOFRQ0[2:0]	Target Frequency (FHOCO)
0x1E8	000b	16 MHz
0x225	001b	18 MHz
0x262	010b	20 MHz
0x1E8	100b	32 MHz
0x1E8	111b	48 MHz

9.7.2 Note on Oscillation Keep in Software Standby mode

By using the oscillation keep function of the HOCO in Software Standby mode, the HOCO oscillation stabilization wait time after canceling Software standby mode can be shortened.

To keep the HOCO oscillation in Software Standby mode, make the following settings:

- Enable the HOCO Standby Oscillation Keep function in the HOCOSCR register. (HOCOSCR.HOCOSOKP bit is set to 1)
- The setting of power control for HOCO in HOCOLDOCR.

If MCU enters Software Standby mode under the above settings and under the HOCO is operating, the HOCO will keep oscillating in Software Standby mode. When canceling Software Standby mode in this state, the HOCO clock can be used without the oscillator stabilization wait time.

If using this function, confirm that the setting value is written to the register before entering the Software standby mode.

However, even if this function is enabled, the clock supply to the system and peripheral modules during Software Standby mode is stopped.

9.7.3 Note on using as TRACE clock source

If HOCO is selected as Trace clock source with TRCKCRSEL bit on TRCKCR register, and then Trace clock function is enable (TRCKCR.TRCKEN=1), HOCO oscillates regardless of the value of HCSTP bit on HOCOCR register. In this situation, HOCO will continue to oscillate even if MCU is in Software Standby mode, Deep Software Standby mode, or Reset (except POR and PVD0) to keep supplying the clock to the TRACE module.

If HOCO is used as Trace clock source, Trace clock should be enabled with TRCKEN bit on TRCKCR register while HOCO is stopped (OFS1(_SEC).HOCOEN=1 and HOCOCR.HCSTP=1) or HOCO is oscillating and stable (OSCSF.HOCOSF=1).

9.8 PLL Circuit

The PLL circuit has a function to multiply the frequency from the oscillator.

Figure 9.7 shows the block diagram of PLL1 circuit, and Figure 9.8 shows the PLL2 circuit. PLL1 and PLL2 have the same circuit configuration. The PLL1 circuit is explained the following as an example.

The input clock source selection circuit selects the clock source input to the PLL.

The input clock divider circuit divides the input clock and generates the PLL reference clock.

The frequency control circuit controls the VCO clock to be generated from the PLL reference clock according to the frequency multiplier setting. The frequency multiplication ratio is determined by the combination of the PLLMUL bits and PLLMULNF bits. The PLLMUL bit determines the integer part of the multiplication ratio, and the PLLMULNF bit determines the fractional part of the multiplication ratio. For example, if the PLLMUL bit setting is 26 and the PLLMULNF bit setting is 0.33, the multiplication ratio is 26.33.

The PLL circuit has the 3 output clocks. The output clock divider circuit divides the VCO clock to generate each PLL clock output. The division ratio for generating each output clock can be set independently with the PLODIVP, PLODIVQ, and PLODIVR bits.

The PLL Input clock frequency, the PLL Reference clock frequency, the VCO clock frequency, and each Output clock frequency must be within the specified in section 70, Electrical Characteristics.

The PLL frequency setting can only be changed the PLL circuit is stopped. Do not change the PLL frequency setting when PLL stop control bit (PLLSTP) is 0.

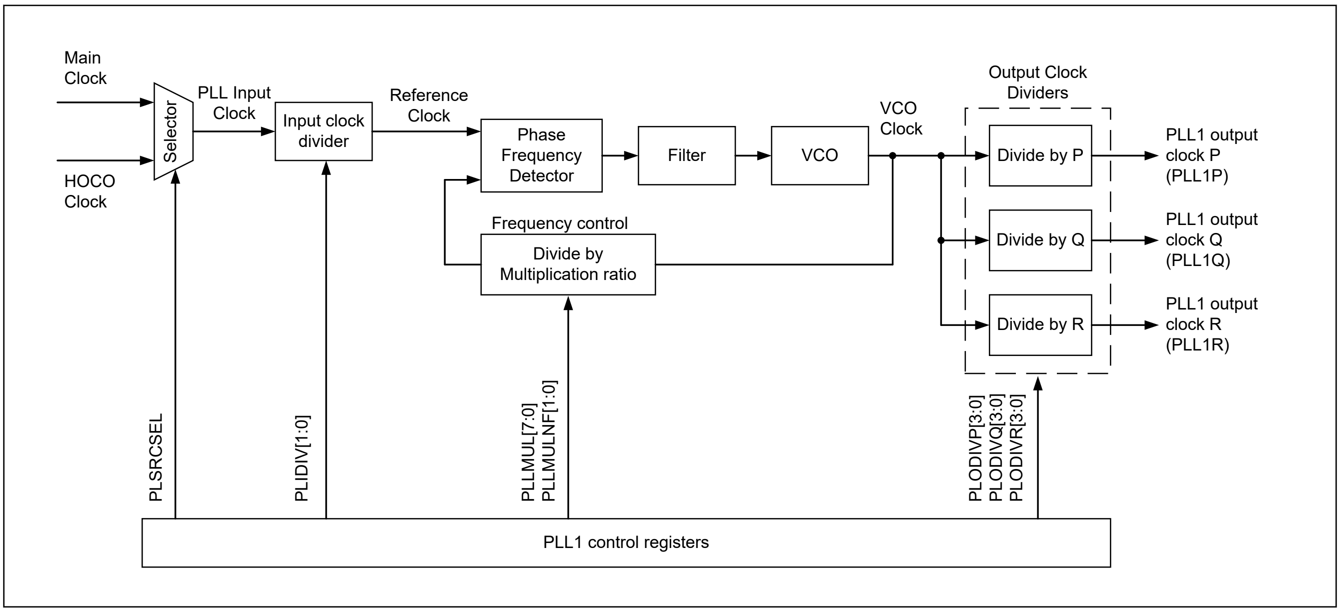


Figure 9.7 PLL1 block diagram

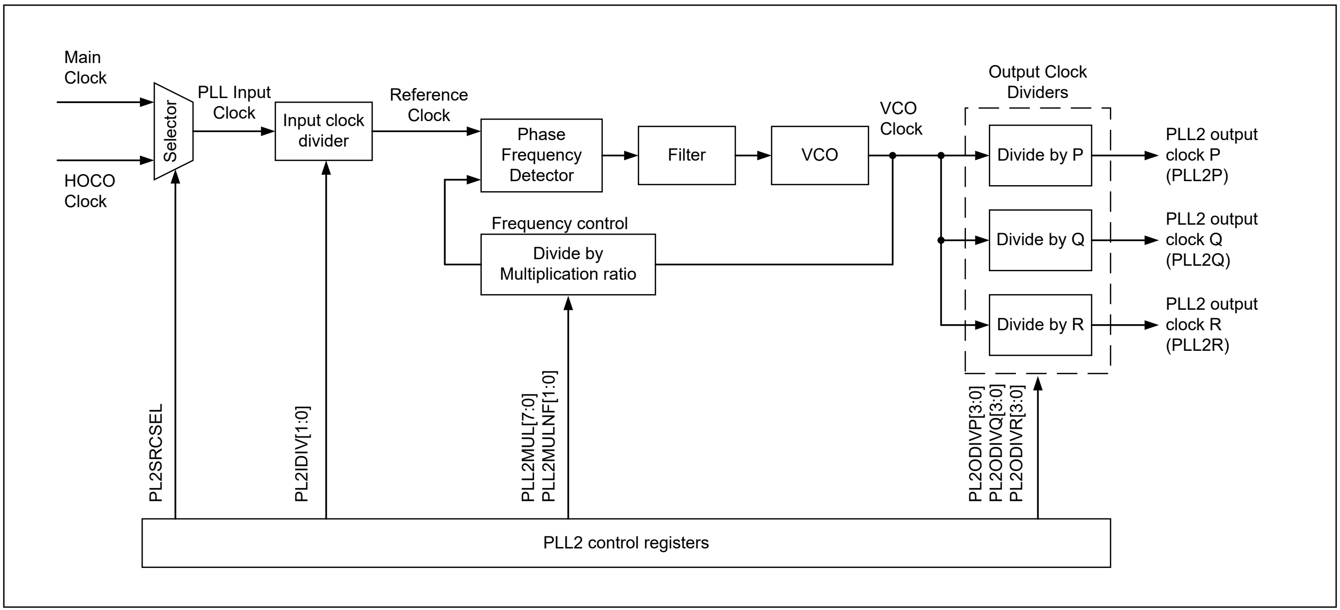


Figure 9.8 PLL2 block diagram

9.9 Oscillation Stop Detection Function

9.9.1 Oscillation Stop Detection and Operation after Detection

The oscillation stop detection function detects the main clock oscillator stop. When oscillation stop is detected, the system clock source switches as follows:

- If an oscillation stop is detected with SCKSCR.CKSEL[2:0] = 011b (system clock source = MOSC), the system clock source switches to the MOCO clock.
- If an oscillation stop is detected with PLLCCR.PLSRCSEL = 0 (PLL1 source clock = MOSC) and SCKSCR.CKSEL[2:0] = 101b (system clock source = PLL1P), PLL1 clock remains the system clock source. The frequency becomes free-running, and the setting in the SCKSCR.CKSEL[2:0] bits does not change.

An oscillation stop detection interrupt request can be generated when an oscillation stop is detected. In addition, the General PWM Timer (GPT) output can be forced to stop state on detection.

The main clock oscillation stop is detected when the input clock remains at 0 or 1 for a certain period, for example, when a malfunction occurs in the main clock oscillator. See section 70, Electrical Characteristics.

Switching between the main clock oscillator and the MOCO clock or between the PLL1 clock and PLL1 free-running clock is controlled by the Oscillation Stop Detection Flag (OSTDSR.OSTDF).

OSTDF controls the switched clock as follows:

- When SCKSCR.CKSEL[2:0] = 011b (system clock source = MOSC):
  - When OSTDF changes from 0 to 1, the clock source switches to the MOCO clock.
  - When OSTDF changes from 1 to 0, the clock source switches back to MOSC.
- When PLLCCR.PLSRCSEL = 0 (PLL1 source clock = MOSC) and SCKSCR.CKSEL[2:0] = 101b (System clock source = PLL1P):
  - When OSTDF changes 0 to 1, the clock source switches to the PLL1 free-running oscillation clock.
  - When OSTDF changes 1 to 0, the clock source switches back to PLL1.

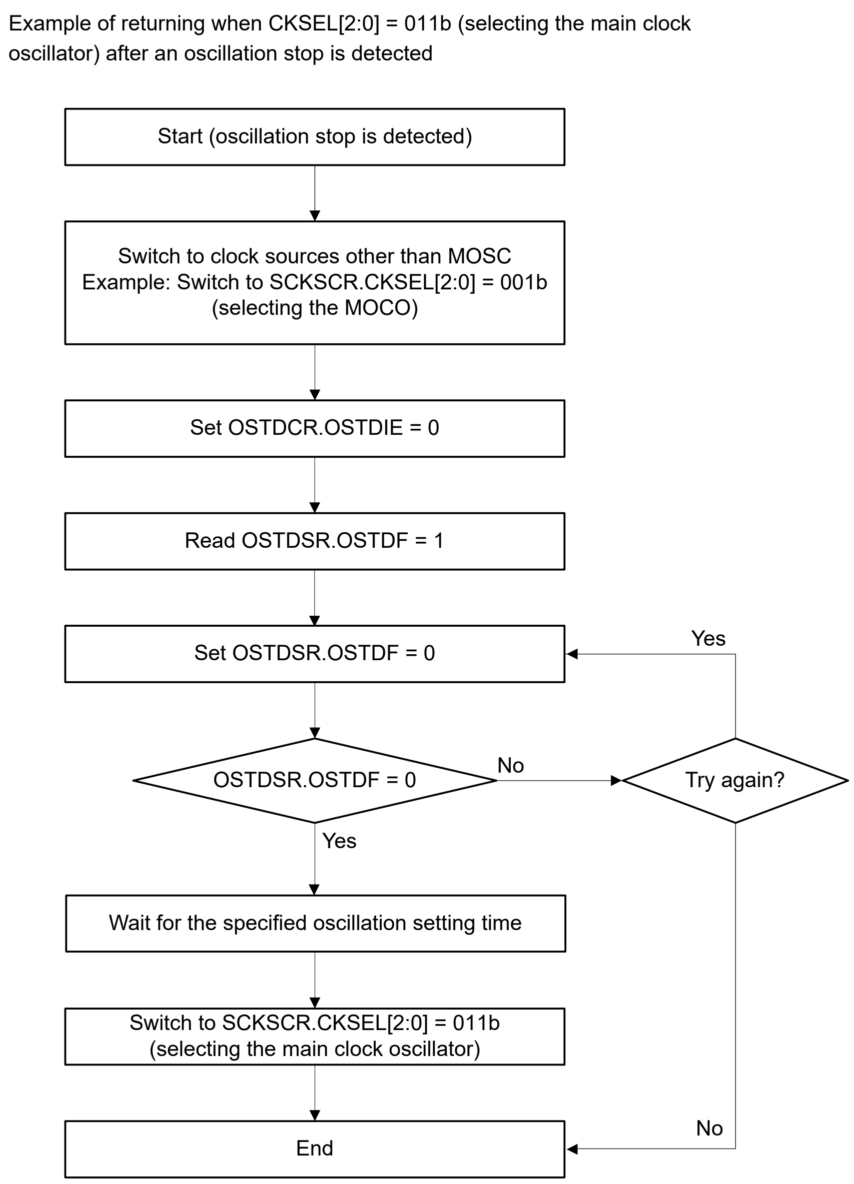
To switch the clock source to the main clock or PLL1 clock again after the oscillation stop detection, set the CKSEL[2:0] bits to a clock source other than the main clock or PLL1 clock and clear the OSTDF flag to 0. Also, check that the OSTDF flag is not 1, then set the CKSEL[2:0] bits to the main clock or PLL1 clock after the specified oscillation stabilization time elapses.

After a reset release, the main clock oscillator is stopped and the oscillation stop detection function is disabled. To enable the oscillation stop detection function, activate the main clock oscillator and write 1 to the oscillation stop detection function enable bit (OSTDCR.OSTDE) after a specified oscillation stabilization time elapses.

The oscillation stop detection function detects when the main clock is stopped by an external cause.

The oscillation stop detection function switches the following clocks to the MOCO clock (when the system clock source is MOSC) or the PLL1 free-running clock (when the system clock source is PLL1P) :

- All clocks that can be selected as the MOSC clock or PLL1 except CLKOUT, USBMCLK, MIPIMCLK and CACMCLK.
- The system clock (ICLK) frequency during the MOCO (when the system clock source is MOSC) or PLL1 free-running (when the system clock source is PLL1P) operation is specified by the MOCO oscillation frequency and the division ratio set by the system clock select bits (SCKDIVCR.ICK[3:0])



Note: On return from the oscillation-stopped state, the factor responsible for stopping the main clock oscillation circuit must be removed from the system to allow oscillation to resume.

Figure 9.9 Flow of recovery on detection of oscillator stop

9.9.2 Oscillation Stop Detection Interrupts

An oscillation stop detection interrupt (MOSC_STOP) is generated when the Oscillation Stop Detection Flag (OSTDSR.OSTDF) is 1 and the Oscillation Stop Detection Interrupt Enable bit in the Oscillation Stop Detection Control Register (OSTDCR.OSTDIE) is 1 (enabled). The Port Output Enable for GPT (POEG) is notified of the main clock oscillator stop. On receiving the notification, the POEG sets the Oscillation Stop Detection Flag in the POEG Group n Setting Register (POEGGn.OSTPF) to 1 (n = A, B, C, D).

After the oscillation stop is detected, wait at least 10 PCLKB clock cycles before writing to the POEGGn.OSTPF flag. When the OSTDSR.OSTDF flag requires clearing, do so after clearing the Oscillation Stop Detection Interrupt Enable bit in the Oscillation Stop Detection Control Register (OSTDCR.OSTDIE). Wait at least 2 PCLKB clock cycles before setting the OSTDCR.OSTDIE bit to 1 again. A longer PCLKB wait time might be required, depending on the number of cycles required to read a given I/O register.

The oscillation stop detection interrupt is a non-maskable interrupt. Because non-maskable interrupts are disabled in the initial state after a reset release, enable non-maskable interrupts through software before using oscillation stop detection interrupts. For details, see section 14, Interrupt Controller Unit (ICU).

9.9.3 Sub-clock Oscillation Stop Detection and Operation after Detection

The sub-clock oscillation stop detection function is used to detect the sub-clock oscillator stop. When an oscillation stop is detected, the system clock switches as follows:

- If an oscillation stop is detected with SCKSCR.CKSEL[2:0] = 100b (system clock source = SOSC), the system clock source switches to the MOCO/256 clock.
  An oscillation stop detection interrupt request can be generated when an oscillation stop is detected.
  The sub-clock oscillation stop is detected when the input clock remains to be 0 or 1 for a certain period, for example, when a malfunction occurs in the sub-clock oscillator. See Electrical Characteristics.
  Switching between the sub-clock oscillator and the MOCO/256 clock is controlled by the Sub-clock Oscillation Stop Detection flag (SOSTDSR.SOSTDF).
  The SOSTDF flag controls the switched clock as the follows:
- SCKSCR.CKSEL[2:0] = 100b (System clock source = SOSC):
  - When SOSTDF changes from 0 to 1, the clock source switches to the MOCO/256 clock.
  - When SOSTDF changes from 1 to 0, the clock source switches to SOSC again.

To switch the clock source to the sub-clock oscillator again after the oscillation stop detection, set the CKSEL[2:0] bits to a clock source other than the sub-clock, and clear the SOSTDF flag to 0. Also, check that the SOSTDF flag is not 1, and then set the CKSEL[2:0] bits to the sub-clock oscillator after the specified oscillation stabilization time elapses.

After a VBAT_POR release, the sub-clock oscillator is stopped and the sub-clock oscillation stop detection function is disabled. To enable the sub-clock oscillation stop detection function, activate the sub-clock oscillator and write 1 to the Sub-clock Oscillation Stop Detection Function Enable bit (SOSTDCR.SOSTDE) after the specified oscillation stabilization time elapses.

After that, be sure to clear the Sub-clock Oscillation Stop Detection Flag (SOSTDSR.SOSTDF) to 0 after the start-up time of the sub-clock oscillation stop detector (Tsosdup) elapses.

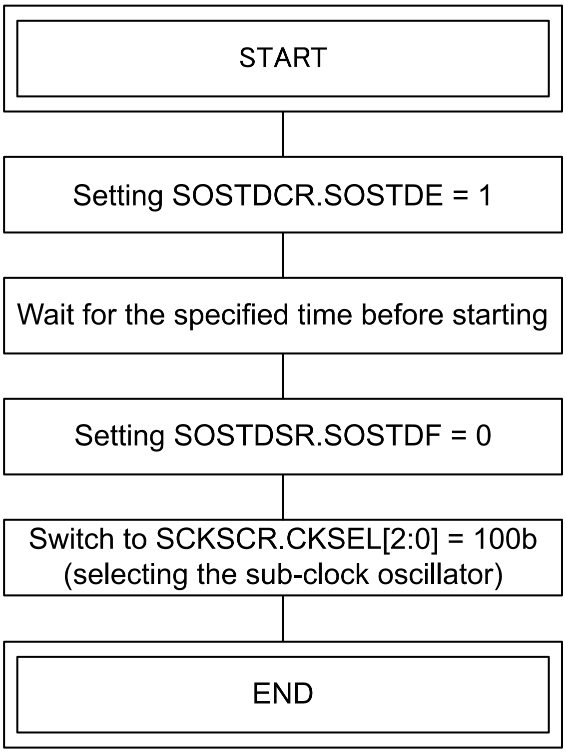
When selecting the sub-clock oscillator for the system clock, complete the above settings before switching.

The sub-clock oscillation stop detection function detects when the Sub-Clock Oscillator is stopped by an external cause.

The sub-clock oscillation stop detection function switches the following clocks to the MOCO/256 clock.

- All clocks that can be selected as the SOSC clock except CACSCLK.
- The system clock (ICLK) frequency during MOCO/256 operation is specified in the MOCO oscillation frequency and the division ratio set in the System Clock Select bits (SCKDIVCR.ICK[3:0]).

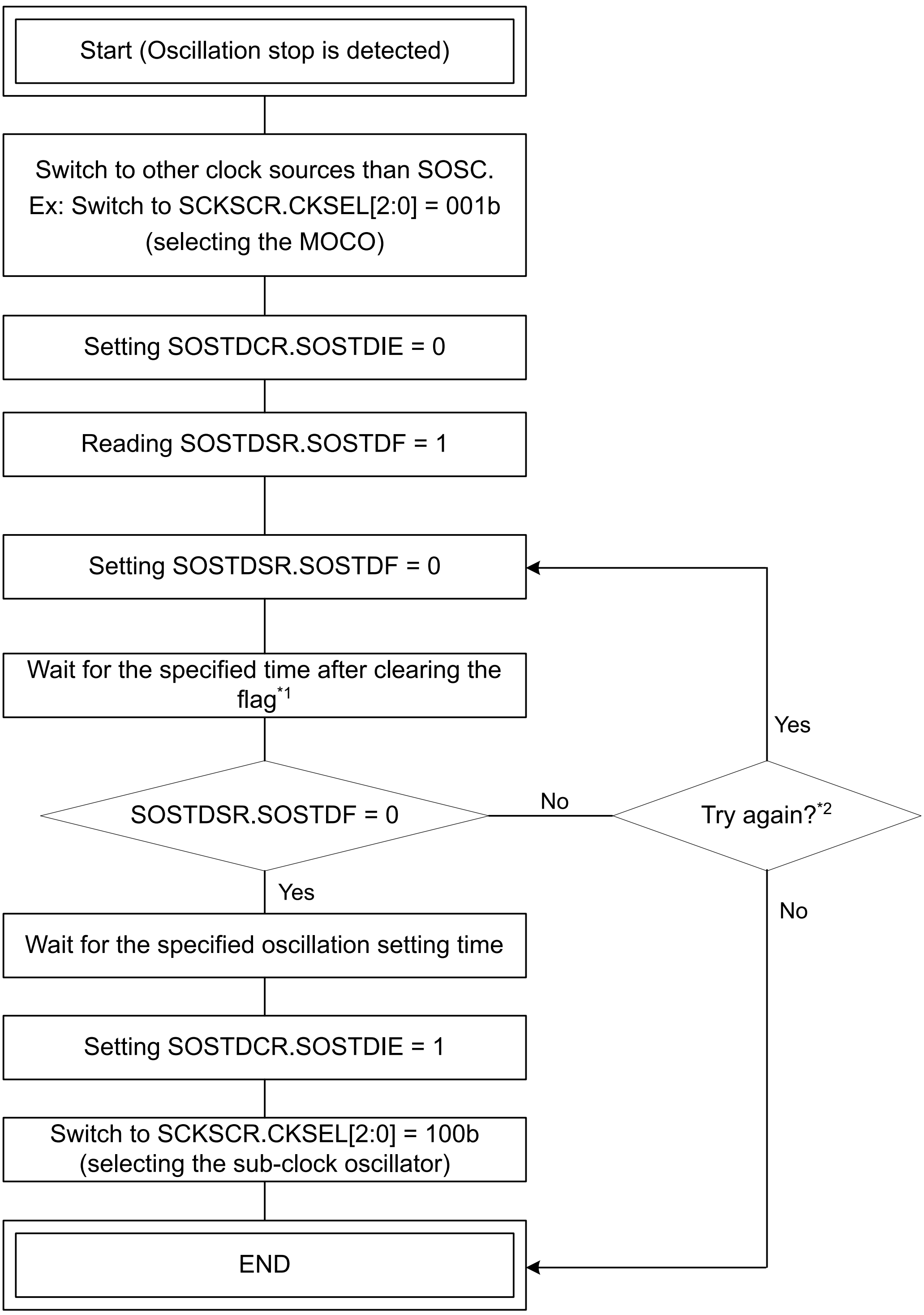
Example of activation of sub-clock oscillation stop detection function when CKSEL[2:0] = 100b (selecting the sub-clock oscillator)



Note: Set the system clock to the other clock sources than the sub-clock oscillator before executing this flow.

Figure 9.10 Flow of Activation of Sub Clock Oscillation Stop Detection Function

Example of re-returning when CKSEL[2:0]=100b (selecting the sub-clock oscillator) after an oscillation stop is detected.



Note: On return from the oscillation-stopped state, the factor responsible for stopping the sub-clock oscillation circuit must be removed on the user system to allow the return of oscillation.

Note 1. If the SOSTDSR.SOSTDF flag is cleared, wait at least 1.5 μs before reading the next SOSTDSR.SOSTDF flag.

Note 2. Retry at least once.

Figure 9.11 Flow of Recovery from Detection of Sub Clock Oscillator Stop

9.9.4 Sub-clock Oscillation Stop Detection Interrupts

A sub-clock oscillation stop detection interrupt (SOSTD) is generated when the Sub-clock Oscillation Stop Detection flag (SOSTDSR.SOSTDF) is 1 and the Sub-clock Oscillation Stop Detection Interrupt Enable bit in the Sub-clock Oscillation Stop Detection Control Register (SOSTDCR.SOSTDIE) is 1 (enabled).

When the SOSTDSR.SOSTDF flag requires clearing, do so after clearing the Sub-clock Oscillation Stop Detection Interrupt Enable bit in the Sub-clock Oscillation Stop Detection Control Register (SOSTDCR.SOSTDIE). Make sure SOSTDF is cleared before setting the SOSTDIE bit to 1 again.

9.10 Internal Clock

Clock sources for the internal clock signals include:

- Main clock oscillator
- Sub-clock oscillator
- HOCO clock
- MOCO clock
- LOCO clock
- PLL1 clock (PLL1P, PLL1Q and PLL1R)
- PLL2 clock (PLL2P, PLL2Q and PLL2R)
- External clock input for JTAG
- External clock input for SWD

The following internal clocks are produced from these sources.

- Operating clock of the CPU0 : CPU0 clock (CPUCLK0)
- Operating clock of the CPU1 : CPU1 clock (CPUCLK1)
- Operating clock of the NPU : NPU clock (NPUCLK)
- Operating clock of the DMAC, DTC, SRAM, System Bus, I/O Port, ICU and DRW: System clock (ICLK)
- Operating clock of the Debug Subsystem : Debug clock (DCLK)
- Operating clock for MRAM code bus : MRAM code bus clock (MRICLK)
- Operating clocks of peripheral modules : Peripheral module clocks (PCLKA, PCLKB, PCLKC, PCLKD, and PCLKE)
- Operating clock for the MRAM peripheral bus : MRAM peripheral bus clock (MRPCLK)
- Clock for the external bus controller and external pin output: External bus clock (BCLK, EBCLK)
- Clock for the external bus controller and external pin output for the SDRAM : SDRAM clock (SDCLK)
- Clock for the trace function and external pin output : Trace clock (TRCLK)
- Operating clock for the SCI : SCI clock (SCICLK)
- Operating clock for the SPI : SPI clock (SPICLK)
- Operating clock for the Octal SPI : Octal-SPI clock (OCTACLK, OCTADIVCLK)
- Operating clock for the CANFD Core : CANFD Core clock (CANFDCLK)
- Operating clock for the USBFS and USBHS : USB clock (USBCLK)
- Operating clock for the USBHS : USB clock (USB60CLK)
- Operating clock for the ADC16H : ADC clock (ADCCLK)
- Operating clock for the GPT : GPT clock (GPTCLK)
- Operating clock for the LCD : LCD clock (LCDCLK)
- Operating clock for the I3C : I3C clock (I3CCLK)
- Clock for the external bus controller (Asynchronous) : External bus clock (BCLKA)
- Operating clock for the EtherSW : ESWM clock (ESWCLK)

- Operating clock for the EtherSW-PHY : ESWM clock (ESWPHYCLK)
- Operating clock for the ETHPHYCLK pin output : Ether-PHY clock (ETHPHYCLK)
- Clock for external pin output : Clock/Buzzer output (CLKOUT)
- Operating clock for the USBHS : MOSC clock for USBHS (USBMCLK)
- Operating clock for the MIPI : MIPI clock (MIPIMCLK)
- Operating clock for the CAN : CAN clock (CANMCLK)
- Operating clocks for the ULPT : ULPT LOCO clock (ULPTLCLK) and ULPT sub-clock (ULPTSCLK)
- Operating clock for the AGT : AGT LOCO clock (AGTLCLK) and AGT sub-clock (AGTSCLK)
- Operating clocks for the CAC : CAC clock (CACCLK)
- Operating clocks for the RTC : RTC LOCO clock (RTCLCLK) and RTC Sub-clock (RTCSCLK)
- Operating clock for the PDMIF : PDMIF clock (PDMIFCLK)
- Operating clock for the IWDT : IWDT clock (IWDTCLK)
- Operating clock for the SysTick timer : SysTick timer clock (SYSTICKCLK)
- Operating clock for the JTAG : JTAG clock (JTAGTCK)

For details on the registers used to set the frequencies of the internal clocks, see section 9.10.6. System Clock (ICLK) to section 9.10.37. JTAG Clock (JTAGTCK)

If the value of any of these bits is changed, subsequent operation is at the frequency determined by the new value.

9.10.1 CPU0 Clock (CPUCLK0)

The CPU0 clock (CPUCLK0) is the operating clock for the CPU0. Specify the frequency in the following bits:

- CPUCK0[3:0] bits in SCKDIVCR2
- CKSEL[2:0] bits in SCKSCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], and PLODIVP[3:0] bits in PLLCCR and PLLCCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)[*1]

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.2 CPU1 Clock (CPUCLK1)

The CPU1 clock, CPUCLK1, is the operating clock for the CPU1. Specify the frequency in the following bits:

- CPUCK1[3:0] bits in SCKDIVCR2
- CKSEL[2:0] bits in SCKSCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], and PLODIVP[3:0] bits in PLLCCR and PLLCCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)[*1]

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.3 NPU Clock (NPUCLK)

The NPU clock, NPUCLK, is the operating clock for NPU. Specify the frequency in the following bits:

- NPUCK[3:0] bits in SCKDIVCR2
- CKSEL[2:0] bits in SCKSCR

- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], and PLODIVP[3:0] bits in PLLCCR and PLLCCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.4 MRAM Code Bus Clock (MRICLK)

The MRAM code bus clock, MRICLK, is the operating clock for the MRAM code bus interface. Specify the frequency in the following bits:

- MRICK[3:0] bits in SCKDIVCR2
- CKSEL[2:0] bits in SCKSCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], and PLODIVP[3:0] bits in PLLCCR and PLLCCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.5 MRAM Peripheral Bus Clock (MRPCLK)

The MRAM peripheral bus clock, MRPCLK, is the operating clock for the MRAM peripheral bus interface. Specify the frequency in the following bits:

- MRPCK[3:0] bits in SCKDIVCR
- CKSEL[2:0] bits in SCKSCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], and PLODIVP[3:0] bits in PLLCCR and PLLCCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.6 System Clock (ICLK)

The system clock (ICLK) is the operating clock of the DMAC, DTC, SRAM, System Bus, I/O Port, ICU and DRW. Specify the frequency in the following bits:

- ICK[3:0] bits in SCKDIVCR
- CKSEL[2:0] bits in SCKSCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0] and PLODIVP[3:0] bits in PLLCCR and PLLCCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

When the ICLK clock source is switched, the duration of the ICLK clock cycle becomes longer during the clock source transition period. See Figure 9.12 and Figure 9.13.

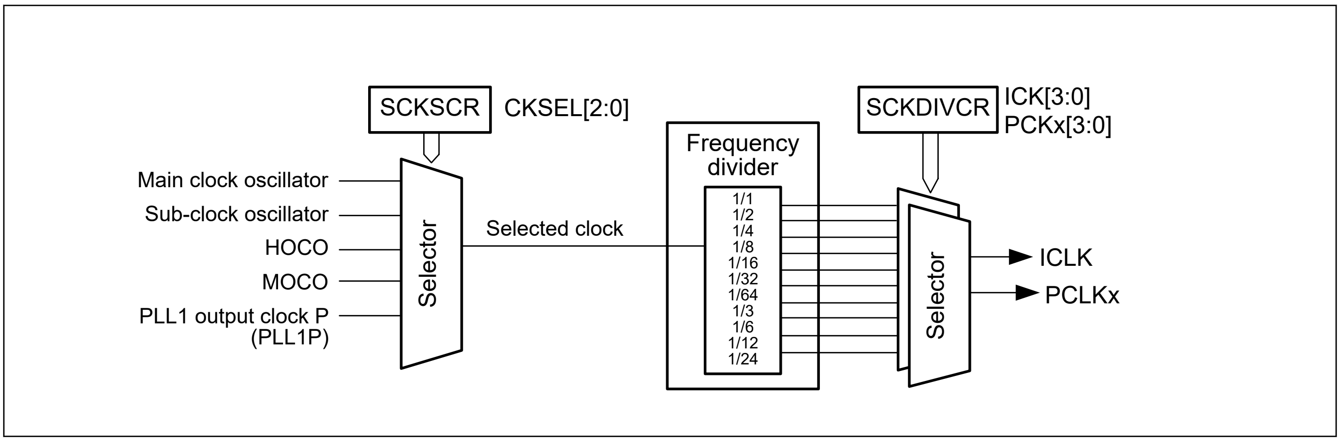


Figure 9.12 Block diagram of clock source selector

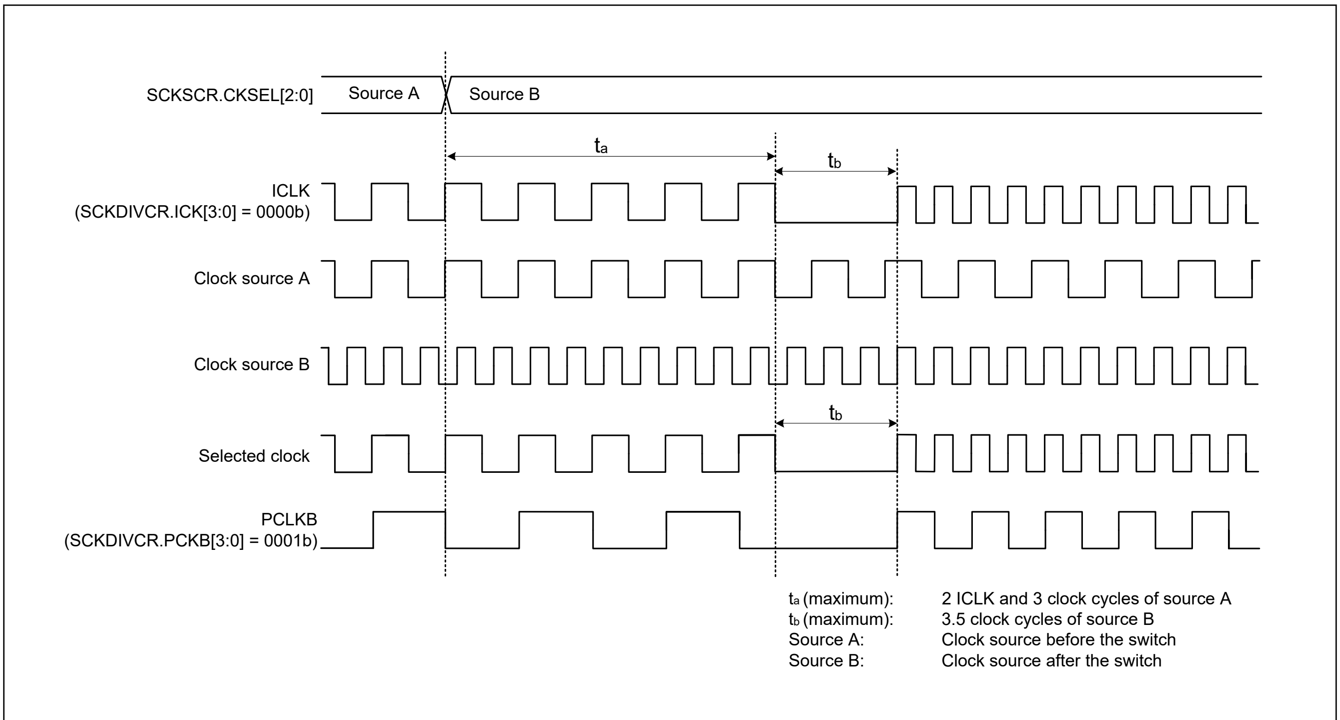


Figure 9.13 Timing of clock source switching

9.10.7 Debugger Clock (DCLK)

The Debugger clock (DCLK) is the operating clock for the Debug Subsystem. The frequency is same as ICLK.

9.10.8 Peripheral Module Clock (PCLKA, PCLKB, PCLKC, PCLKD, PCLKE)

The peripheral module clocks (PCLKA, PCLKB, PCLKC, PCLKD and PCLKE) are the operating clocks for the peripheral modules.

The frequency of the given clock is specified in the following bits:

- PCKA[3:0], PCKB[3:0], PCKC[3:0], PCKD[3:0] and PCKE[3:0]bits in SCKDIVCR
- CKSEL[2:0] bits in SCKSCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0] and PLODIVP[3:0] bits in PLLCCR and PLLCCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

When the clock source of the peripheral module clock is switched, the duration of the peripheral module clock cycle becomes longer during the clock source transition period. See Figure 9.12 and Figure 9.13.

9.10.9 External Bus Clock (BCLK, EBCLK)

The external bus clock (BCLK) is an operating clock for the external bus controller. It is also output externally from the EBCLK pin for the external connection bus.

BCLK can be output from the EBCLK pin by setting the EBCKOCR.EBCKOEN bit to 1 and setting the PmnPFS.PSEL[4:0] to 01011b. Make sure that modification of the PmnPFS.PSEL[4:0] to 01011b must always be performed while the EBCKOCR.EBCKOEN bit is 0.

When the BCKCR.BCLKDIV bit is set to 1, the BCLK clock divided by 2 is output from the EBCLK pin.

Specify the frequency in the following bits:

- BCK[3:0] bits in SCKDIVCR
- CKSEL[2:0] bits in SCKSCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0] and PLODIVP[3:0] bits in PLLCCR and PLLCCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

A frequency higher than the system clock (ICLK) should not be set for the BCLK.

9.10.10 SDRAM Clock (SDCLK)

The SDRAM clock (SDCLK) is an operating clock for the external bus controller. It is output externally from the SDCLK pin for the SDRAM that is connected to the external bus. To output SDCLK on the SDCLK pin, set the SDCKOCR.SDCKOEN bit to 1 and set the PmnPFS.PSEL[4:0] bits to 01011b(enabling SDCLK output). Only change the value in the PmnPFS.PSEL[4:0] bits when the SDCKOCR.SDCKOEN bit is 0. Specify the frequency in the following bits:

- BCK[3:0] bits in SCKDIVCR
- CKSEL[2:0] bits in SCKSCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0] and PLODIVP[3:0] bits in PLLCCR and PLLCCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

Do not set SDCLK to a frequency higher than that of the system clock (ICLK).

9.10.11 Trace Clock (TRCLK)

The trace clock (TRCLK) is the CPU tracing clock for the On-Chip Debugger function.

The TRCLK frequency is specified by the TRCK[3:0] bits and TRCKSEL bit in TRCKCR (The Trace Clock frequency division ratio and the Trace Clock Source select). When changing the frequency of the Trace Clock, the Trace Clock must be stopped (TRCKCR.TRCKEN = 0).

9.10.12 SCI Clock (SCICLK)

The SCI clock (SCICLK) is the operating clock for the SCI module.

Specify the frequency in the following bits:

- SCICKDIV[3:0] bits in SCICKDIVCR
- SCICKSEL[3:0] bits in SCICKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLL2CCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.13 SPI Clock (SPICLK)

The SPI clock (SPICLK) is the operating clock for the SPI module.

Specify the frequency in the following bits:

- SPICKDIV[3:0] bits in SPICKDIVCR
- SPICKSEL[3:0] bits in SPICKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLL2CCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.14 Octal-SPI Clock (OCTACLK, OCTADIVCLK)

The Octal-SPI clock (OCTACLK) is the operating clock for the Octal-SPI module.

Specify the frequency in the following bits:

- OCTACKDIV[3:0] bits in OCTACKDIVCR
- OCTACKSEL[3:0] bits in OCTACKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLL2CCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

OCTADIVCLK always outputs 2-division of OCTACLK.

9.10.15 CANFD Core Clock (CANFDCLK)

The CANFD Core clock (CANFDCLK) is the operating clock for the CANFD module.

Specify the frequency in the following bits:

- CANFDCKDIV[3:0] bits in CANFDCKDIVCR

- CANFDCKSEL[3:0] bits in CANFDCKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLL2CCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.16 USB Clock (USBCLK)

The USB clock (USBCLK) is the operating clock for the USBFS and USBHS module.

A 48-MHz clock must be supplied when using the USBFS module or when using the USBHS module in CL-Only mode. USBCLK does not need to be supplied when not using USBHS module in CL-Only mode.

The USBCLK frequency is specified in the following bits:

- USBCKDIV[3:0] bits in USBCKDIVCR
- USBCKSEL[3:0] bits in USBCKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0] and PLODIVP[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLL2CCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.17 USB Clock (USB60CLK)

The USB clock (USB60CLK) is the operating clock for the USBHS module. A 60-MHz clock must be supplied when using the USBHS module in CL-Only mode. USB60CLK does not need to be supplied when not using USBHS in CL-Only mode.

Specify the frequency in the following bits:

- USB60CKDIV[3:0] bits in USB60CKDIVCR
- USB60CKSEL[3:0] bits in USB60CKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLL2CCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.18 MOSC Clock for USBHS (USBMCLK)

The MOSC clock for USBHS (USBMCLK) is the operating clock for the USBHS when not using the module in CL-Only mode. USBMCLK does not need to be supplied when using USBHS in CL-Only mode.

The USBMCLK frequency is 12 MHz or 20 MHz or 24 MHz or 48 MHz supplied from the main clock oscillator.

9.10.19 ADC Clock (ADCCLK)

The ADC clock, ADCCLK, is the operating clock for the ADC16H module. Specify the frequency in the following bits:

- ADCCKDIV[3:0] bits in ADCCKDIVCR
- ADCCKSEL[3:0] bits in ADCCKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLLCCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.20 GPT Clock (GPTCLK)

The GPT clock, GPTCLK, is the operating clock for the GPT module. Specify the frequency in the following bits:

- GPTCKDIV[3:0] bits in GPTCKDIVCR
- GPTCKSEL[3:0] bits in GPTCKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLLCCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.21 LCD Clock (LCDCLK)

The LCD clock (LCDCLK) is the operating clock for the LCD module.

Specify the frequency in the following bits:

- LCDCKDIV[3:0] bits in LCDCKDIVCR
- LCDCKSEL[3:0] bits in LCDCKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLL2CCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.22 I3C Clock (I3CCLK)

The I3C clock (I3CCLK) is the operating clock for the I3C module.

Specify the frequency in the following bits:

- I3CCKDIV[3:0] bits in I3CCKDIVCR

- I3CCKSEL[3:0] bits in I3CCKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLL2CCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.23 Asynchronous External Bus Clock (BCLKA)

The Asynchronous external bus clock, BCLKA, is the operating clock for the external bus controller. Specify the frequency in the following bits:

- BCKACKDIV[3:0] bits in BCKADIVCR
- BCKACKSEL[3:0] bits in BCKACR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLLCCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.24 EtherSW Clock (ESWCLK)

The EtherSW clock, ESWCLK, is the operating clock for the ESWM module. Specify the frequency in the following bits:

- ESWCKDIV[3:0] bits in ESWCKDIVCR
- ESWCKSEL[3:0] bits in ESWCKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLLCCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.25 EtherSW-PHY Clock (ESWPHYCLK)

The EtherSW-PHY clock, ESWPHYCLK, is the operating clock for the ESWM module. Specify the frequency in the following bits:

- ESWPCKDIV[3:0] bits in ESWPCKDIVCR
- ESWPCKSEL[3:0] bits in ESWPCKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2

- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLLCCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.26 Ether-PHY Clock (ETHPHYCLK)

The Ether-PHY clock, ETHPHYCLK, is the operating clock for the ETHPHYCLK pin output. Specify the frequency in the following bits:

- ETHPCKDIV[3:0] bits in ETHPCKDIVCR
- ETHPCKSEL[3:0] bits in ETHPCKCR
- PLLMUL[8:0], PLLMULNF[1:0], PLIDIV[1:0], PLODIVP[3:0], PLODIVQ[3:0] and PLODIVR[3:0] bits in PLLCCR and PLLCCR2
- PLL2MUL[8:0], PLL2MULNF[1:0], PL2IDIV[1:0], PL2ODIVP[3:0], PL2ODIVQ[3:0] and PL2ODIVR[3:0] bits in PLLCCR and PLL2CCR2
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.27 CAN Clock (CANMCLK)

The CAN clock (CANMCLK) is the operating clock for the CAN module. CANMCLK is generated by the main clock oscillator.

9.10.28 MIPI Clock (MIPIMCLK)

The MIPI clock (MIPIMCLK) is the operating clock for the MIPI module. MIPIMCLK is generated by the main clock oscillator.

9.10.29 ULPT Clock (ULPTLCLK, ULPTSCLK)

The ULPT clocks (ULPTLCLK and ULPTSCLK) are the operating clock for the ULPT module. ULPTLCLK is generated by the LOCO clock and ULPTSCLK is generated by the Sub-clock oscillator.

9.10.30 CAC Clock (CACCLK)

The CAC clock (CACCLK) is the operating clock for the CAC. CACCLK includes the following clocks:

- CAC Main clock (CACMCLK) is generated by the Main clock oscillator
- CAC Sub oscillator clock (CACSCLK) is generated by the Sub-clock oscillator
- CAC HOCO clock (CACHCLK) is generated by the HOCO clock
- CAC MOCO clock (CACMOCLK) is generated by the MOCO clock
- CAC LOCO clock (CACLCLK) is generated by the LOCO clock

9.10.31 RTC Clock (RTCSCLK, RTCLCLK)

The RTC clock (RTCSCLK, RTCLCLK) is the operating clock for the RTC.

RTCSCLK is generated by the sub-clock oscillator, and RTCLCLK is generated by the LOCO clock.

9.10.32 IWDT Clock (IWDTCLK)

The IWDT clock (IWDTCLK) is the operating clock for the IWDT.

IWDTCLK is internally generated by the LOCO clock which is always divided by 2.

9.10.33 AGT Clock (AGTSCLK, AGTLCLK)

The AGT clocks (AGTSCLK and AGTLCLK) are the operating clocks for the AGT. AGTSCLK is generated by the sub-clock oscillator, and AGTLCLK is generated by the LOCO clock.

9.10.34 PDMIF Clock (PDMIFCLK)

The PDMIF clock, PDMIFCLK, are the operating clock for the PDMIF.

PDMIFCLK is generated by the MOCO clock.

9.10.35 SysTick Timer Clock (SYSTICKCLK)

The SysTick timer clock (SYSTICKCLK) is the operating clock for the SysTick timer. SYSTICKCLK is generated by the MOCO clock which is always divided by 8.

9.10.36 External Pin Output Clock (CLKOUT)

The CLKOUT is output externally from the CLKOUT pin for the clock or buzzer output. The CLKOUT is output to the CLKOUT pin when the CKOCR.CKOEN bit is set to 1. Only change the value in the CKODIV[2:0] bits or CKOSEL[2:0] bits in CKOCR when the CKOCR.CKOEN bit is 0.

The CLKOUT clock frequency is specified in the following bits:

- CKODIV[2:0] bits or CKOSEL[2:0] bits in CKOCR
- HOCOFRQ0[2:0] bits in OFS1(_SEC)*1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1(_SEC).HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits.

9.10.37 JTAG Clock (JTAGTCK)

The JTAG clock (JTAGTCK) is the clock for the JTAG.

JTAGTCK is generated by the JTAG external clock (TCK).

9.11 Clock Setting

9.11.1 System Clock Setting

9.11.1.1 Initial System Clock Setting

An example of the initial system clock setting procedure is shown in Table 9.7.

Table 9.7 Example of the initial system clock setting procedure after reset release/after Deep Software Standby cancellation (1 of 2)

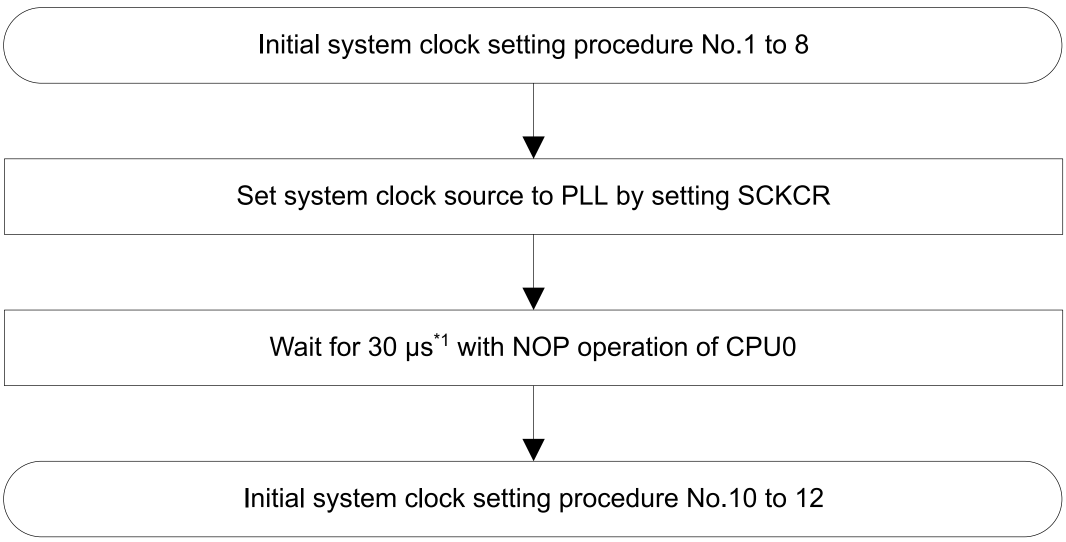
No.	Step	Description
1	Start	After reset release / Deep software standby cancellation (MOCO is selected as system clock source)
2	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1 *1*2bit in PRCR register
3	Change Operating Power Control Mode to High-Speed mode*1	Change to High-Speed mode with OPCCR register. *1 For details, see section 11, Low Power Mode.
4	Set a clock source to oscillate	Set a clock source that switches to the system clock source to oscillate.

Table 9.7 Example of the initial system clock setting procedure after reset release/after Deep Software Standby cancellation (2 of 2)

No.	Step	Description
5	Set MRAM and SRAM Wait Cycle	Set an appropriate access wait count for the MRAM and SRAM according to the frequency at which it operates after the system clock (ICLK), MRICLK and MRPCLK are set. For details, see section 60, MRAM and section 59, SRAM.
6	Set voltage scaling	Set an appropriate target voltage of VDD according to the frequency of each clocks after switching the system clock. For details, see section 11, Low Power Mode.
7	Set internal clock division ratio	Set the internal clock division ratio with SCKDIVCR and SCKDIVCR2 registers.
8	Set External Bus clock output / SDRAM clock output*3	Set the EBCLK pin output setting with BCKCR. Set the SDCLK pin output setting with SDCKOCR.
9	Switch the System Clock source	Switch the system clock source with SCKCR register. Before switching the system clock source, confirm that the clock source oscillation is stable.
10	Change Operating Power Control Mode	Change the Operating Power Control Mode with OPCCR register. *2
11	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1 *1*2bit in PRCR register
12	End	System clock setting is completed.

Note 1. Setting is not necessary if the operating power control mode is already the High-Speed mode.
Note 2. Setting is not necessary if the operating power control mode is not changed.
Note 3. Setting is not necessary if the External bus or SDRAM are not used.

When the PLL1 is selected as the system clock source, additional procedure is required for procedure No.9. This additional procedure is shown in Figure 9.14. It is recommended to use software to measure the wait time. Be sure to consider the worst-case conditions to ensure that the required wait time elapses. If an interrupt is unavoidably generated during the wait time, retry the measurement after return from the interrupt.



Note 1. 30 μs is in DCDC mode, 10 μs in External VDD mode.

Figure 9.14 Additional flow of initial system clock setting when PLL is used for the system clock source (assuming all module stop bits are initial value)

9.11.1.2 System Clock Setting for Faster Frequency

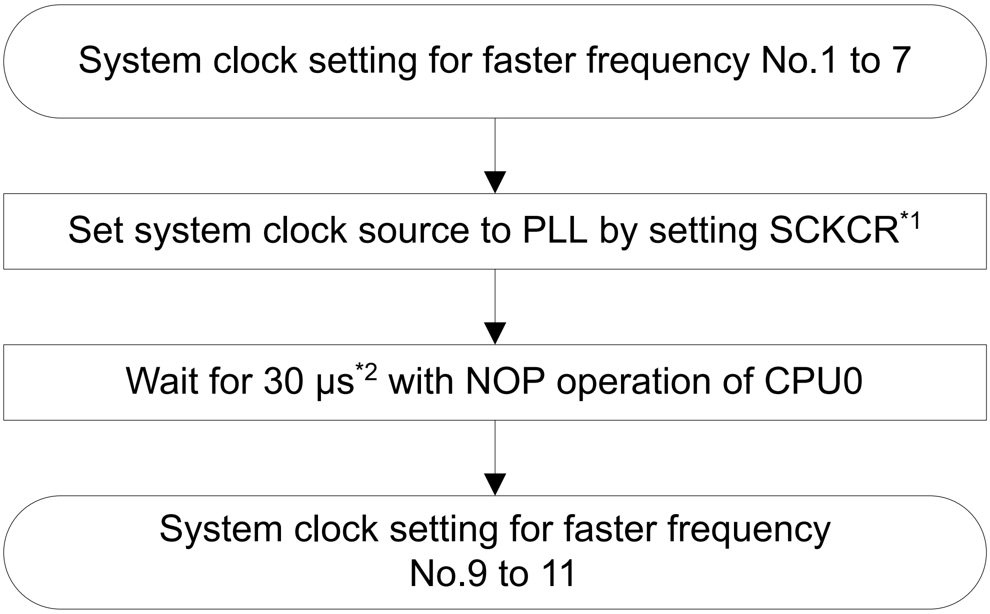
An example of setting procedure when changing the current system clock frequency to a faster frequency is shown in Table 9.8

Table 9.8 Example of setting procedure when changing the current system clock frequency to a faster frequency

No.	Step	Description
1	Start	
2	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1*1*2 bit in PRCR register
3	Change Operating Power Control Mode to High-Speed mode*1	Change to High-Speed mode with OPCCR register.*1 For details, see section 11, Low Power Mode.
4	Set a clock source to oscillate*3	Set a clock source that switches to the system clock source to oscillate.*4
5	Set MRAM and SRAM Wait Cycle	Set an appropriate access wait count for the MRAM and SRAM according to the frequency at which it operates after the system clock (ICLK), MRICLK and MRPCLK are set. For details, see section 60, MRAM and section 59, SRAM.
6	Set voltage scaling	Set an appropriate target voltage of VDD according to the frequency of each clocks after switching the system clock. For details, see section 11, Low Power Mode.
7	Set internal clock division ratio	Set the internal clock division ratio with SCKDIVCR and SCKDIVCR2 registers.
8	Switch the System Clock source*3	Switch the system clock source with SCKCR register. Before switching the system clock source, confirm that the clock source oscillation is stable.
9	Change Operating Power Control Mode	Change the Operating Power Control Mode with OPCCR register. *2
10	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1*1*2 bit in PRCR register
11	End	System clock is changed to a Faster frequency.

Note 1. Setting is not necessary if the operating power control mode is already the High-Speed mode.
Note 2. Setting is not necessary if the operating power control mode is not changed.
Note 3. Setting is not necessary if the system clock source is not changed.
Note 4. To change the frequency of the PLL selected as the system clock source, it is necessary to change the system clock source to another clock source, stop the PLL, and then change the PLL setting.

When the PLL1 is selected as the system clock source, additional procedure is required for procedure No.8. This additional procedure is shown in Figure 9.15. It is recommended to use software to measure the wait time. Be sure to consider the worst-case conditions to ensure that the required wait time elapses. If an interrupt is unavoidably generated during the wait time, retry the measurement after return from the interrupt.



Note 1. Do not set SCKCR when the PLL is already selected.
Note 2. 30 μs is in DCDC mode, 10 μs in External VDD mode.

Figure 9.15 Additional flow of system clock setting when PLL is used for the system clock source and CPUCLK0 to be faster frequency

9.11.1.3 System Clock Setting for Slower frequency

An example of setting procedure when changing the current system clock frequency to a slower frequency is shown in Table 9.9.

Table 9.9 Example of setting procedure when changing the current system clock frequency to a slower frequency (1 of 2)

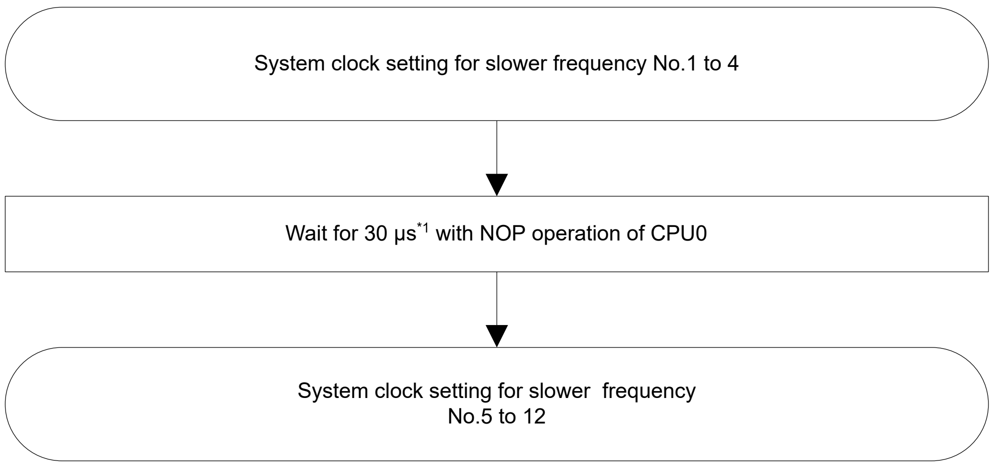
No.	Step	Description
1	Start	
2	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1[*1*2] bit in PRCR register
3	Change Operating Power Control Mode to High-Speed mode[*1]	Change to High-Speed mode with OPCCR register.[*1] For details, see section 11, Low Power Mode.
4	Set a clock source to oscillate[*3]	Set a clock source that switches to the system clock source to oscillate.[*4]
5	Set internal clock division ratio	Make the following settings so that the frequency does not exceed the electrical characteristics range when the system clock source is switched in the next step. ● Set the internal clock division ratio larger than the current division ratio. ● Set the internal clock division ratio so that the frequency after switching the system clock source is slower than the current frequency.[*3]
6	Switch the System Clock source[*3]	Switch the system clock source with SCKCR register. Before switching the system clock source, confirm that the clock source oscillation is stable.
7	Re-set internal clock division ratio	Re-set the internal clock division ratio with SCKDIVCR and SCKDIVCR2 registers according to the frequency after switching the system clock source.
8	Set voltage scaling	Set an appropriate target voltage of VDD according to the frequency of each clocks after switching the system clock. For details, see section 11, Low Power Mode.

Table 9.9 Example of setting procedure when changing the current system clock frequency to a slower frequency (2 of 2)

No.	Step	Description
9	Set MRAM and SRAM Wait Cycle	Set an appropriate access wait count for the MRAM and SRAM according to the frequency at which it operates after the system clock (ICLK), MRICLK and MRPCLK are set. For details, see section 60, MRAM and section 59, SRAM.
10	Change Operating Power Control Mode	Change the Operating Power Control Mode with OPCCR register. [*2]
11	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1 [*1*2] bit in PRCR register
12	End	System clock is changed to a Slower frequency.

Note 1. Setting is not necessary if the operating power control mode is already the High-Speed mode.
Note 2. Setting is not necessary if the operating power control mode is not changed.
Note 3. Setting is not necessary if the system clock source is not changed.
Note 4. To change the frequency of the PLL selected as the system clock source, it is necessary to change the system clock source to another clock source, stop the PLL, and then change the PLL setting.

When the PLL1 is selected as the system clock source before switching the system clock source, additional procedure is required before procedure No.5. This additional procedure is shown in Figure 9.16. It is recommended to use software to measure the wait time. Be sure to consider the worst-case conditions to ensure that the required wait time elapses. If an interrupt is unavoidably generated during the wait time, retry the measurement after return from the interrupt.



Note 1. 30 μs is in DCDC mode, 10 μs in External VDD mode.

Figure 9.16 Additional flow of system clock setting when PLL is used for the system clock source and CPUCLK0 to be slower frequency

9.11.2 Peripheral Module-dedicated Clock Setting

This section describes the setting procedure of the operating clock for the peripheral module. The target clocks are as follows : SCICLK, SPICLK, OCTACLK, CANFDCLK, USBCLK,USB60CLK, ADCCLK, GPTCLK, LCDCLK, I3CCLK, BCLKA, ESWCLK, ESWPHYCLK and ETHPHYCLK.

9.11.2.1 Initial Peripheral Module-dedicated Clock Setting

An example of the initial clock setting procedure for peripheral module is shown in Table 9.10.

Table 9.10 Example of the setting procedure of the operating clock for peripheral module after reset release/after Deep Software Standby cancellation (1 of 2)

No.	Step	Description
1	Start	After reset release / Deep software standby cancellation (MOCO is selected as system clock source)

Table 9.10 Example of the setting procedure of the operating clock for peripheral module after reset release/after Deep Software Standby cancellation (2 of 2)

No.	Step	Description
2	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1 *1*2bit in PRCR register
3	Change Operating Power Control Mode to High-Speed mode*1	Change to High-Speed mode with OPCCR register. *1 For details, see section 11, Low Power Mode.
4	Set CKSREQ and wait for setting CKSRDY	Write 1 to CKSREQ bit in each CKCR register. Polling until CKSRDY bit in each CKCR register is read as 1 While CKSRDY is 1, the clock supply stops.
5	Set a clock source to oscillate	Set a clock source that switches to the peripheral module-dedicated clock source to oscillate.
6	Set the clock division ratio and switch the clock source	Write the setting value to CKDIV[3:0] in each CKDIVCR register. Write the setting value to CKSEL[3:0] in each CKCR register.
7	Clear CKSREQ and wait for clearing CKSRDY	Write 0 to CKSREQ bit in each CKCR register. Polling until CKSRDY bit in each CKCR register is read as 0. After CKSRDY is 0, the clock supply starts.
8	Cancel the Module Stop Control	Cancel the Module Stop Control in MSTPCRn register (n = A to E) and supply clock to peripheral module.
9	Change Operating Power Control Mode	Change the Operating Power Control Mode with OPCCR register.*2
10	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1 bit in PRCR register
11	End	Clock setting is completed. And then, set the peripheral modules.

Note 1. Setting is not necessary if the operating power control mode is already the High-Speed mode.
Note 2. Setting is not necessary if the operating power control mode is not changed.

9.11.2.2 Peripheral Module-dedicated Clock Source Changing

Table 9.11 shows an example of the setting change procedure when switching the clock source of the operating clock for peripheral module in use.

Table 9.11 Example of the setting change procedure when switching the clock source of the operating clock for peripheral module in use (1 of 2)

No.	Step	Description
1	Start	The operating clock for the peripheral module is being used.
2	Stop the peripheral module	Stop the operation of the peripheral module that the operating clock is to be changed.
3	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1*1*2bit in PRCR register
4	Change Operating Power Control Mode to High-Speed mode *1	Change to High-Speed mode with OPCCR register.*1 For details, see section 11, Low Power Mode.
5	Set CKSREQ and wait for setting CKSRDY	Write 1 to CKSREQ bit in each CKCR register. Polling until CKSRDY bit in each CKCR register is read as 1. While CKSRDY is 1, the clock supplying stops.
6	Set a clock source to oscillate	Set a clock source that switches to the peripheral module-dedicated clock source to oscillate.
7	Set the clock division ratio and switch the clock source	Write the setting value to CKDIV[3:0] in each CKDIVCR register. Write the setting value to CKSEL[3:0] in each CKCR register.
8	Clear CKSREQ and wait for clearing CKSRDY	Write 0 to CKSREQ bit in each CKCR register. Polling until CKSRDY bit in each CKCR register is read as 0. After CKSRDY is 0, the clock supplying starts.
9	Change Operating Power Control Mode	Change the Operating Power Control Mode with OPCCR register. *2
10	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1 bit in PRCR register
11	Restart the peripheral module	Change the peripheral module settings according to the operating clock after the settings are changed. After that, restart the operation of the peripheral module.

Table 9.11 Example of the setting change procedure when switching the clock source of the operating clock for peripheral module in use (2 of 2)

No.	Step	Description
12	End	Clock setting is completed.

Note 1. Setting is not necessary if the operating power control mode is already the High-Speed mode.
Note 2. Setting is not necessary if the operating power control mode is not changed.

9.11.2.3 Peripheral Module-dedicated Clock Division Ratio Changing

Table 9.12 shows an example of the setting change procedure when switching the division ratio of the operating clock for peripheral module in use.

Table 9.12 Example of the setting change procedure when switching the division ratio of the operating clock for peripheral module in use (without the clock source change)

No.	Step	Description
1	Start	The operating clock for the peripheral module is being used.
2	Stop the peripheral module	Stop the operation of the peripheral module that the operating clock is to be changed.
3	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1 *1*2 bit in PRCR register
4	Change Operating Power Control Mode to High-Speed mode *1	Change to High-Speed mode with OPCCR register. *1 For details, see section 11, Low Power Mode.
5	Set the clock division ratio	Write the setting value to CKDIV[3:0] in each CKDIVCR register.
6	Change Operating Power Control Mode	Change the Operating Power Control Mode with OPCCR register.*2
7	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1 bit in PRCR register
8	Restart the peripheral module	Change the peripheral module settings according to the operating clock after the settings are changed. After that, restart the operation of the peripheral module.
9	End	Clock setting is completed.

Note 1. Setting is not necessary if the operating power control mode is already the High-Speed mode.
Note 2. Setting is not necessary if the operating power control mode is not changed.

9.11.3 Main Clock Oscillator Setting

Table 9.13 shows an example of the initial setting procedure for the Main Clock Oscillator.

Table 9.13 Example of the initial setting procedure for Main Clock Oscillator after reset release / after Deep Software Standby cancellation

No.	Step	Description
1	Start	The Main Clock Oscillator is stopped after reset release / Deep software standby cancellation
2	Cancel Register Write Protection	Set 1 to PRC0 bit in PRCR register
3	Set the mode of Main Clock Oscillator	Set the following Main Clock Oscillator mode with MOMCR register ● Resonator or Clock input ● Drive Capability
4	Set the oscillation stabilization wait time	Set the Main Clock Oscillator Wait Time in MOSCWTCR register.
5	Set the oscillation keep in Software Standby mode	Set MOSCSCR register if Main Clock Oscillator keeps oscillation in Software Standby mode.
6	Set the Main Clock Oscillator to operate	Set the Main Clock Oscillator to start oscillating with the MOSCCR register.
7	Wait for Main Clock oscillation to stabilize	Polling until MOSCSF bit in OSCSF register is read as 1 (Oscillation is stable)
8	Apply Register Write Protection	Set 0 to PRC0 bit in PRCR register
9	End	The Main Clock Oscillator setting is completed. The Main Clock is available.

9.11.4 Sub-Clock Oscillator Setting

Table 9.14 shows an example of the initial setting procedure for the Sub-Clock Oscillator.

Table 9.14 Example of the initial setting procedure for Sub-Clock Oscillator after VBAT Power On Reset release

No.	Step	Description
1	Start	The Sub-Clock Oscillator is stopped after VBAT Power On Reset release.
2	Cancel Register Write Protection	Set 1 to PRC0 bit in PRCR register
3	Setting the mode of Sub-Clock Oscillator	Set the following Sub-Clock Oscillator mode with SOMCR register • Resonator or Clock input • Drive Capability
4	Set the Sub-Clock Oscillator to operate	Set the Sub-Clock Oscillator to start oscillating with the SOSCCR register.
5	Wait for Sub-Clock oscillation to stabilize	Wait for sub-clock oscillation stabilization time (tSUBOSCWT). For the sub-clock oscillation stabilization time, see section 70, Electrical Characteristics.
6	Apply Register Write Protection	Set 0 to PRC0 bit in PRCR register
7	End	The Sub-Clock Oscillator setting is completed. The Sub-Clock is available.

9.11.5 High-speed On Chip Oscillator Setting

9.11.5.1 HOCO Initial Clock Setting

Table 9.15 shows an example of the initial setting procedure for the HOCO.

Table 9.15 Example of the HOCO initial setting procedure after reset release / after Deep Software Standby cancellation (OFS1(_SEC).HOCOEN = 1, without FLL)

No.	Step	Description
1	Start	The HOCO is stopped after reset release / Deep software standby cancellation when OFS1(_SEC).HOCOEN is 1.
2	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1 bit in PRCR register
3	Check the HOCO power supply*1	Check the following bit in the HOCOLDOCR register • LDOSTP bit is 0 (LDO is enabled)
4	Set the HOCO frequency	Set the HOCO frequency with HOCOCR2 register
5	Set the oscillation keep in Software Standby mode	If HOCO keeps oscillation in Software Standby mode, set the following: • HOCOSOKP bit in HOCOSCR register • SKEEP bit in HOCOLDOCR register
6	Set HOCO to operate	Set HOCO to start oscillating with the HOCOCR register.
7	Wait for HOCO clock oscillation to stabilize	Polling until HOCOSF bit in OSCSF register is read as 1 (Oscillation is stable)
8	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1 bit in PRCR register
9	End	HOCO clock setting is completed.HOCO Clock is available.

Note 1. When HOCOLDOCR.LDOSTP bit is 1, do not operate HOCO. If HOCOLDOCR.LDOSTP is switched from 1 to 0, waiting for the HOCO power supply to stabilize is required. See section 70, Electrical Characteristics on waiting time for the HOCO power supply to stabilize.

9.11.5.2 HOCO Setting with FLL Function

Table 9.16 shows an example of the initial setting procedure for HOCO with FLL function.

Table 9.17 shows the FLL function setting flow for the Software Standby mode transition and cancellation.

Table 9.16 Example of the HOCO setting procedure with FLL function after reset release / after Deep Software Standby cancellation (OFS1(_SEC).HOCOEN = 1)

No.	Step	Description
1	Start	The HOCO is stopped after reset release / Deep software standby cancellation when OFS1(_SEC).HOCOEN is 1.
2	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1 bit in PRCR register
3	Check the HOCO power supply*1	Check the following bit in the HOCOLDOCR register. ● LDOSTP bit is 0 (LDO is enabled)
4	Set the HOCO frequency	Set the HOCO frequency with HOCOCR2 register
5	Set the FLL function to enable*2	Set the FLL Multiplication Control with FLLCR2 register. Set the FLL function to enable with FLLCR1 register.
6	Set HOCO to operate	Set HOCO to start oscillating with the HOCOCR register.
7	Wait for HOCO clock oscillation to stabilize	Polling until HOCOSF bit in OSCSF register is read as 1 (Oscillation is stable)
8	Wait for FLL stabilization	Wait for FLL stabilization wait time (t_{FLLWT}), or wait until the HOCO clock is measured to confirm that the frequency accuracy is stable.
9	Check the HOCO stabilization	Check that HOCOSF bit in OSCSF register is read as 1
10	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1 bit in PRCR register
11	End	FLL setting is completed. HOCO Clock is available.

Note 1. When HOCOLDOCR.LDOSTP bit is 1, do not operate HOCO.
If HOCOLDOCR.LDOSTP is switched from 1 to 0, waiting for the HOCO power supply to stabilize is required. See section 70, Electrical Characteristics on waiting time for the HOCO power supply to stabilize.
Note 2. Sub-Clock Oscillator must be oscillating with the stabilization.

Table 9.17 FLL setting flow for Software Standby mode transition and cancellation

No.	Step	Description
1	Start	The HOCO is oscillating with FLL is enabled
2	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1 bit in PRCR register
3	Stop HOCO*1	Set HOCO to stop with HOCOCR register.
4	Disable FLL	Set the FLL function to disable with FLLCR1 register.
5	WFI instruction*2	Transition to Software Standby mode with WFI instruction
6	Software Standby mode*2	MCU is in Software Standby mode
7	Cancellation Software Standby mode*2	Software Standby mode is cancelled
8	Set the FLL function to enable*3	Set the FLL Multiplication Control with FLLCR2 register. Set the FLL function to enable with FLLCR1 register.
9	Set HOCO to operate	Set HOCO to start oscillating with the HOCOCR register.
10	Wait for FLL stabilization	Wait for FLL stabilization wait time (t_{FLLWT}), or wait until the HOCO clock is measured to confirm that the frequency accuracy is stable.
11	Check the HOCO stabilization	Check that HOCOSF bit in OSCSF register is read as 1.
12	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1 bit in PRCR register
13	End	HOCO Clock is available.

Note 1. If HOCO is used as the system clock source or the clock source of PLL, these clock source must be changed to another clock before HOCO is stopped.
Note 2. See section 11, Low Power Mode for detail of Software Standby mode.
Note 3. Sub-Clock Oscillator must be oscillating with the stabilization.

9.11.6 PLL Setting

9.11.6.1 PLL Initial Clock Setting

Table 9.18 shows an example of the initial setting procedure for the PLL.

Table 9.18 Example of the PLL initial setting procedure after reset release / after Deep Software Standby cancellation (PLL1 or PLL2)

No.	Step	Description
1	Start	The PLL1 is stopped after reset release / Deep software standby cancellation. The PLL2 is stopped after reset release / Deep software standby cancellation.
2	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1 bit in PRCR register.
3	Change Operating Power Control Mode to High-Speed mode*1	Change to High-Speed mode with OPCCR register. For details, see section 11, Low Power Mode.
4	Check the PLL power supply*2	[PLL1] Check the following bit in the PLL1LDOCR register. ● LDOSTP bit is 0 (LDO is enabled). [PLL2] Check the following bit in the PLL2LDOCR register. ● LDOSTP bit is 0 (LDO is enabled).
5	Configure the PLL settings	[PLL1] Configure the following PLL1 settings in PLLCCR and PLLCCR2 registers. [PLL2] Configure the following PLL2 settings in PLL2CCR and PLL2CCR2 registers. [Setting Items] ● PLL input frequency division ratio ● PLL clock source ● Frequency Multiplication Factor ● Frequency Multiplication Fractional Factor ● PLL Output Frequency Division Ratio (P/Q/R)
6	Set PLL to operate	[PLL1] Set PLL1 to start oscillating with the PLLCR register. [PLL2] Set PLL2 to start oscillating with the PLL2CR register.
7	Wait for PLL clock oscillation to stabilize	[PLL1] Polling until PLLSF bit in OSCSF register is read as 1 (Oscillation is stable). [PLL2] Polling until PLL2SF bit in OSCSF register is read as 1 (Oscillation is stable).
8	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1 bit in PRCR register
9	End	PLL1 or PLL2 clock setting is completed. [PLL1] PLL1P, PLL1Q, PLL1R clocks are available. [PLL2] PLL2P, PLL2Q, PLL2R clocks are available.

Note 1. Setting is not necessary if the operating power control mode is already the High-Speed mode.
Note 2. When PLL1LDOCR.LDOSTP bit is 1, do not operate PLL1.
If PLL1LDOCR.LDOSTP is switched from 1 to 0, waiting for the PLL1 power supply to stabilize is required. See section 70, Electrical Characteristics on waiting time for the PLL1 power supply to stabilize.
When PLL2LDOCR.LDOSTP bit is 1, do not operate PLL2.
If PLL2LDOCR.LDOSTP is switched from 1 to 0, waiting for the PLL2 power supply to stabilize is required. See section 70, Electrical Characteristics on waiting time for the PLL2 power supply to stabilize.

9.11.6.2 PLL Clock Setting Change

Table 9.19 shows an example of the initial setting procedure for the PLL.

Table 9.19 Example of the PLL setting change procedure (PLL1 or PLL2) (1 of 2)

No.	Step	Description
1	Start	PLL1 is oscillating and the PLL1 clocks (PLL1P, PLL1Q, PLL1R) are being used. PLL2 is oscillating and the PLL2 clocks (PLL2P, PLL2Q, PLL2R) are being used.
2	Cancel Register Write Protection	Set 1 to PRC0 bit and PRC1 bit in PRCR register.

Table 9.19 Example of the PLL setting change procedure (PLL1 or PLL2) (2 of 2)

No.	Step	Description
3	Stop PLL*1	[PLL1] Set PLL1 to stop with PLLCR register. [PLL2] Set PLL2 to stop with PLL2CR register.
4	Change the PLL settings	[PLL1] Configure the following PLL1 settings in PLLCCR and PLLCCR2 registers. [PLL2] Configure the following PLL2 settings in PLL2CCR and PLL2CCR2 registers. [Setting Items] ● PLL input frequency division ratio ● PLL clock source ● Frequency Multiplication Factor ● Frequency Multiplication Fractional Factor ● PLL Output Frequency Division Ratio (P/Q/R)
5	Set PLL to operate	[PLL1] Set PLL1 to start oscillating with the PLLCR register. [PLL2] Set PLL2 to start oscillating with the PLL2CR register.
6	Wait for PLL clock oscillation to stabilize	[PLL1] Polling until PLLSF bit in OSCSF register is read as 1 (Oscillation is stable). [PLL2] Polling until PLL2SF bit in OSCSF register is read as 1 (Oscillation is stable).
7	Apply Register Write Protection	Set 0 to PRC0 bit and PRC1 bit in PRCR register.
8	End	PLL1 or PLL2 clock setting change is completed. [PLL1] PLL1P, PLL1Q, PLL1R clocks are available. [PLL2] PLL2P, PLL2Q, PLL2R clocks are available.

Note 1. If PLL1P clock is used as the system clock source, the system clock source must be changed to another clock source before PLL1 is stopped.

9.12 Usage Notes

9.12.1 Notes on Clock Generation Circuit

The frequency of the following clocks supplied to each module changes according to the setting of the SCKDIVCR and SCKDIVCR2 register:

- CPU0 clock (CPUCLK0)
- CPU1 clock (CPUCLK1)
- NPU clock (NPUCLK)
- MRAM code bus clock (MRICLK)
- System clock (ICLK)
- Peripheral module clocks (PCLKA, PCLKB, PCLKC, PCLKD and PCLKE)
- MRAM peripheral bus clock (MRPCLK)
- external bus clock (BCLK)

Each frequency must meet the following conditions:

- Each frequency must be selected within the operation-guaranteed range of the operating frequency (f) specified in the AC characteristics. See section 70, Electrical Characteristics.
- Each clock must be set according to Table 9.2.

- Do not change the clock frequency during external bus access. Furthermore, when access via the external bus is to start after a change to the clock frequency, only start access via the bus after confirming that the change to the frequencies has been completed.

To ensure correct processing after the clock frequency changes, first write to the relevant Clock Control register to change the frequency, then read the value from the register, and finally perform the subsequent processing.

9.12.2 Notes on Resonator

Because various resonator characteristics relate closely to your board design, adequate evaluation is required before use. See the resonator connection example in Figure 9.5. The circuit constants for the resonator depend on the resonator to be used and the stray capacitance of the mounting circuit. Therefore, consult the resonator manufacturer when determining the circuit constants. The voltage to be applied between the resonator pins must be within the absolute maximum rating.

9.12.3 Notes on Board Design

When using a crystal resonator, place the resonator and its load capacitors as close to the XTAL and EXTAL pins as possible. Other signal lines should be routed away from the oscillation circuit as shown in Figure 9.17 to prevent electromagnetic induction from interfering with correct oscillation. Figure 9.17 shows the case which the main clock oscillator is used. In case of sub-clock oscillator, it is also same as Figure 9.17.

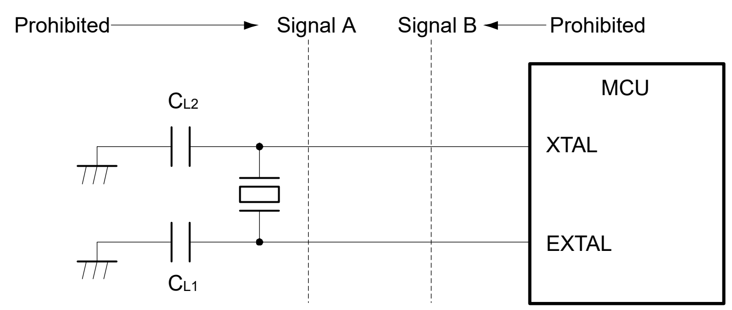


Figure 9.17 Signal routing in board design for oscillation circuit

9.12.4 Notes on Resonator Connect Pin

When the main clock is not used, the EXTAL and XTAL pins can be used as general ports. When these pins are used as general ports, the main clock must be stopped (MOSCCR.MOSTP bit should be set to 1).

When the sub-clock is not used, the XCIN and XCOUT pins can be used as general ports. When these pins are used as the general ports, the sub-clock must be stopped (SOSCCR.SOSTP bit should be set to 1).

9.12.5 Notes on Using Sub-Clock Oscillator

The output of the P212 (EXTAL), P213 (XTAL), P408 (for BGA-289pin) and P415 (for BGA-224pin) pins may affect the oscillation by the sub-clock oscillator.

If the sub-clock oscillator is used, implement board design so as not to affect the oscillation. Renesas strongly recommends setting the PmnPFS.DSCR[1:0] bits to 00b or 01b when using the P212 (EXTAL), P213 (XTAL), P408 (for BGA-289pin) and P415 (for BGA-224pin) as output pins and using the sub-clock oscillator.

In addition, when using the sub-clock oscillator in Low power mode drive capability (SOMCR.SODRV[1:0]= 01b, 10b and 11b), Renesas recommends setting the PmnPFS.DSCR[1:0] bits to 00b when using the P212 (EXTAL), P213 (XTAL), P408 (for BGA-289pin) and P415 (for BGA-224pin) as output pins and using the sub-clock oscillator.

9.12.6 Notes on Reset Conditions of SOSC and LOCO

RTC is not initialized by any reset source.

SOSC and LOCO can be selected as the clock source of RTC. The followings are the states of SOSC and LOCO when reset occurs.

Table 9.20 Reset Behavior of SOSC

Clock source	Item	Reset source	
		VBATT_POR reset	Other
SOSC	Enable or disable Drive capability	Initialized to enable	continue the state before the reset occurs

Table 9.21 Reset Behavior of LOCO

Clock source	Item	Reset source	
		Power-on reset/ Voltage-Monitoring 0, 4, 5 reset/ Core Voltage Monitor reset/ Deep software standby 2, 3 reset	Other
LOCO	Enable or disable	Initialized to enable	
	Oscillation accuracy*1	Initialized to accuracy before trimming by LOCOUTCR (accuracy : ±15 %)	Continue the accuracy that was trimmed by LOCOUTCR

Note 1. If the LOCO that is trimmed by LOCOUTCR is selected as the RTC source clock, please be careful that LOCO oscillation accuracy will be initialized by Power-on reset, Voltage-Monitor 0, 4, 5 reset, Core Voltage-Monitor reset and Deep software standby 2, 3 reset.

When a reset is canceled, a reset exception handling starts.

For details on the reset exception handling, see section 6, Resets